Emergence of spinor symmetry and supersymmetry from mixing of two bosonic times in generic AdS_5

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(Dated: April 6, 2026)

We study when the $\mathbb{Z}_2$ holonomy identified in the limited-scope two-time theorem is realized as a genuine odd symmetry on the AdS_5 boundary. The setting has two bosonic times—physical time and renormalization-group time—and assumes a local Schwinger–Keldysh bicomplex on sufficiently local observables, the limited-scope holonomy package, an RG-monotone/anomalous-descent package, parity transport together with a same-tick selected-line obstruction supplement and a delayed selected-line supplement at the successor step, a Peierls bracket, energy/covariance control, and a UV conformal common sector. Under these hypotheses, RG time produces an odd first-descendant class on the boundary. The limited-scope square-loop is transported to a sign action on a same-tick selected quotient image, while the delayed selected-line supplement furnishes a nonzero quotient point at the successor step. These two selected-line interfaces supply the nonvanishing input needed for canonicalization, so the odd class is nonzero and admits a unique canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$. Its Peierls Hamiltonian action defines a nontrivial local Γ-odd derivation $\delta_{\Lambda_k} = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}$. Using AQFT reconstruction together with the Haag–Łopuszański–Sohnius theorem, we obtain a finite-dimensional Weyl-spinor odd charge module on the UV vacuum representation. Combining this odd sector with the common conformal sector yields the full four-dimensional superconformal algebra on the same Hilbert space: $\mathfrak{su}(2,2|N)$ for $N \neq 4$, and $\mathfrak{psu}(2,2|4)$ in the $N = 4$ case after the standard central quotient. A separate strengthened set of assumptions provides an optional criterion selecting $N = 4$. The result gives a theorem-level route from bosonic two-time data to boundary spinorial and superconformal structure.

I. INTRODUCTION

We propose a framework in which the questions "why anything would happen" (against the constant von Neumann entropy wall of unitary AdS/CFT) and "why anything (in the sense of matter origin) would exist" can be considered together. This paper should be read together with its companion paper on Two Times Limited Scope Theorem (2T-LS) [1], which we recommend reading first.

The common structural theme arises when AdS_5/CFT_4 satisfies the assumptions of Two Times Main Theorem (2T-Main). The theorem then shows that, in such a two-time setting, the assumptions force structural components that induce an intrinsic odd sector whose realization has both dynamical consequences for time and symmetry consequences for matter. It is shown in 2T-LS that the classical ledger within broad unitary microscopic dynamics induces both fundamental Z_2-gradedness with cochain-level $d^2 = 0$ and a genuine distinction between "before" and "after". In this paper, we show that 2T-Main establishes spinor symmetry and superconformal boundary structure and, conjecturally, points toward a microscopic Type IIB realization. The further layout addresses the theorem-level realization problem inside this broader framework.

At theorem level the proposed 2T program has two stages. The first asks whether a two-time structure carries an intrinsic odd holonomy at all. The Limited Scope theorem answers that question affirmatively: the square-loop has odd parity, the induced $\mathbb{Z}_2$-cover does not split, and a detector makes the holonomy operationally distinguishable. The second stage asks when that odd holonomy survives the boundary reduction steps and becomes a genuine local odd channel with symmetry consequences. That second-stage realization problem is the subject of the present paper.

This distinction is essential. The Limited Scope theorem is a broad obstruction theorem. By itself it does not produce a sufficiently local odd generator, a Hilbert-space odd implementer, a Weyl-spinor transformation law, or a superalgebra. The theorem proved here is more physical: it singles out the class of AdS_5 boundary theories in which the LS odd holonomy reaches the renormalization-group anomaly sector, survives the local descendant quotient, remains visible on local probes, and can therefore be promoted to a genuine odd symmetry derivation.

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A. Landscape

The present work sits at the intersection of several literatures that are usually discussed separately. Standard AdS/CFT and holographic renormalization relate radial evolution to boundary renormalization-group structure and organize how conformal anomalies, counterterms, and boundary data are encoded in the bulk/boundary dictionary [2–7]. Lorentzian holography and Schwinger–Keldysh methods, meanwhile, make precise how real-time propagation, contour doubling, and BRST/cohomological structures enter holographic calculations [8–12]. Local RG work gives a complementary anomaly-based language for Weyl consistency and flow data [13, 14]. These developments organize physical time, RG time, and anomaly structure, but they do not relate a multiplicity of physical times to physical symmetries at all.

On the symmetry side, classification results such as Coleman–Mandula, Haag–Łopuszański–Sohnius, and Nahm tell us what extensions of spacetime and conformal symmetry are allowed once suitable odd generators and Hilbert-space hypotheses are already present [15–17]. Those results are indispensable for the endpoint of the present argument, but they are endpoint theorems: they do not explain how a relevant odd generator is created from bosonic bulk/boundary data, nor how it survives locality, quotient, and activity constraints.

There is also a substantial literature on emergent supersymmetry at boundaries and quantum critical points [18–20]. That literature is important because it shows that supersymmetry need not be a microscopic input and because it exhibits nontrivial obstructions to emergence. Its mechanisms are, however, different from the one studied here: they typically begin with fermionic degrees of freedom or tuned Yukawa/critical systems. Our starting point is instead a purely bosonic two-time setting equipped with the 2T-Limited Scope odd holonomy obstruction.

A distinct earlier program on two-time physics was developed by Bars and collaborators, where the second time is controlled by gauge symmetry and different one-time systems arise as gauge-fixed or "holographic" pictures of a higher-dimensional parent theory [21–23]. Within that framework there are explicit field-theoretic, gravitational, and supersymmetric constructions, including gauge/gravity backgrounds, $N = 1$ supersymmetric field theories, and superconformal/twistor-string realizations [24–27]. This line of work is important context for the present paper because it shows that two-time setups can consistently host rich supersymmetric structure. Its mechanism is nevertheless different from ours: the odd or supersymmetric sector is built into the microscopic two-time framework from the start, whereas here we ask when a purely bosonic two-time setting is forced to realize a local odd boundary channel with superconformal closure.

To our knowledge, the missing piece in this landscape is a theorem-level route from the LS odd holonomy to a local, physically active boundary odd derivation with superconformal closure on a single UV Hilbert space. The companion 2T-LS paper establishes the holonomy side [1]. The present paper studies the realization side: when and how that $\mathbb{Z}_2$-gradedness survives the boundary reduction steps and becomes symmetry.

B. Our Proposal

Our proposal is that the LS odd holonomy is not merely a square-loop obstruction. In an admissible two-time AdS_5 setting it can be carried all the way to the UV boundary and realized there as a concrete odd symmetry channel. The Main theorem identifies a realization chain with two explicit selected-line interfaces: a same-tick obstruction package controlling the square-loop no-extra-collapse step, and a delayed selected-line package furnishing a nonzero quotient point at the successor ledger step. RG time produces the odd descendant class; these selected-line interfaces and the activity package prevent its collapse or darkness; canonicalization and Hamiltonization turn it into a local odd derivation; and Hilbert-space plus common-sector closure produce Weyl-spinor charges and same-$\mathcal{H}$ superconformal symmetry.

In plain language, the paper does not say that any theory with two bosonic times becomes supersymmetric. It singles out the narrower class in which RG time leaves a genuine odd trace, that trace survives quotienting, remains locally visible, and can actually be promoted to symmetry. The result is therefore a criterion for when two bosonic times give rise to a nontrivial odd boundary channel with superconformal consequences.

The introduction-level roadmap is summarized in Fig. 1. Its nodes name the objects the theorem constructs, while its arrows name the operations that move from one stage to the next.

Here $[\Xi^{(1)}]$ denotes the odd first-descendant class produced by the RG interface, $\mathcal{J}^{\mathrm{can}}$ its derived canonical representative, and δ the resulting Hamiltonized odd derivation. Only this Hamiltonized derivation enters Haag–Łopuszański–Sohnius.

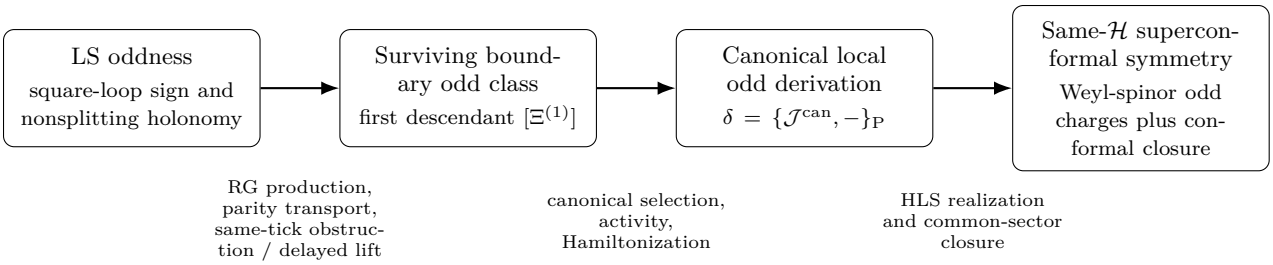

FIG. 1: Introduction-level roadmap for the Main theorem. The nodes indicate what kind of odd object is available at each stage; the arrows indicate what the theorem must still do in order to pass to the next stage.

The two bosonic times play fundamentally different roles. Physical time organizes real-time propagation, descent, and Ward identities. RG time organizes coarse-graining and produces the odd anomaly insertion that the theorem tracks. There is no general reason why these two times should yield odd boundary symmetry. The theorem isolates the required mechanism: RG time must produce a genuine odd shadow, and that shadow must survive the relevant reduction steps and remain active in the even sector.

The theorem excludes theories in which the odd shadow is absent, scheme-trivial, destroyed by the relevant quotient, or present only as a dynamically dark sector. This will be unpacked later in Section V; for now the important point is that the theorem identifies a bridge from the LS oddness to superconformal symmetry, not a universal statement about every two-time model.

The base theorem stops at same-$\mathcal{H}$ superconformal closure. That endpoint is already strong: the odd charges and conformal generators act on one and the same UV Hilbert space and close to the full four-dimensional superconformal algebra. The stronger claim forcing $\mathcal{N} = 4$ belongs to a separate optional criterion with additional assumptions. It should not be confused with the base realization theorem.

One further warning already matters at the level of orientation: ghost parity, physical Γ-parity, and the LS holonomy character ω must not be identified. Their precise roles are fixed in Section II before the main statement is given.

Accordingly, the proposal does not assume an initial odd AQFT derivation, a Hilbert-space odd implementer supplied at the start, Weyl-spinor generators or superalgebra closure inserted by hand, an a priori superconformal multiplet, or an identification of the physical-time BRST differential Q_t with a microscopic supercharge.

The paper is organized as follows. Section II formulates the two-time boundary setting, lays out the relevant failure modes, organizes the assumption packages by role, and states the main theorem. Section III then gives the transport chain itself, from the LS sign action to the Hamiltonized odd derivation. Section IV separates the base same-$\mathcal{H}$ superconformal closure theorem from the stronger optional maximality criterion. Sections V and VI are not afterthoughts: they explain what class of theories the theorem actually selects, what the LS input contributes inside that class, why the ledger question matters, and how a microscopic realization is expected to fit into the broader program.

II. SETTING, ASSUMPTIONS, AND MAIN THEOREM

We now formulate the theorem as a statement about a generic AdS_5 UV boundary equipped with two bosonic times. The assumptions are presented in package form because the theorem is not a one-step implication. It begins with the LS odd holonomy, passes through RG descendant formation and local selection, and ends only after Hamiltonization and Hilbert-space symmetry closure have been established.

A. Two bosonic times and one local target

We work with a five-dimensional AdS_5 background whose conformal boundary admits a four-dimensional Lorentzian local quantum field theory on a Minkowski UV chart. Physical time t and radial/RG time u are treated as two independent monotonic directions. Physical time acts through ordinary Lorentzian

evolution. RG time acts through the radial/Wilsonian direction. The Schwinger–Keldysh completion is assumed to make both time directions visible on the same sufficiently local class of boundary observables, so that the analysis can be carried out inside one local framework. This common framework is not a convenience. If physical time and RG time lived only on different descriptions, their odd descendants could fail to communicate and the theorem would have no coherent local target.

Within this framework the SK construction supplies a two-time BRST bicomplex (Q_t, Q_u) on sufficiently local observables. The operator Q_t belongs to the real-time SK structure and organizes descent and Ward identities. The operator Q_u belongs to the RG direction and is the source of the odd insertion that the theorem tracks. The two differentials are therefore not interchangeable. The theorem follows the RG-produced odd descendant $Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}$, not the physical-time differential Q_t itself.

Next point must be kept sharp: the LS chain bicomplex and the boundary BRST bicomplex are not identified. The LS theorem supplies an odd holonomy and a sign action; the Main theorem shows how that sign action reaches the local boundary descendant quotient. Only after parity transport, nonvanishing, and canonical selection does one obtain the Hamiltonized odd derivation $\delta = \{\mathcal{J}^{\mathrm{can}}, -\}_{\mathrm{P}}$ that enters the Hilbert-space symmetry theorem.

B. Failure modes and package architecture

At this stage neither supersymmetry nor a local odd charge is assumed. The problem is instead to determine whether the two-time data force a genuine odd boundary channel with physical content. Four obstructions can prevent this from happening:

(i) RG time may fail to produce a genuine odd first descendant once scheme contributions are removed;

(ii) the LS sign action may fail to survive transport into the local descendant quotient;

(iii) a surviving odd class may remain physically dark, acting trivially on all sufficiently local even probes;

(iv) even a nontrivial active odd derivation may fail to satisfy the locality, covariance, energy, or implementability requirements needed for a Hilbert-space symmetry theorem and conformal closure.

The package stack is organized exactly around these failure modes. One part of the stack supplies the LS holonomy input, a second part produces and stabilizes the local odd descendant class, a third part promotes that class to a genuine derivation, and a final part places the resulting derivation inside the Hilbert-space and conformal framework where HLS and same-$\mathcal{H}$ closure apply.

C. Package synopsis and main statement

The assumption packages are best read in four gates rather than as one undifferentiated bundle.

- **LS oddness.** The package $\mathbf{2T}_{\mathrm{LS}}$ supplies the parity cocycle, the nonsplitting $\mathbb{Z}_2$-cover, the detector-based distinguishability statement, and the square-loop sign action. This is the oddness that the Main theorem attempts to realize on the boundary.

- **Descendant formation.** The package $\mathbf{2T}_{\mathrm{RG}}$ provides an even sufficiently local representative $\mathcal{O}^{(0)}$, decomposes the RG odd operator into physical and scheme parts, and isolates the genuine odd insertion $Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}$. Together with $\mathbf{2T}_{\mathrm{char}}^{\mathrm{weak}}$, it places the resulting odd data in the affine descendant class $[\Xi_{\Lambda_k}^{(1)}]$ where one can ask whether a genuine odd direction survives.

- **Survival and visibility.** The supplement $\mathbf{2T}_{\mathrm{sel}}$ provides the same-tick obstruction lane on the abstract selected quotient, while $\mathbf{2T}_{\mathrm{sel}}^{\mathrm{delay}}$ furnishes the delayed quotient-first route at the successor ledger step. Together they keep the selected odd channel alive at the right abstract and physical interfaces. The supplement $\mathbf{2T}_{\mathrm{char}}^{\mathrm{act}}$ requires this selected odd direction to be visible to at least one sufficiently local even probe, so that the surviving odd channel is not physically dark. The Peierls package $\mathbf{2T}_{\mathrm{Peierls}}$ then converts the selected sufficiently local representative into a genuine derivation on observables.

- **Boundary symmetry closure.** The energy-bounds/covariance package $\mathbf{2T}_{\mathrm{EBC}}$ supplies the UV locality, covariance, vacuum, and energy-control hypotheses needed for AQFT reconstruction and for the verification of the Haag–Łopuszański–Sohnius interface. The conformal common-sector package $\mathbf{2T}_{\mathrm{conf}}$ ensures that the conformal generators and the odd charges act on one vacuum representation and one common invariant core, so that the odd symmetry sector closes with the $SO(4,2)$ generators on a single Hilbert space.

Two structural remarks are crucial. First, $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ is not a separate primitive assumption. It is derived from $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ and supplies the Hilbertizing form used to select the canonical representative $\mathcal{J}^{\mathrm{can}}$. Second, the BRST, AQFT, and HLS structures are not independent axioms inserted by hand. The SK construction supplies the BRST bicomplex, the UV locality/covariance data supply the AQFT representation, and the Hamiltonized odd derivation is then shown to satisfy the HLS hypotheses. The odd derivation and the superconformal algebra are therefore derived consequences of the package hypotheses.

Remark II.1 (Three $\mathbb{Z}_2$ parities (do not mix))**.** We keep three distinct $\mathbb{Z}_2$ parities throughout the analysis: *ghost parity* (SK/BRST cohomological degree mod 2), *physical parity* Γ (an involutive $*$-automorphism on the observable net, defining Γ-even and Γ-odd sectors), and the *Limited-Scope holonomy* $\omega \in \mathbb{Z}_2$ of the procedure cover, represented on observables by Γ^{ω}. Ghost parity is bookkeeping, whereas Γ and ω are physical. The LS chain bicomplex $(Q^{\mathrm{ch}}_t, Q^{\mathrm{ch}}_u)$ and the boundary BRST bicomplex (Q_t, Q_u) are not identified; they interact only through the parity-transport step described in Appendix G.

Theorem II.1 (2T-Main theorem, informal)**.** *Consider a generic AdS_5 setup whose UV boundary admits a four-dimensional local quantum field theory together with the assumption packages*

$$\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}_{\mathrm{RG}} + \mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}_{\mathrm{EBC}} + \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}} + \mathbf{2T}^{\mathrm{act}}_{\mathrm{char}} + \mathbf{2T}_{\mathrm{sel}} + \mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}} + \mathbf{2T}_{\mathrm{conf}},$$

and suppose that the two-time Schwinger–Keldysh construction furnishes the corresponding BRST bicomplex on sufficiently local observables. Then the following statements hold.

(i) *The RG package produces a genuine odd first-descendant class $[\Xi^{(1)}_{\Lambda_k}]$, and the LS square-loop is transported by parity to a sign action on its selected quotient image $\overline{\Xi}^{(1)}_{\Lambda_k}$.*

(ii) *The same-tick selected-line supplement provides the obstruction route on that abstract quotient, while the delayed selected-line supplement supplies a nonzero quotient point at the successor ledger step. Together they provide the nonvanishing input used by the derived calibration package to select a unique nonzero canonical odd representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$.*

(iii) *The activity supplement shows that this canonical odd line is not locally dark, and the Peierls bracket Hamiltonizes $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ into a nontrivial local Γ-odd derivation*

$$\delta_{\Lambda_k} = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}.$$

(iv) *The AQFT and Haag–Łopuszański–Sohnius hypotheses are satisfied by this Hamiltonized derivation. Hence the UV boundary carries a finite-dimensional Weyl-spinor odd charge module on the same vacuum representation, with indexed generators Q^I_α and $\bar{Q}_{\dot\alpha I}$ satisfying*

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2\,\delta^I{}_J\,\sigma^\mu_{\alpha\dot\beta} P_\mu, \qquad \{Q^I_\alpha, Q^J_\beta\} = \{\bar{Q}_{\dot\alpha I}, \bar{Q}_{\dot\beta J}\} = 0;$$

(v) *Combining the odd charge module with the common conformal sector yields the full four-dimensional superconformal algebra on the same Hilbert space. More precisely, the resulting operator Lie superalgebra is $\mathfrak{su}(2,2|N)$ for $N \neq 4$, while in the $N = 4$ case the standard simple form is obtained after quotienting the central $\mathfrak{u}(1)$, giving $\mathfrak{psu}(2,2|4)$.*

The base conclusion of the theorem is therefore same-$\mathcal{H}$ superconformal closure. The odd sector is not inserted by hand. It is reconstructed from the RG descendant, stabilized by the same-tick obstruction and delayed selected-line supplements, made dynamically meaningful by activity and Hamiltonization, and only then promoted to Hilbert-space symmetry by HLS.

This is the natural endpoint of the base realization theorem. Forcing $N = 4$ is a separate strengthening problem. It uses additional structural assumptions on the UV fixed point and should not be folded back into the assumption burden of the base theorem.

Proposition II.1 (Strengthened maximality criterion)**.** *Assume, in addition to the hypotheses of Theorem II.1, that the UV fixed point is non-gravitational with a unique stress tensor, genuinely interacting in the sense that the SK RG monotone has a nondegenerate UV extremum and no decoupled free SCFT sector is present, and carries no higher-spin enhancement. Assume also the standard spin-2 saturation input that a unitary four-dimensional superconformal theory with $N > 4$ and a unique stress tensor must contain a graviton and therefore cannot occur in the non-gravitational setting, and assume that interacting non-gravitational SCFTs exist for each $N \in \{1, 2, 3, 4\}$.*

Finally impose the following maximality principle*: among SK-admissible UV fixed points compatible with the same macroscopic data (dimension, locality, spin range, and the standing two-time structure), the RG dynamics selects those with the largest available unbroken symmetry. In particular, once the allowed superconformal algebras have been reduced to $\mathfrak{su}(2, 2|N)$ with $1 \leq N \leq 4$, the realized fixed point takes the largest admissible N; a smaller value of N cannot be compensated by adding further continuous symmetries without changing the universality class.*

Then the superconformal multiplicity parameter is forced to be $N = 4$. This yields a separate optional criterion for maximal $\mathcal{N} = 4$ supersymmetry, as summarized in Proposition K.2.

Section III now proves the transport-and-realization chain announced above. Section IV establishes the exact closure endpoint and keeps it separate from maximality. Only after that theorem spine is in place do Sections V and VI return to the broader interpretive questions: what class of theories is really being selected, what the LS input contributes inside that class, and what a concrete ledger realization would have to supply.

III. FROM LS HOLONOMY TO A BOUNDARY ODD DERIVATION

The core logic of the theorem is best read as a transport-and-realization chain. The LS theorem gives an odd sign action. The Main theorem shows how that sign action reaches the selected descendant quotient, forces nonvanishing, produces a canonical odd representative, and finally becomes a local odd derivation with symmetry consequences. Each step removes a distinct obstruction.

(1) **LS parity transport.** The Limited Scope theorem provides the square-loop sign action at the record/process level. This is only the beginning. The Main theorem still has to transport that sign action to the boundary descendant sector. The parity-transport step identifies the LS square-loop action with physical parity on first descendants, but this transport is meaningful only after one specifies the selected local quotient in which the boundary odd channel is to live.

(2) **RG time produces the odd descendant class.** The package $\mathbf{2T}_{\mathrm{RG}}$ provides an even sufficiently local representative $\mathcal{O}^{(0)}$, decomposes the RG odd operator into physical and scheme parts, and isolates the genuine odd insertion $Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}$. The anomalous descent relation

$$Q_u^{\mathrm{phys}}\mathcal{O}^{(0)} + Q_t\Xi^{(1)} = \mathcal{A}^{(1)}$$

is the first place where bosonic RG time produces the odd class $[\Xi^{(1)}_{\Lambda_k}]$ used by the theorem. This is why the theorem follows the RG anomaly sector rather than treating Q_t as the symmetry generator.

(3) **The same-tick obstruction package realizes the LS sign action on the abstract selected quotient.** The supplement $\mathbf{2T}_{\mathrm{sel}}$ isolates a nonzero odd line in the descendant ambiguity group and provides the obstruction contract required to transport the LS sign action to the selected quotient image $\overline{\Xi}^{(1)}_{\Lambda_k}$. This is the precise abstract bridge from the LS holonomy landscape to the Main descendant quotient.

(4) **The delayed selected-line package gives the delayed-step nonvanishing route.** The supplement $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$ furnishes a nonzero quotient point at the successor step together with the delayed split-to-descendant map and the separate dominant-response gate. This is the physically relevant odd-to-even export interface. With the same-tick obstruction route and the delayed route both present, the odd class cannot collapse away before canonicalization.

(5) **The nonzero class yields a canonical odd representative and a local odd derivation.** The derived calibration supplement supplies the Hilbertizing form needed to choose the unique norm-minimizing

representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$. The activity supplement then guarantees that this distinguished odd direction is not locally dark. The Peierls package Hamiltonizes it to

$$\delta_{\Lambda_k} = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}.$$

This is the point at which the odd content stops being a quotient class and becomes a local odd derivation on observables.

(6) **HLS promotes the Hamiltonized derivation to a Weyl-spinor odd charge module.** The energy-bounds/covariance package, together with the UV locality data, places the theory in the setting where AQFT reconstruction and Haag–Łopuszański–Sohnius apply. HLS does not act on Q_t; it acts on the Hamiltonized derivation δ. The conclusion is a finite-dimensional Weyl-spinor odd charge module with indexed generators Q^I_α and $\bar{Q}_{\dot\alpha I}$.

(7) **The odd charge module closes with the conformal sector on the same Hilbert space.** The conformal common-sector package ensures that the conformal generators and the odd charges act on one vacuum representation and one common invariant core. The result is same-$\mathcal{H}$ superconformal closure. Only after this base closure theorem is in place does it make sense to ask whether stronger assumptions force $N = 4$.

Two remarks summarize the mechanism. First, the theorem's central generator is not Q_t but δ. The role of Q_t is to organize descent and Ward identities. Second, the same-tick obstruction supplement, the delayed selected-line supplement, and the activity supplement solve different problems. The first gives the abstract no-extra-collapse route, the second gives the quotient route through the successor ledger step, and the third prevents the surviving odd channel from remaining physically dark. The logic is therefore rigid: removing any one of these steps breaks a different part of the chain.

IV. SUPERCONFORMAL CLOSURE VERSUS MAXIMALITY

The intrinsic endpoint of the base theorem is same-$\mathcal{H}$ superconformal closure. This means that the conformal generators inherited from the AdS_5 boundary geometry and the odd generators obtained from the Hamiltonized descendant sector act on one and the same UV Hilbert space. The result is already strong: it is not merely the existence of an odd charge module, but the closure of that module with the full four-dimensional conformal algebra.

At this stage the theorem yields the precise distinction

$$\mathfrak{su}(2,2|N) \quad (N \neq 4), \qquad \mathfrak{psu}(2,2|4) \quad (N = 4).$$

For $N = 4$ the simple superalgebra is obtained only after quotienting the central $\mathfrak{u}(1)$, which is why the natural final form is $\mathfrak{psu}(2,2|4)$.

This closure theorem is already substantial even before any maximality claim is made. It says that the odd channel reconstructed from the RG descendant is not an isolated algebraic artifact; it integrates coherently with the full four-dimensional conformal symmetry on the physical UV Hilbert space.

It is mathematically essential to separate this closure theorem from the stronger criterion forcing $N = 4$. The stronger criterion uses additional assumptions: non-gravitationality, spin-2 saturation, interaction, absence of higher-spin enhancement, and the existence and maximality conditions recorded in the strengthening analysis. These assumptions restrict the UV fixed point in ways that go beyond the odd-channel realization problem itself.

For this reason the theorem naturally has two layers of conclusion.

(i) The base theorem yields same-$\mathcal{H}$ superconformal closure and determines the superconformal algebra up to the multiplicity parameter N.

(ii) The strengthened maximality criterion imposes additional structural assumptions and then forces $N = 4$.

These layers provide clear distinction in assumption burden so the logical structure of the result is transparent.

V. PHYSICAL MEANING AND SCOPE

A. What class of theories is selected?

These assumptions do not single out "supersymmetric theories" in any direct or circular way. What they single out is a narrower and more precise class: theories in which the bosonic two-time structure supports a nontrivial odd shadow that survives all reduction steps and remains physically active on the UV boundary. The restrictive assumptions are therefore concentrated in three places: the RG package, the selected-line supplement, and the activity supplement.

(1) **The theorem excludes a purely bosonic RG sector.** If the RG monotone can be described, after removing scheme terms, without a genuine odd insertion $Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}$, then the theory lies outside the admissible class. The point is not that every bosonic theory is ruled out, but that the class treated here must have an odd cohomological shadow attached to its RG time.

(2) **The theorem excludes a scheme-trivial odd shadow.** If the odd RG insertion is killed by Q_t-exactness or can be absorbed entirely into scheme redefinitions, then the odd channel carries no invariant physical content and the theorem does not apply. Thus the odd descendant must encode more than a bookkeeping artifact of the renormalization scheme.

(3) **The theorem excludes collapse before the local descendant quotient.** If a candidate odd line exists only in a coarser raw packet but dies when one passes to the descendant quotient, then the relevant selected-line package fails: either the same-tick obstruction route $\mathbf{2T}_{\mathrm{sel}}$ is not available on the abstract selected carrier, or the delayed route $\mathbf{2T}_{\mathrm{sel}}^{\mathrm{delay}}$ does not produce the quotient point at the successor ledger step. The theorem therefore requires not just abstract odd data, but odd data that survives the precise quotient procedure relevant to the local boundary theory.

(4) **The theorem excludes a physically dark odd sector.** If the selected odd line survives abstractly but acts trivially on every even sufficiently local probe, then $\mathbf{2T}_{\mathrm{char}}^{\mathrm{act}}$ fails and the theory is excluded. This is a minimal nondegeneracy requirement: the odd direction must be dynamically visible somewhere in the local even sector.

The physical slogan is therefore the following. The theorem applies to theories in which bosonic RG time produces a genuine odd descendant, one selected odd component survives the ambiguity and collapse steps, and that surviving component is detectable through local even observables. This is a considerable restriction, but it is still weaker than assuming a fermionic symmetry from the outset.

The selected class may therefore be described as proto-fermionic rather than supersymmetric in the usual input sense. These theories are distinguished not because they begin with a supercharge, but because their two-time structure is rich enough to support an odd sector that remains stable under all the physically relevant reductions and can no longer be ignored dynamically.

B. What does the LS input do inside the selected class?

Entering the admissible class, the LS input supplies the physical picture of two times in our paradigm. It introduces a *classical ledger of records*, a reversible microscopic time evolution between ledger checkpoints, and an *RG/coarse-graining operation that forgets fine-grained information* returning the world to the ledger point. These two times are ledger-mediated: one carries microscopic evolution between checkpoint, while the other changes how much of that evolution survives in the recorded history.

LS then isolates the key fact about the interaction of these two times. A ledger by itself is not enough: if everything microscopic were already fixed by the visible record, then "live, then forget" and "forget, then live" would be the same story in two notations. But the two-time square to be genuinely nontrivial; it means there must be a *completion sector* beyond the ledger, carrying microscopic information that is not recorded at the checkpoint.

On the ledger itself, the square still closes: the same endpoint record can be reached either way. But one must also specify how the unrecorded microscopic completion data are attached to that common corner. That completion rule is what LS calls a *filler*. LS singles out two canonical fillers, and their difference

is exactly the physical point: they agree on the same visible ledger step, so they must be invisible there, yet they are not the same microscopic realization. Their mismatch survives in the completion sector as an irreducible $\mathbb{Z}_2$ sign. That sign is objective. It is not a matter of notation, state relabeling, or endpoint bookkeeping.

LS does not select the theory class. Instead, it provides the underlying source of oddness: a genuine mismatch between two filler-related microscopic completions of one and the same visible ledger history. Main then asks for the narrower class in which this hidden completion-sector difference survives the reduction chain and reaches the UV boundary as an active odd channel.

C. But what is the ledger?

The ledger is not the full microscopic state. It is the accepted classical record sector: the commuting facts that survive coarse-graining and define what counts as the same visible history at a checkpoint. Abstractly, what these records are made of can be ignored. We use only the structural claim that such a classical record layer exists and is rich enough so the Universe can perform RG/CPTP coarse-graining step, and that the completion sector carries the microscopic data that the ledger forgets.

These abstract claim level have considerable implications from 2T-LS and 2T-Main.

- Wheeler's slogan "it from bit" becomes theorem-level statement [28];
- $\mathbb{Z}_2$-gradedness surfaces in two deeply linked appearances:
  - objective distinction between before and after *compatible with unitary evolution*; hence the real arrow of time that circumvents the hard wall of the constant von Neumann entropy.
  - *spinor symmetry* originating at the UV boundary; hence existence of the matter-rich Universe.
- confluence of two appearances at a conformal boundary, conjecturally producing IIb microscopic theory in the standard AdS_5/CFT_4 setup.

This is why the ledger question matters so much. If nature really contains such a classical record layer, then the same structure would underwrite the whole chain from highest-level principle to microscopic theory.

The present article stops one step short of identifying the ledger concretely. That is a cosmology theory question, and it is deferred to the discussion.

VI. DISCUSSION

A. Toward a cosmology realization

A concrete answer to the ledger question requires a source theory that identifies, in one microscopic construction, the record sector, the forgetting/coarse-graining operation, and the completion data whose mismatch carries the LS sign.

One proposed theory constructs ledger as a Weyl anomaly floor sector of a pure vacuum Karch-Randall AdS_4 brane, which acts as a receiver that accumulates Brown-York flux from the incoming zero-area repeating pulses so that brane is physically holographically renormalized to a higher cutoff, moving in radial direction. [1]

The remaining anomaly floor is (a) classical, in the sense of being a local function on the brane and not an interacting field, and (b) preserves already stored facts due to widths band-locking. [2] A Fisher-odd direction feeds complementary sector that carries the odd completion residue not visible in the current reduced receiver. Such a split af action into receiver wall-move, anomaly floor influx, and completion sector

[1] The governing principle is the least SK influence action that is uniquely minimized with the information export pulses of the temporally/spatially LoG/LoG shapes and receiver cutoff-bandlocked widths.

[2] Notably, AdS_5 with three spatial dimensions is the minimal space which can contain a ledger of this type: even-d spaces does not contain a Weyl anomaly, while AdS_3 anomaly floor is not a proper physical information storage, but rather a gauge/bookkeeping direction, and it explains how dimension-independent $\mathbb{Z}_2$-gradedness is realized in spinor symmetry only at $d \geq 3$.

update makes two latter targets independent: the two filler-related realizations in the completion sector remain indistinguishable in the receiver anomaly floor imprint.

However, unlike the ledger, the completion sector influx is affine and bears increasing cost, so eventually the real wall-moving solution ceases to exist and completion/receiver complex splits. In this split, that completion sector becomes the newborn source dS_4 brane. [3]

This sequence is CCC-like: the chain of dS_4 observable Universes with an approximately conformal inheritance. Its ledger step is *aeon*: the period from pre-split to pre-split finals of Universes, when the source coherences is maximally exhausted. So the accumulated $\mathbb{Z}_2$ difference is eventually read and appears in the ledger, but on the *next step*. This is the *delayed zig-zag* picture: invisible on the current ledger, carried through the completion sector, and first readable at the next receiver. Such a picture can be understood as a rough mirror of the 2T-LS ledger requirement: it provides the concrete physical mechanism by which two times mix and create the true holonomy over history.

Readers who want the full implementation package, including the 2T-LS and 2T-Main interface implementations, exact audit reports, and 2T-LS formal proof in Agda, should consult the archival Zenodo workspace release [29].

B. Outlook

In the light of 2T-Main results, several directions are proposed.

1. Obvious direction is strenghtning of the optional $\mathcal{N} = 4$ step. It is expected that one could be resolved on the level of an implementation theory, hence search for such a theory can solidify the vertical chain proposed in Section V C.

2. Another direction is strengthening of Main proof strategy. Currently, the post-Haag–Łopuszański–Sohnius same-$\mathcal{H}$ conformal closure step is existential only. Envisioned post-HLS transfer theorem could show that the package $Q, \bar{Q}, S, \bar{S}$ itself is induced from the reduced data of implemention theory+2T-Main.

3. Finaly, and this direction could be partially following from the first one, the post-HLS rigidity theorem could show that after transfer there is no hidden modulus beyond the candidate sign involution.

If such a program is finished, we could state that the microscopic-level Universe is $\mathbb{Z}_2$-lift of its own history, accepted to the ledger.

The main conclusion, however, is already clear. In the presence of the assumption packages described above, two bosonic times can do more than generate an abstract odd holonomy. They can force that oddness into a surviving, locally effective boundary channel, and that channel can be promoted to a superconformal symmetry algebra on the UV Hilbert space.

That is the sense in which the result should be read. It is not a universal statement about every theory with two times, and it is not a hidden assumption of supersymmetry in alternative language. It is a structural theorem identifying a precise bridge from bosonic two-time data to odd boundary symmetry. Once that bridge exists, the symmetry consequences are far more rigid than the microscopic starting point would have suggested.

Appendix A: Guide to the logic

This appendix isolates the minimal interfaces and implications needed to reach the HLS classification step from the standing interface packages. The guide below follows the proof dependence rather than the raw appendix order: preparatory subprograms are listed where they first enter the argument.

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[3] *Source* here means that the oddness of completion sector surfaces as a brane coherence stock created by the split that feeds the next receiver due to QND-type leaks to the dS_4 canonical commutant, which is geometrically situated between source and receiver branes along with the new completion sector.

(i) **Standing package bundle.** The standing assumptions are packaged as $\mathbf{2T}_{\mathrm{Main}}$ (Definition B.20), namely

$$\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}_{\mathrm{RG}} + \mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}_{\mathrm{EBC}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}} + \mathbf{2T}^{\mathrm{act}}_{\mathrm{char}} + \mathbf{2T}_{\mathrm{sel}} + \mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}} + \mathbf{2T}_{\mathrm{conf}}.$$

The record interface $\mathbf{2T}_{\mathrm{rec}}$ and detector interface $\mathbf{2T}_{\mathrm{det}}$ enter only through the LS summand, while the derived BRST package will be added below to form $\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}}$.

(ii) **Subprogram 1a (derived BRST).** Starting from the Lorentzian SK path integral, Subprogram 1a constructs the two-time BRST/bicomplex differentials Q_t, Q_u and their Ward identities on SK–admissible states, thereby producing the interface $\mathbf{2T}_{\mathrm{BRST}}$ on the sufficiently local class $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. At that point the appendix spine works with

$$\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}} := \mathbf{2T}_{\mathrm{Main}} + \mathbf{2T}_{\mathrm{BRST}}.$$

(iii) **Subprogram 2a (Results from the 2T Limited Scope Theorem).** We import the LS parity cocycle, the nonsplitting $\mathbb{Z}_2$-cover, the detector witness, and the record-process-chain bicomplex identities. This isolates the abstract two-time obstruction data that later feed the parity-transport step.

(iv) **Subprogram 2b (Cosmology theory Peierls bracket and swap invariance).** We record the abstract Peierls package on sufficiently local boundary observables, its swap-equivariant Green operators, and the resulting Poisson automorphism property of the boundary swap involution.

(v) **Subprogram 2c (Parity transport and physical $\mathbb{Z}_2$ grading).** Combining the LS import with the swap-equivariant boundary structures, we identify the physical parity involution Γ on first descendants and transport the LS square-loop sign to the boundary ambiguity sector.

(vi) **Subprogram 3a (selected-line nonvanishing, canonicalization, and characteristic activity).** This subsection packages the finite-cutoff canonical odd representative together with the same-tick obstruction lane, the delayed selected-line lane, the resulting nonvanishing statements, and the characteristic-algebra/activity consequences that follow once the odd descendant class survives.

(vii) **Subprogram 3b (RG monotone and Hamiltonized odd derivation).** The RG package $\mathbf{2T}_{\mathrm{RG}}$ supplies the monotone and its anomalous descent source, while the Peierls and characterization packages are used to Hamiltonize the canonical odd representative and establish the local physically odd derivation δ_{Λ_k}.

(viii) **Subprogram 1b (local net, GNS reconstruction, and implementation template).** Subprogram 1b supplies the vacuum Wightman/AQFT representation $(\mathcal{H}, \pi, U, \Omega)$, the net $O \mapsto \mathfrak{A}(O)$, and the dense implementation core on the Minkowski boundary. These Hilbert-space inputs first enter when the argument passes from the local odd derivation to the AQFT/HLS step.

(ix) **Subprogram 3c (AQFT net and HLS on the Minkowski boundary).** Using $\mathbf{2T}_{\mathrm{EBC}}$ together with the Subprogram 1b vacuum representation, Subprogram 3c extends δ to the polynomial core, proves implementability of the odd derivation, verifies the HLS interface $\mathbf{2T}_{\mathrm{HLS}}$, and identifies the finite-dimensional Weyl-spinor odd charge module on the UV vacuum representation.

(x) **Subprogram 4a (dilatation weight $\frac{1}{2}$ of the odd charges on the common UV sector).** Combining the HLS odd module from Subprogram 3c, the HLS completeness statement, and the conformal common-sector package $\mathbf{2T}_{\mathrm{conf}}$, Subprogram 4a places the odd charges on the common UV sector and forces their dilatation weight to be $\frac{1}{2}$.

(xi) **Subprogram 4b (same-$\mathcal{H}$ conformal closure of the HLS odd sector).** With the weight-$\frac{1}{2}$ normalization from Subprogram 4a in hand, Subprogram 4b takes the finite conformal closure of the HLS odd seed, proves same-$\mathcal{H}$ implementability of that odd conformal closure, recovers the degree-zero internal algebra, and identifies the resulting operator superalgebra with the standard 4D superconformal algebra.

(xii) **Subprogram 4c (optional maximality strengthening).** This final section is not part of the base Main theorem. It imports the superconformal closure from Subprogram 4b and studies what additional assumptions would force maximal supersymmetry, i.e. the optional $N = 4$ conclusion at the UV fixed point.

Appendix B: Master packages and interface assumptions

1. Master package block

a. Input Interfaces

Definition B.1 (Record two–time interface $\mathbf{2T}_{\mathrm{rec}}$)**.** A *record two–time interface* consists of:

(R1) SK and records. Sets $\mathcal{S}_{\mathrm{SK}}$ (SK states) and $\mathcal{S}_{\mathrm{rec}}$ (record states), a map $\mathrm{diag} : \mathcal{S}_{\mathrm{rec}} \to \mathcal{S}_{\mathrm{SK}}$, and an involution $\mathsf{swap} : \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{SK}}$ such that $\mathsf{swap}^2 = \mathrm{id}_{\mathcal{S}_{\mathrm{SK}}}$ and $\mathsf{swap} \circ \mathrm{diag} = \mathrm{diag}$.

(R2) Two arrows. A group T and the additive monoid $(\mathbb{R}_{\geq 0}, +, 0)$, with actions $\mathrm{act}_T : T \times \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{SK}}$ and $\mathrm{act}_U : \mathbb{R}_{\geq 0} \times \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{SK}}$ satisfying the action axioms and *weak commutativity* on $\mathcal{S}_{\mathrm{SK}}$:

$$\mathrm{act}_U(u, \mathrm{act}_T(t, x)) = \mathrm{act}_T(t, \mathrm{act}_U(u, x)).$$

(R3) Strict RG diagonalization. For each $u > 0$ a map $\Pi_u : \mathcal{S}_{\mathrm{SK}} \to \mathcal{S}_{\mathrm{rec}}$ and a record RG action $R_u : \mathcal{S}_{\mathrm{rec}} \to \mathcal{S}_{\mathrm{rec}}$ such that

$$\mathrm{act}_U(u, x) = \mathrm{diag}(\Pi_u(x)), \qquad \Pi_u(\mathrm{diag}\, r) = R_u(r) \quad (u > 0),$$

with $R_0 = \mathrm{id}$ and $R_{u_2+u_1} = R_{u_2} \circ R_{u_1}$.

(R4) Chosen RG sector (reachability). A section $\mathrm{sec}_\Pi : \mathcal{S}_{\mathrm{rec}} \to \mathcal{S}_{\mathrm{SK}}$ with $\Pi_1(\mathrm{sec}_\Pi(r)) = r$.

(R5) No mixed anomaly at the boundary of squares. A bisimplicial set $\mathcal{C}^{\mathrm{SK}}_{m,n}$ with horizontal and vertical face maps d^H_i, d^V_j satisfying simplicial relations in each direction and mixed commutation $d^H_i d^V_j = d^V_j d^H_i$ [30].

(R6) Record reflection. Diagonal cancellation:

$$\mathrm{diag}(r) = \mathrm{diag}(r') \Rightarrow r = r'.$$

Remark B.1. The discrete RG-step case is recovered by restricting $u \in \mathbb{N} \subset \mathbb{R}_{\geq 0}$.

Definition B.2 (Detector interface $\mathbf{2T}_{\mathrm{det}}$)**.** A *detector interface* is any additional structure that implies the $\mathbb{Z}_2$–action on the difference object $\mathsf{Diff} = \mathrm{Cofib}(\mathrm{diag})$ is nontrivial, i.e. $\mathsf{flip} \neq \mathrm{id}_{\mathsf{Diff}}$ or equivalently there exists $y \in \mathsf{Diff}$ with $\mathsf{flip}(y) \neq y$.

Definition B.3 (Limited Scope package $\mathbf{2T}_{\mathrm{LS}}$)**.** Define the *2T Limited Scope package* by

$$\mathbf{2T}_{\mathrm{LS}} := \mathbf{2T}_{\mathrm{rec}} + \mathbf{2T}_{\mathrm{det}}.$$

Definition B.4 (Two independent $\mathbb{Z}_2$ gradings)**.** We distinguish:

(a) *Cohomological/BRST parity* $|\cdot|_{\mathrm{ch}} \in \mathbb{Z}_2$ on the cochain algebra $\mathcal{F}^\Lambda_{\mathrm{s.loc}} = (\mathcal{F}^\Lambda_{\mathrm{s.loc}})_{\bar{0}} \oplus (\mathcal{F}^\Lambda_{\mathrm{s.loc}})_{\bar{1}}$, with $|Q_t|_{\mathrm{ch}} = |Q_u|_{\mathrm{ch}} = 1$. Accordingly, $H^0(\mathcal{F}^\Lambda_{\mathrm{s.loc}}, Q_t)$ and $H^1(\mathcal{F}^\Lambda_{\mathrm{s.loc}}, Q_t)$ denote even/odd cohomology.

(b) *Physical parity* $|\cdot|_\Gamma \in \mathbb{Z}_2$ induced by the involution Γ (SK swap), giving a decomposition into Γ–even/odd parts $\mathcal{F}^\Lambda_{\mathrm{s.loc}} = {\mathcal{F}^\Lambda_{\mathrm{s.loc}}}^+ \oplus {\mathcal{F}^\Lambda_{\mathrm{s.loc}}}^-$.

Oddness in the AQFT/HLS steps always refers to $|\cdot|_\Gamma$.

Definition B.5 (RG–monotone interface $\mathbf{2T}_{\mathrm{RG}}$)**.** An *RG–monotone interface* consists of:

(M1) Monotone and representative. A real monotone $M(u)$ near the UV ceiling u_0, a family of SK–admissible states ρ_u, and a quasi–local family $\mathcal{O}^{(0)}(u) \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ with

$$M(u) = \langle \mathcal{O}^{(0)}(u) \rangle_{\rho_u}, \qquad Q_t \mathcal{O}^{(0)}(u) = 0, \qquad [\mathcal{O}^{(0)}(u_0)] \neq 0 \in H^0(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t).$$

(M2) RG Ward identity (odd form, refined). On the monotone representative $\mathcal{O}^{(0)}(u)$, decompose the u–differential as

$$Q_u = Q_u^{\text{phys}} + Q_u^{\text{scheme}},$$

where Q_u^{scheme} encodes scheme/reparametrisation effects (local counterterm and coupling redefinition ambiguities). We assume that Q_u^{phys} and Q_u^{scheme} are odd derivations on $\mathcal{F}^{\Lambda}_{\text{s.loc}}$. We additionally assume Q_u^{phys} and Q_u^{scheme} are *presheaf derivations*: for every double cone O they restrict to odd derivations

$$Q_u^{\text{phys}} : \mathcal{F}^{\Lambda}_{\text{s.loc}}(O) \to \mathcal{F}^{\Lambda}_{\text{s.loc}}(O), \qquad Q_u^{\text{scheme}} : \mathcal{F}^{\Lambda}_{\text{s.loc}}(O) \to \mathcal{F}^{\Lambda}_{\text{s.loc}}(O).$$

Finally, we assume they anticommute with Q_t on this class (so $Q_u^{\text{phys}} \mathcal{O}^{(0)}$ is Q_t–closed whenever $Q_t \mathcal{O}^{(0)} = 0$), and that for u near u_0,

$$\frac{d}{du} M(u) = -\big\langle (Q_u^{\text{phys}} \mathcal{O}^{(0)})(u) \big\rangle_{\rho_u}, \qquad \big\langle (Q_u^{\text{scheme}} \mathcal{O}^{(0)})(u) \big\rangle_{\rho_u} = 0,$$

up to expectation values of Q_t–exact terms (which vanish for SK–admissible states by **2T**$_{\text{BRST}}$-**(B2)**).

Cohomological triviality of scheme terms on the monotone sector. For every $X \in \mathcal{F}^{\Lambda}_{\text{s.locmon}}$ with $Q_t X = 0$, we assume

$$[\, Q_u^{\text{scheme}} X \,] = 0 \quad \text{in} \quad H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t).$$

(Optional packaging.) Equivalently, one may encode the same information by an even derivation L_{RG} on $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ (Callan–Symanzik plus anomaly/contact terms) with $\frac{d}{du} M(u) = \langle (L_{\text{RG}} \mathcal{O}^{(0)})(u) \rangle_{\rho_u}$.

(M3) Anomalous first descent (no Ward trivialization). There exist sufficiently local raw first-descendant potentials $\Xi^{(1)}(u) \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ and an anomaly insertion $\mathcal{A}^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\text{s.loc}})_{\bar{1}}$ such that

$$Q_u^{\text{phys}} \mathcal{O}^{(0)}(u) \;+\; Q_t \Xi^{(1)}(u) \;=\; \mathcal{A}^{(1)}(u), \qquad Q_t \mathcal{A}^{(1)}(u) = 0. \tag{B1}$$

We interpret (B1) as an identity in the locality quotient (Definition B.7); in particular, $\mathcal{A}^{(1)}(u)$ represents the Q_t–cohomology class $\big[\, Q_u^{\text{phys}} \mathcal{O}^{(0)}(u) \,\big] \in H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t)$ (Definition B.8). For SK–admissible states, $\langle Q_t X \rangle_{\rho_u} = 0$ implies $\langle (Q_u^{\text{phys}} \mathcal{O}^{(0)})(u) \rangle_{\rho_u} = \langle \mathcal{A}^{(1)}(u) \rangle_{\rho_u}$ and hence $\frac{d}{du} M(u) = -\langle \mathcal{A}^{(1)}(u) \rangle_{\rho_u}$. This replaces an anomaly–free descent $Q_u^{\text{phys}} \mathcal{O}^{(0)} + Q_t \Xi^{(1)} = 0$, which would force $\frac{d}{du} M(u) = 0$ by the Q_t Ward identity. We furthermore require that the chosen first-descendant potential at the UV ceiling is not a c-number on the vacuum sector (i.e. not constant on the connected vacuum component of the reduced phase space).

(M4) Nontrivial UV slope. $\partial_u M(u)|_{u_0} < 0$ away from fixed points / exactly marginal loci.

Definition B.6 (Monotone sector of sufficiently local observables)**.** Fix the UV ceiling u_0 and choose $\varepsilon > 0$. For each double cone O, define $\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O) \subset \mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ to be the smallest $\mathbb{R}$–subspace such that:

(i) for all $u \in (u_0 - \varepsilon, u_0 + \varepsilon)$, the chosen monotone representatives $\mathcal{O}^{(0)}(u)$ restricted to O lie in $\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)$;

(ii) the associated anomaly insertions $\mathcal{A}^{(1)}(u)$ and chosen raw first-descendant potentials $\Xi^{(1)}(u)$ (whenever defined) restricted to O lie in $\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)$;

(iii) $\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)$ is Γ–invariant and Q_t–stable: $\Gamma(\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)) \subset \mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)$ and $Q_t(\mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)) \subset \mathcal{F}^{\Lambda}_{\text{s.locmon}}(O)$.

Here Γ denotes the swap-induced involution on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ introduced in Section G. We write $\mathcal{F}^{\Lambda}_{\mathrm{s.loc\,mon}} := \bigcup_O \mathcal{F}^{\Lambda}_{\mathrm{s.loc\,mon}}(O)$.

Definition B.7 (Locality quotient relation)**.** Fix a double cone O. On $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ define an equivalence relation $\simeq$ generated by the following elementary moves:

(a) $F \mapsto F + Q_t X$ for any $X \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$;

(b) $F \mapsto F + (Q_u^{\mathrm{scheme}} Y)$ for any $Y \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc\,mon}}(O)$;

(c) $F \mapsto F + \partial_\mu K^\mu$ for any sufficiently local current K^μ supported in O (so the term vanishes after smearing in double cones);

(d) $F \mapsto F + \Delta_{\mathrm{ct}}$ where $\Delta_{\mathrm{ct}} \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ is a local counterterm coboundary (supported in O) representing a scheme change.

Write $[F]_{\mathrm{loc}}$ for the $\simeq$–class of F.

Definition B.8 (Anomaly insertion as the Q_t–cohomology class)**.** Assume $Q_t \mathcal{O}^{(0)}(u) = 0$ and that $\{Q_t, Q_u^{\mathrm{phys}}\}\mathcal{O}^{(0)}(u) = 0$ for u near u_0 (equivalently, $Q_t(Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}(u)) = 0$). Define the anomaly class

$$[\mathcal{A}^{(1)}(u)] \;:=\; \big[\, Q_u^{\mathrm{phys}} \mathcal{O}^{(0)}(u) \,\big] \;\in\; H^1(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t),$$

and choose a sufficiently local representative $\mathcal{A}^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\mathrm{s.loc}})_{\bar{1}}$ of this class. Scheme/reparametrisation contributions are absorbed into the locality quotient $\simeq$ (Definition B.7) and are assumed cohomologically trivial on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc\,mon}}$ by Definition B.5-**(M2)**.

Definition B.9 (Peierls interface $\mathbf{2T}_{\mathrm{Peierls}}$)**.** A *Peierls interface* consists of the following properties:

(P1) Renormalized SK differentiability. The renormalized SK generating functional $W[J_+, J_-]$ exists and is twice differentiable near $J_a = 0$ on a specified space of boundary sources (including the gravitational sources).

(P2) Causal Green structure. The gauge–fixed linearized renormalized bulk operator is Green–hyperbolic with the chosen AdS boundary conditions, so advanced/retarded propagators exist. The induced boundary retarded kernel

$$G_{\mathrm{ret}}(x, y) := \frac{\delta^2 W}{\delta J_a(x)\, \delta J_r(y)}\Big|_{J_a = 0}$$

has retarded support, and G_{adv} is defined analogously.

(P3) Renormalized symplectic form. The renormalized covariant symplectic current ω_{ren} exists and yields a finite conserved symplectic form Ω_{ren} compatible with the causal propagator $\Delta := G_{\mathrm{ret}} - G_{\mathrm{adv}}$.

(P4) Gauge reduction and independence on invariants. Pure gauge directions are exactly the kernel of Ω_{ren}, so Ω_{ren} descends to a symplectic form on the reduced phase space. The induced Peierls bracket on gauge–invariant functionals is independent of the gauge–fixing choice (equivalently, it is canonical on the reduced phase space).

(P5) Closed locality class. There exists a specified algebra $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ of gauge–invariant "sufficiently local" boundary functionals, together with a locality presheaf $O \mapsto \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ (closed under smearing with smooth compact support kernels), such that the Peierls bracket is well-defined and satisfies Leibniz/Jacobi on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ (and restricts to each $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$).

(P6) Pointer-real swap equivariance. There exists a swap-induced involution Γ_* on the chosen linearized field, test-source, and boundary-data spaces such that

$$P_{\mathrm{gf,ren}}\Gamma_* = \Gamma_* P_{\mathrm{gf,ren}}, \qquad B_\Lambda \Gamma_* = \Gamma_* B_\Lambda,$$

the source insertion and boundary extraction maps are Γ_*–equivariant, and the pairing used in the Peierls formula is Γ_*–invariant.

Remark B.2 (Meaning of $\mathbf{2T}_{\mathrm{Peierls}}$)**.** The package $\mathbf{2T}_{\mathrm{Peierls}}$ means that the gauge-fixed linearized bulk problem admits the causal Green structure needed for advanced and retarded propagators, with admissible boundary and junction conditions; that the renormalized symplectic current yields a finite reduced symplectic form with no extra degeneracies beyond pure gauge; that the resulting Peierls bracket is well-defined on the sufficiently local boundary class; and that the pointer-real swap involution acts equivariantly on the linearized renormalized IBVP and its Peierls pairing. These are the healthiness and symmetry properties encoded abstractly by **(P2)**–**(P6)**.

Definition B.10 (Energy bounds and covariance interface $\mathbf{2T}_{\mathrm{EBC}}$ (EB1–EB3 + Poincaré covariance))**.** An *energy bounds and covariance interface* $\mathbf{2T}_{\mathrm{EBC}}$ consists of the following data and properties on a Minkowski UV chart (uniformly in the UV cutoff):

(0i) a vacuum state ϖ on the local Borchers–Uhlmann net generated by $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$, with vacuum GNS triple $(\mathcal{H}, \pi, \Omega)$;

(0ii) a strongly continuous positive-energy unitary representation U_g of the Poincaré group implementing an induced action α_g on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ and fixing the vacuum vector, $U_g\Omega = \Omega$, with time-translation generator $H := P^0$.

In this vacuum representation, the following estimates and covariance identities hold:

(i) **EB1 (local generators).** For every double cone $O \subset \mathbb{R}^{1,3}$ and every $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$, there exist $s \geq 0$ and $C_{F,O} > 0$ such that on the vacuum core $\mathcal{D}_0$ (Definition D.3) one has

$$\|\pi(\Phi(F))\psi\| \;\leq\; C_{F,O}\,\|(1+H)^s\psi\| \qquad (\psi \in \mathcal{D}_0).$$

(ii) **EB2 (Sobolev continuity).** The causal operator Δ extends to a bounded map between the Sobolev spaces controlling functional derivatives on each O.

(iii) **EB3 (δ energy bound; dependence on the generator).** Fix a first-descendant potential $\mathcal{J} \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O_{\mathcal{J}})$ and define $\delta_{\mathcal{J}}(F) := \{\mathcal{J}, F\}_{\mathrm{P}}$ on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. (We extend $\delta_{\mathcal{J}}$ to the Borchers–Uhlmann core by $\delta_{\mathcal{J}}(\Phi(F)) := \Phi(\delta_{\mathcal{J}}F)$ and the graded Leibniz rule.) Then for every double cone O there exist $s \geq 0$ and a constant $C_{O,\mathcal{J}} > 0$ such that for all $A \in \mathfrak{A}_{\mathrm{fin}} \cap \mathfrak{A}_{\mathrm{BU}}(O)$ one has

$$\|\pi(\delta_{\mathcal{J}}(A))\Omega\| \;\leq\; C_{O,\mathcal{J}}\,\|(1+H)^s\,\pi(A)\Omega\|.$$

(iv) **Covariance (Peierls bracket).** In the Minkowski chart, the Peierls bracket is Poincaré covariant:

$$\alpha_g(\{F,G\}_{\mathrm{P}}) \;=\; \{\alpha_g F, \alpha_g G\}_{\mathrm{P}} \qquad (F, G \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}),$$

where α_g is the induced action on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ by pushforward of test kernels and fields.

Remark B.3 (Uniformity in EB3)**.** In the proof of EB3, the bound is *uniform in the cutoff* and *uniform in the tested observable* once $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ is measured by the same family of locality seminorms (then transported to energy norms via EB1/EB2). However, the constant depends on the chosen Hamiltonian generator through finitely many Sobolev/seminorm controls on its first functional derivative. Accordingly, one may either (i) fix $\mathcal{J}$ once and for all and write $C_{O,\mathcal{J}}$, or (ii) restrict to an admissible class of $\mathcal{J}$ with uniform seminorm bounds, in which case one may choose C_O uniformly within that class.

Remark B.4. The interface $\mathbf{2T}_{\mathrm{EBC}}$ is formulated on a Minkowski UV chart with the cutoff-uniform energy bounds, Poincaré covariance, and package theorem needed below.

Definition B.11 (EB3-admissible first-descendant potentials)**.** Fix $\sigma > \frac{5}{2}$ and a double cone $O_{\mathcal{J}}$. A first-descendant potential $\mathcal{J} \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O_{\mathcal{J}})$ is called *EB3-admissible* if its first functional derivative exists and satisfies a cutoff-uniform Sobolev bound

$$\mathcal{J}^{(1)} \in H^{\sigma}_0(O_{\mathcal{J}}), \qquad \|\mathcal{J}^{(1)}\|_{H^{\sigma}(O_{\mathcal{J}})} \;\leq\; C^{(\sigma)}_{\mathcal{J},O_{\mathcal{J}}},$$

where $C^{(\sigma)}_{\mathcal{J},O_{\mathcal{J}}}$ depends only on the defining locality seminorms/kernels of $\mathcal{J}$ and is independent of the UV cutoff for all sufficiently large cutoff.

Lemma B.1 (Locality ⇒ EB3-admissibility of $\mathcal{J}$)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$*-**(P5)** and let* $\mathcal{J} \in \mathcal{F}^{\Lambda}_{\text{s.loc}}(O_{\mathcal{J}})$*. Then* $\mathcal{J}$ *is EB3-admissible in the sense of Definition B.11.*

Proof. By $\mathbf{2T}_{\text{Peierls}}$**-(P5)**, the sufficiently local class $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O_{\mathcal{J}})$ is closed under the functional-derivative operations used in the Peierls construction (in particular, first functional derivatives exist and are supported in the same double cone). Thus the first functional derivative $\mathcal{J}^{(1)}$ exists, is supported in $O_{\mathcal{J}}$, and is controlled by finitely many locality seminorms of $\mathcal{J}$ uniformly in the UV cutoff (this is the locality input used in the EB3 estimate). Translating these locality seminorm bounds into Sobolev norms yields, for the fixed $\sigma > \frac{5}{2}$ of Definition B.11,

$$\mathcal{J}^{(1)} \in H^{\sigma}(O_{\mathcal{J}}), \qquad \|\mathcal{J}^{(1)}\|_{H^{\sigma}(O_{\mathcal{J}})} \le C^{(\sigma)}_{\mathcal{J},O_{\mathcal{J}}},$$

with $C^{(\sigma)}_{\mathcal{J},O_{\mathcal{J}}}$ independent of the cutoff. Since $\mathcal{J}^{(1)}$ is supported in $O_{\mathcal{J}}$, it lies in $H^{\sigma}_0(O_{\mathcal{J}})$, proving EB3-admissibility. □

Definition B.12 (Weak polarized characterization interface $\mathbf{2T}^{\text{weak}}_{\text{char}}$)**.** A *weak polarized characterization interface* on the ledger-step skeleton $\{\Lambda_k\}_{k\in K_{\text{acc}}}$ consists, for each ledger step k (with K_{acc} denoting the retained ledger-step index set), of the following data:

(C1) Odd generator space. A real vector space

$$E^{-}_{\Lambda_k} := \{F \in \mathcal{F}^{\Lambda}_{\text{s.loc}} : \Gamma_{\Lambda_k} F = -F\}$$

of physically odd sufficiently local generators.

(C2) Descendant ambiguity subspace. A linear subspace

$$\mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k}.$$

(C3) Admissible odd descendant class. An affine odd class

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k},$$

where $J_{0,\Lambda_k} \in E^{-}_{\Lambda_k}$ is any chosen representative.

(C4) Orientation functional. A continuous linear functional

$$\chi_{\Lambda_k} : E^{-}_{\Lambda_k} \to \mathbb{R}$$

such that

$$\chi_{\Lambda_k}(A) = 0 \qquad \forall\, A \in \mathcal{A}^{-}_{\Lambda_k},$$

and χ_{Λ_k} is constant and nonzero on the affine class $[\Xi^{(1)}_{\Lambda_k}]$.

(C5) Orientation sign. A sign

$$\varepsilon_{\Lambda_k} \in \{\pm 1\}$$

such that

$$\chi_{\Lambda_k}(J) = \varepsilon_{\Lambda_k} \qquad \forall\, J \in [\Xi^{(1)}_{\Lambda_k}].$$

Definition B.13 (Calibration supplement $\mathbf{2T}^{\text{cal}}_{\text{char}}$)**.** A *calibration supplement* over $\mathbf{2T}^{\text{weak}}_{\text{char}}$ consists, for each ledger step k, of a symmetric positive definite bilinear form

$$\mathsf{B}_{\Lambda_k} : E^{-}_{\Lambda_k} \times E^{-}_{\Lambda_k} \to \mathbb{R}$$

whose induced norm Hilbertizes $E^{-}_{\Lambda_k}$, and for which $\mathcal{A}^{-}_{\Lambda_k}$ is closed in the induced Hilbert topology. In particular, the affine class $[\Xi^{(1)}_{\Lambda_k}]$ is a closed affine subspace of the associated Hilbert completion.

Definition B.14 (Characterization-activity supplement $\mathbf{2T}^{\mathrm{act}}_{\mathrm{char}}$)**.** A *characterization-activity supplement* over $\mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ consists, for each ledger step k, of a linear subspace

$$\mathcal{P}^{+}_{\Lambda_k} \subset E^{+}_{\Lambda_k} := \{A \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}} : \Gamma_{\Lambda_k} A = A\}$$

of physically even sufficiently local probes such that for every $F \in E^{-}_{\Lambda_k}$,

$$\{F, A\}_{\mathrm{P},\Lambda_k} = 0 \qquad \forall\, A \in \mathcal{P}^{+}_{\Lambda_k}$$

implies

$$\chi_{\Lambda_k}(F) = 0.$$

Remark B.5 (Operational reading of $\mathbf{2T}^{\mathrm{act}}_{\mathrm{char}}$)**.** Definition B.14 says that the characterization functional χ_{Λ_k} is not merely an orientation on the odd ambiguity quotient: it is detected by a sufficiently local even probe sector. Equivalently, if $F \in E^{-}_{\Lambda_k}$ satisfies $\chi_{\Lambda_k}(F) \neq 0$, then there exists $A \in \mathcal{P}^{+}_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ with

$$\{F, A\}_{\mathrm{P},\Lambda_k} \neq 0.$$

This is a targeted activity statement for the distinguished odd direction, not a global Poisson-separation axiom on the full reduced phase space.

Definition B.15 (Characterization-algebra supplement $\mathbf{2T}_{\mathrm{alg}}$)**.** A *characterization-algebra supplement* consists, for each ledger step k, of a distinguished subalgebra

$$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$$

attached to the derived odd generator.

Definition B.16 (Aggregate polarized characterization interface $\mathbf{2T}_{\mathrm{char}}$)**.** Define the aggregate characterization package by

$$\mathbf{2T}_{\mathrm{char}} := \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}} + \mathbf{2T}_{\mathrm{alg}}.$$

Remark B.6 (Scope of $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$)**.** Definition B.12 contains only the weak characterization data for the descendant class. Additional ambiguity-control or reduction statements may be formulated separately, but they are not part of $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ itself.

Remark B.7 (Role of the weak characterization interface)**.** The weak interface $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ specifies the abstract descendant class

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k}.$$

It does not identify a preferred odd generator representative. The calibration form B_{Λ_k} may be any admissible Hilbertizing form on $E^{-}_{\Lambda_k}$ compatible with this weak interface. At the level of the base theorem, orientation is introduced canonically only after nonvanishing, via Proposition H.3. The characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$ is isolated separately as the supplement $\mathbf{2T}_{\mathrm{alg}}$. Stronger identifications among the orientation functional, the characterization algebra, and active-generator data belong to an optional stronger supplement, not to the base interface.

Theorem B.1 (Every weak characterization interface admits a calibration supplement)**.** *Assume the weak polarized characterization interface* $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ (Definition B.12)*. Then the calibration supplement* $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ (Definition B.13) *holds.*

Proof. Fix a ledger step k. Since $\mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k}$ is a linear subspace, choose an algebraic complement C_{Λ_k} so that

$$E^{-}_{\Lambda_k} = \mathcal{A}^{-}_{\Lambda_k} \oplus C_{\Lambda_k}.$$

Choose Hamel bases of $\mathcal{A}^-_{\Lambda_k}$ and C_{Λ_k}, declare the union of those bases orthonormal, and extend bilinearly. This gives a symmetric positive definite bilinear form B_{Λ_k} on $E^-_{\Lambda_k}$.

By construction, the induced norm identifies $E^-_{\Lambda_k}$ with a pre-Hilbert space whose Hilbert completion contains $\mathcal{A}^-_{\Lambda_k}$ as a closed subspace, because $\mathcal{A}^-_{\Lambda_k}$ is orthogonal to C_{Λ_k}. Therefore the affine class $[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k}$ is a closed affine subspace in the induced Hilbert topology. This is exactly the statement of Definition B.13. □

Definition B.17 (Selected-line supplement package $\mathbf{2T}_{\mathrm{sel}}$)**.** Fix a ledger step k, and let $\mathcal{V}^1$ be the first-descendant ambiguity group of Definition G.3. The supplementary package

$$\mathbf{2T}_{\mathrm{sel}}$$

consists of:

(i) a nonzero selected odd line

$$L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k} \subset \mathcal{V}^1;$$

(ii) a selected representative

$$J^{\mathrm{sel}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}] \cap E^-_{\Lambda_k};$$

(iii) the selected-line compatibility implication

$$q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}) = 0 \quad \Longrightarrow \quad [J^{\mathrm{sel}}_{\Lambda_k}]_{\mathcal{V}^1} = 0. \tag{B2}$$

Remark B.8 (Same-tick versus delayed role of $\mathbf{2T}_{\mathrm{sel}}$)**.** The package $\mathbf{2T}_{\mathrm{sel}}$ supplies the same-tick selected-line obstruction contract used in the square-loop no-extra-collapse argument. It governs the same-tick selected quotient. The corresponding delayed passage to the successor step is carried by the separate supplement $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$.

Remark B.9 (Why $\mathbf{2T}_{\mathrm{sel}}$ is separate)**.** The packages $\mathbf{2T}_{\mathrm{sel}}$ and $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$ govern different selected-line steps and are therefore kept as separate summands of $\mathbf{2T}_{\mathrm{Main}}$. The former supplies the same-tick obstruction input used in Theorem H.3; the latter supplies the delayed lift to the successor step together with its nondegeneracy clause.

Definition B.18 (Delayed selected-line supplement package $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$)**.** Fix consecutive steps k and $k+1$. The delayed selected-line supplement package consists of:

(i) a delayed selected line

$$\widehat{L}^{\mathrm{sel,delay}}_k = \mathbb{R}\,\widehat{\ell}^{\mathrm{delay}}_k \subset \tau^{\mathrm{comp,delay}}_k;$$

(ii) a delayed quotient line

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} = \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}};$$

(iii) a linear delayed split-to-descendant map

$$\mathfrak{L}^{\mathrm{split}}_{k\to k+1} : \widehat{L}^{\mathrm{sel,delay}}_k \longrightarrow \mathfrak{D}^-_{\Lambda_{k+1}}$$

satisfying

$$\mathfrak{L}^{\mathrm{split}}_{k\to k+1}(\widehat{\ell}^{\mathrm{delay}}_k) = \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}};$$

(iv) a descendant representative

$$J^{\text{split}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^{-}_{\Lambda_{k+1}}$$

with

$$q_{\Lambda_{k+1}}(J^{\text{split}}_{\Lambda_{k+1}}) = \overline{\ell}^{\text{Main}}_{\Lambda_{k+1}};$$

(v) dominant activity / nondegeneracy clause on the delayed image:

$$\overline{\rho}^{\text{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\text{Main}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}} \neq 0.$$

Definition B.19 (UV conformal common-sector interface $\mathbf{2T}_{\text{conf}}$)**.** A *UV conformal common-sector interface* on the same vacuum representation $(\mathcal{H}, \pi, \Omega)$ as $\mathbf{2T}_{\text{EBC}}$ consists of:

(CF1) a dense UV-local conformal core $\mathcal{D}_{\text{conf}} \subset \mathcal{H}$ and UV-local subalgebras

$$\mathfrak{A}_{\text{conf}}(O) \subset \mathfrak{A}(O)$$

for double cones $O \subset \mathbb{R}^{1,3}$, with $\pi(\mathfrak{A}_{\text{conf}}(O))\Omega \subset \mathcal{D}_{\text{conf}}$;

(CF2) a strongly continuous unitary representation $U_{\text{conf}}(g)$ of the connected conformal group $\text{Conf}_0(\mathbb{R}^{1,3})$ on $\mathcal{H}$, fixing the vacuum, extending the Poincaré representation of $\mathbf{2T}_{\text{EBC}}$, and acting geometrically on the UV-local core:

$$U_{\text{conf}}(g)\, \pi(\mathfrak{A}_{\text{conf}}(O))\, U_{\text{conf}}(g)^{-1} \;=\; \pi(\mathfrak{A}_{\text{conf}}(gO));$$

(CF3) a common Stone core $\mathcal{D}_{\text{Stone}} \subset \mathcal{D}_{\text{conf}}$ for the selfadjoint generators $P_\mu, M_{\mu\nu}, D, K_\mu$ of the translation, Lorentz, dilatation, and special-conformal one-parameter subgroups, satisfying on $\mathcal{D}_{\text{Stone}}$ the standard conformal commutation relations, in particular

$$[D, P_\mu] = iP_\mu, \qquad [D, K_\mu] = -\,iK_\mu, \qquad [K_\mu, P_\nu] = 2i(\eta_{\mu\nu}D - M_{\mu\nu}),$$

together with the usual Lorentz commutators of $M_{\mu\nu}$ with P_ρ, K_ρ, and $M_{\rho\sigma}$;

(CF4) on the compactly smeared scalar UV class, the infinitesimal dilatation subgroup in (CF2) agrees with the physical radial/RG derivation.

Remark B.10 (Why $\mathbf{2T}_{\text{conf}}$ is isolated)**.** The package $\mathbf{2T}_{\text{conf}}$ isolates the UV conformal/common-sector hypothesis. It requires that the conformal generators $P_\mu, M_{\mu\nu}, D, K_\mu$ and the odd charges act on the same vacuum representation and common core, which is the structural input needed for the Lorentz finite-type and superconformal arguments.

Remark B.11 (Physical meaning of $\mathbf{2T}_{\text{conf}}$)**.** The package $\mathbf{2T}_{\text{conf}}$ provides an abstract same-representation conformal interface on the UV vacuum sector. Its content is that the UV-local net carries a conformal enhancement whose dilatation subgroup agrees with the physical radial/RG action on the scalar UV class.

Definition B.20 (Main package $\mathbf{2T}_{\text{Main}}$)**.** Define the *2T Main theorem package* by

$$\mathbf{2T}_{\text{Main}} := \mathbf{2T}_{\text{LS}} + \mathbf{2T}_{\text{RG}} + \mathbf{2T}_{\text{Peierls}} + \mathbf{2T}_{\text{EBC}} + \mathbf{2T}^{\text{cal}}_{\text{char}} + \mathbf{2T}^{\text{act}}_{\text{char}} + \mathbf{2T}_{\text{sel}} + \mathbf{2T}^{\text{delay}}_{\text{sel}} + \mathbf{2T}_{\text{conf}}.$$

b. Spine Interface

Definition B.21 (Local two–time BRST interface $\mathbf{2T}_{\text{BRST}}$)**.** A *local two–time BRST interface* consists of:

(B1) A net (or presheaf) of sufficiently local observable cochains $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ on the UV boundary, equipped with two (cohomological) differentials Q_t (physical time) and Q_u (RG/radial) such that on each O:

$$Q_t^2 = 0, \qquad Q_u^2 = 0, \qquad Q_tQ_u + Q_uQ_t = 0.$$

(B2) A class of *SK–admissible* states ρ such that expectation values of Q_t–exact operators vanish:

$$\langle Q_t X \rangle_\rho = 0.$$

Definition B.22 (Main spine package)**.** Define the *Main spine package* by

$$\mathbf{2T}^{\text{spine}}_{\text{Main}} := \mathbf{2T}_{\text{Main}} + \mathbf{2T}_{\text{BRST}}.$$

Remark B.12. The package $\mathbf{2T}_{\text{BRST}}$ is obtained in Subprogram 1a rather than taken as an independent standing assumption. Thus $\mathbf{2T}^{\text{spine}}_{\text{Main}}$ abbreviates the hypothesis $\mathbf{2T}_{\text{Main}}$ together with the derived bicomplex $(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t, Q_u)$ on which $\mathbf{2T}_{\text{RG}}$ is interpreted.

Remark B.13 (Derived BRST bicomplex)**.** Definition B.21 specifies the local two–time BRST/bicomplex structure used in the descent arguments. The package $\mathbf{2T}_{\text{BRST}}$ is not taken as an independent standing assumption: Subprogram 1a derives the bicomplex and the Q_t Ward identity from the bulk SK construction.

c. Internal Interfaces

Definition B.23 (AQFT interface $\mathbf{2T}_{\text{AQFT}}$)**.** An *AQFT interface* on the Minkowski UV boundary consists of:

(Q1) A Haag–Kastler net $O \mapsto \mathfrak{A}(O)$ of local *-algebras (dense Borchers–Uhlmann core) on $\mathbb{R}^{1,3}$ with isotony and locality.

(Q2) A Poincaré action $\alpha_{(a,\Lambda)}$ by *-automorphisms with strong continuity on the vacuum representation.

(Q3) A vacuum state ϖ on $\mathfrak{A}$ that is Poincaré invariant and yields a GNS triple $(\mathcal{H}, \pi, \Omega)$.

(Q4) Spectrum condition: the translation generators P^μ satisfy $\text{spec}(P) \subset \overline{V}_+$.

(Q5) **(H4) OS/positivity input.** The chosen renormalized counterterm scheme yields reflection positivity / temperedness on the boundary theory so that ϖ is positive and the GNS construction applies.

Remark B.14. We use the standard Haag–Kastler/AQFT setup [31] and treat the BU algebra as a dense core for the local algebras.

Definition B.24 (HLS interface $\mathbf{2T}_{\text{HLS}}$)**.** An *HLS interface* consists of a nontrivial local derivation δ on $\mathfrak{A}_{\text{loc}}$, odd with respect to the physical Γ–grading, such that:

(H1) δ is a graded *-derivation (for the Γ–grading), local, and densely defined.

(H2) **Physical oddness.** δ is odd with respect to the physical Γ–grading:

$$\Gamma \circ \delta \circ \Gamma^{-1} = -\delta.$$

(H3) **Covariance (translations + Lorentz finite type).** δ commutes with translations,

$$\alpha_a \circ \delta \circ \alpha_a^{-1} = \delta \qquad (a \in \mathbb{R}^{1,3}),$$

and its Lorentz orbit spans a finite-dimensional module: writing $\delta_\Lambda := \alpha_\Lambda \circ \delta \circ \alpha_\Lambda^{-1}$ for $\Lambda \in SO^+(1,3)$, the real linear span $\mathscr{D} := \text{span}_{\mathbb{R}}\{\delta_\Lambda\}$ is finite-dimensional.

(H4) $\varpi \circ \delta = 0$ on $\mathfrak{A}^+$ (vacuum invariance on the Γ–even subalgebra), and δ is implementable by closable operators on a common invariant core.

Remark B.15 (Derived AQFT and HLS input)**.** Definitions B.23 and B.24 record the hypotheses needed to apply the HLS classification step. They are not taken as independent parts of $\mathbf{2T}_{\text{Main}}$: Subprogram 1b derives the Wightman/AQFT representation of the UV boundary theory, and Subprogram 3c verifies the HLS admissibility properties for the constructed odd derivation δ.

2. Boundary interface data

At the base level one assumes the weak interface data

$$\big(E^-_{\Lambda_k}, \mathcal{A}^-_{\Lambda_k}, [\Xi^{(1)}_{\Lambda_k}], \chi_{\Lambda_k}, \varepsilon_{\Lambda_k}\big)$$

and in particular the transported odd descendant class $[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^-_{\Lambda_k}$ and the ambiguity subspace $\mathcal{A}^-_{\Lambda_k}$. The full calibration supplement $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ is then derived abstractly from the weak interface by Theorem B.1. The characterization algebra $\mathcal{O}^{\mathrm{char}}_{\Lambda_k}$ is likewise derived from the canonical representative by Theorem H.7. A stronger interface may also supply a normalized dominant-response functional $\nu^{-1}_{\Lambda_k}\rho^{\mathrm{dom}}_{\Lambda_k}$ to witness activity directly, but the base package does not require that datum. A stronger layer may identify this data with a rank-one active generator $J^{\mathrm{act}}_{\Lambda_k}$, a response functional ρ_{Λ_k}, and a one-line stiffness $\widetilde{\alpha}(\Lambda_k)$. Throughout Subprograms 3b and 3c we work on the chosen Minkowski boundary chart on which the Peierls and EBC interfaces are assumed.

Appendix C: Subprogram 1a: Two–time SK bicomplex and Ward identities

This subprogram provides the *local two–time descent machinery* used later in Subprograms 3a and 3b. Starting from the Lorentzian Schwinger–Keldysh (SK) path integral on a two–time base (t, u) [8, 9], we record two odd differentials Q_t, Q_u implementing infinitesimal reparametrisations along t (physical time) and u (radial/RG time), and their Ward identities on SK–admissible states.

1. Two odd differentials from reparametrisations in t and u

a. Definition of Q_t, Q_u. On local bulk SK fields (doubled on the two SK branches) introduce two Grassmann–odd scalar ghosts c_t, c_u and define Q_t, Q_u as Lie derivatives along the odd vector fields $c_t\partial_t$ and $c_u\partial_u$:

$$Q_t(\Psi) := \mathcal{L}_{c_t\partial_t}\Psi, \qquad Q_u(\Psi) := \mathcal{L}_{c_u\partial_u}\Psi,$$

extended by the graded Leibniz rule to local functionals. We assume the base coordinates commute, $[\partial_t, \partial_u] = 0$.

Remark C.1 (Translation BRST convention)**.** In this appendix, Q_t, Q_u encode the BRST differentials for the *abelian* translation symmetries in t and u. Accordingly, we take the ghosts c_t, c_u to be constant odd parameters (so $\partial_t c_t = \partial_u c_u = 0$).

Lemma C.1 (Nilpotency and anticommutation)**.** *The operators Q_t and Q_u are Grassmann–odd derivations satisfying*

$$Q_t^2 = 0, \qquad Q_u^2 = 0, \qquad \{Q_t, Q_u\} := Q_tQ_u + Q_uQ_t = 0.$$

Proof. By the translation convention above, c_t, c_u are constant odd parameters. Hence the odd vector fields $c_t\partial_t$ and $c_u\partial_u$ satisfy $(c_t\partial_t)^2 = (c_u\partial_u)^2 = 0$ and $[c_t\partial_t, c_u\partial_u] = 0$ because $[\partial_t, \partial_u] = 0$. Taking Lie derivatives along these commuting square-zero odd vector fields gives $Q_t^2 = Q_u^2 = \{Q_t, Q_u\} = 0$. □

Definition C.1 (Total two–time differential)**.** Define the total differential

$$\mathbb{Q} := Q_t + Q_u.$$

Then $\mathbb{Q}^2 = 0$ by Lemma C.1.

2. Ward identities and SK–admissible states

Definition C.2 (SK–admissible states)**.** A state is *SK–admissible* if the initial density matrix is vacuum or KMS (or a finite-energy perturbation thereof) and the sources on the two SK branches are equal (physical SK).

Proposition C.1 (Ward identities)**.** *For SK–admissible states, insertions of Q_t–exact operators have vanishing expectation value on the closed time path:*

$$\langle Q_t X\rangle = 0.$$

The Q_u Ward identity holds up to radial boundary/anomaly terms determined by the chosen renormalized variational principle (GHY + counterterms):

$$\langle Q_u X\rangle = (\text{radial boundary/anomaly term}).$$

At the strict UV fixed point on flat boundary data with trivial Weyl factor, this radial term vanishes; for controlled Weyl pulses it reduces to the boundary Weyl-anomaly density, which is the input for the anomaly-floor monotone in Subprogram 3b.

Proof. Write the (renormalized) SK expectation value in the physical SK sector as

$$\langle X\rangle \;:=\; \frac{1}{Z}\int \mathcal{D}\Psi_+\mathcal{D}\Psi_-\, e^{iS_{\mathrm{SK,ren}}[\Psi_+,\Psi_-]}\, X,$$

where $S_{\mathrm{SK,ren}}$ includes the renormalized bulk action, the initial state insertion (vacuum/KMS), and the renormalized boundary terms determined by the variational principle (GHY + counterterms).

a. Q_t Ward identity. Let ε be a constant Grassmann–odd parameter and perform the change of variables $\Psi_\pm \mapsto \Psi_\pm + \varepsilon\, Q_t\Psi_\pm$ in the SK path integral. For SK–admissible states, the contour is closed and the sources are equal on the two branches, so the renormalized SK measure and action are invariant under t–reparametrisations generated by Q_t (no boundary term is picked up on the closed time path). Therefore the variation of the integral vanishes:

$$0 = \varepsilon \int \mathcal{D}\Psi_+\mathcal{D}\Psi_-\, e^{iS_{\mathrm{SK,ren}}}\,(Q_t X),$$

which implies $\langle Q_t X\rangle = 0$.

b. Q_u Ward identity. Similarly, perform $\Psi_\pm \mapsto \Psi_\pm + \varepsilon\, Q_u\Psi_\pm$. Unlike t, the radial/RG direction u has a boundary at the UV cutoff (and possibly at an IR regulator), so the variation of the renormalized action produces a boundary/anomaly term determined by the renormalized variational principle. Denoting the resulting insertion by $\mathcal{A}_u(X)$, one obtains the Ward identity

$$\langle Q_u X\rangle = \langle \mathcal{A}_u(X)\rangle.$$

At a strict UV fixed point on flat boundary data with trivial Weyl factor this insertion vanishes, whereas for controlled Weyl pulses it reduces to the boundary Weyl-anomaly density (the standard radial/Weyl Ward insertion in the Hamilton–Jacobi renormalization scheme). □

Remark C.2. The Q_t Ward identity is the usual SK statement that BRST-exact insertions decouple for equal sources; the Q_u Ward identity is the radial/Weyl analogue, whose nonzero anomaly term is precisely the source of monotonicity away from fixed points.

Proposition C.2 (Identification of $\mathcal{A}^{(1)}$ with the renormalized radial/Weyl anomaly insertion)**.** *In the physical SK sector ($J_+ = J_-$) the renormalized SK functional obeys a radial Ward identity*

$$\langle Q_u X\rangle \;=\; \langle \mathcal{A}_u(X)\rangle,$$

where $\mathcal{A}_u(X)$ is a local boundary/anomaly insertion determined by the renormalized variational principle. On the monotone representative $\mathcal{O}^{(0)}(u)$, the induced insertion defines a local cochain $\mathcal{A}^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\mathrm{s.loc}})_{\bar{1}}$ such that

$$\big[\, Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}(u)\,\big] \;=\; [\mathcal{A}^{(1)}(u)] \in H^1(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t),$$

and the descent equation holds in the locality quotient:

$$\big[\, Q_u^{\mathrm{phys}}\mathcal{O}^{(0)}(u) + Q_t\Xi^{(1)}(u) - \mathcal{A}^{(1)}(u)\,\big]_{\mathrm{loc}} = 0.$$

Proof. By Proposition C.1, the u–reparametrisation BRST operator Q_u obeys a Ward identity in the physical SK sector:

$$\langle Q_u X\rangle \;=\; \langle \mathcal{A}_u(X)\rangle,$$

where $\mathcal{A}_u(X)$ is the boundary/anomaly insertion produced by the renormalized variational principle.

Apply this to the monotone representative $X = \mathcal{O}^{(0)}(u)$. Define $\mathcal{A}^{(1)}(u)$ to be the induced sufficiently local cochain on the UV boundary, i.e. the restriction of $\mathcal{A}_u(\mathcal{O}^{(0)}(u))$ to the locality class $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ guaranteed by $\mathbf{2T}_{\text{Peierls}}$-**(P5)**. Since $Q_t\mathcal{O}^{(0)}(u) = 0$ and $\{Q_t, Q_u^{\text{phys}}\}$ vanishes on the monotone representative (Definition B.8), the cochain $Q_u^{\text{phys}}\mathcal{O}^{(0)}(u)$ is Q_t–closed and defines a class in $H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t)$. The Ward insertion $\mathcal{A}^{(1)}(u)$ represents the same class, i.e.

$$\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)}(u)\,\big] = [\mathcal{A}^{(1)}(u)] \in H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t),$$

with scheme/reparametrisation effects absorbed in the locality quotient (Definition B.7).

Since $\mathcal{A}^{(1)}(u)$ represents the same Q_t–cohomology class as $Q_u^{\text{phys}}\mathcal{O}^{(0)}(u)$, there exists a (sufficiently local) raw first-descendant potential $\Xi^{(1)}(u) \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ such that

$$\mathcal{A}^{(1)}(u) - Q_u^{\text{phys}}\mathcal{O}^{(0)}(u) = Q_t\Xi^{(1)}(u).$$

Interpreting the equality modulo the standard locality ambiguities encoded in Definition B.7 yields the descent equation in the locality quotient:

$$\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)}(u) + Q_t\Xi^{(1)}(u) - \mathcal{A}^{(1)}(u)\,\big]_{\text{loc}} = 0.$$

□

Lemma C.2 (Locality of the anomaly insertion and existence of a sufficiently local potential)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$-***(P5)***. *Then the anomaly insertion $\mathcal{A}^{(1)}(u)$ extracted from the renormalized radial Ward identity lies in $(\mathcal{F}^{\Lambda}_{\text{s.loc}})_{\bar{1}}$, and there exists a first-descendant potential $\Xi^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\text{s.loc}})_{\bar{0}}$ such that the anomalous descent equation holds in the locality quotient:*

$$\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)}(u) + Q_t\Xi^{(1)}(u) - \mathcal{A}^{(1)}(u)\,\big]_{\text{loc}} = 0.$$

Proof. By $\mathbf{2T}_{\text{Peierls}}$-**(P5)**, the sufficiently local class $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ is closed under the locality operations used to define renormalized Ward insertions on the UV boundary. In particular, the radial Ward insertion $\mathcal{A}_u(\mathcal{O}^{(0)}(u))$ extracted from the renormalized variational principle is a gauge-invariant local boundary functional, hence determines a sufficiently local cochain $\mathcal{A}^{(1)}(u) \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$. Since it is the insertion associated to the odd operator Q_u, it has cohomological parity $|\mathcal{A}^{(1)}(u)|_{\text{ch}} = 1$, i.e. $\mathcal{A}^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\text{s.loc}})_{\bar{1}}$.

By Proposition C.2 and Definition B.8, $\mathcal{A}^{(1)}(u)$ represents the same Q_t–cohomology class as $Q_u^{\text{phys}}\mathcal{O}^{(0)}(u)$ in $H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t)$. Therefore there exists a cochain $\Xi^{(1)}(u) \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ such that

$$\mathcal{A}^{(1)}(u) - Q_u^{\text{phys}}\mathcal{O}^{(0)}(u) = Q_t\Xi^{(1)}(u) \qquad \text{in } \mathcal{F}^{\Lambda}_{\text{s.loc}},$$

and $\Xi^{(1)}(u)$ necessarily has even cohomological parity $|\Xi^{(1)}(u)|_{\text{ch}} = 0$, i.e. $\Xi^{(1)}(u) \in (\mathcal{F}^{\Lambda}_{\text{s.loc}})_{\bar{0}}$. Rewriting yields

$$Q_u^{\text{phys}}\mathcal{O}^{(0)}(u) + Q_t\Xi^{(1)}(u) - \mathcal{A}^{(1)}(u) = 0.$$

Interpreting this identity modulo the standard scheme/contact/counterterm ambiguities encoded in the locality quotient (Definition B.7) gives the stated descent equation in $[\,\mathcal{F}^{\Lambda}_{\text{s.loc}}\,]_{\text{loc}}$. □

3. Restriction to the UV boundary and the BRST interface

Theorem C.1 (Boundary BRST interface)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$-***(P5)*** *so that the sufficiently local observable class $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ is available on the UV boundary. Then on each bounded region O the differentials Q_t, Q_u restrict to $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ and satisfy*

$$Q_t^2 = Q_u^2 = \{Q_t, Q_u\} = 0,$$

and on SK–admissible states one has $\langle Q_t X\rangle = 0$. In other words, the interface $\mathbf{2T}_{\text{BRST}}$ (Definition B.21) holds on the locality class used in the later Hamiltonization step.

Proof. By $\mathbf{2T}_{\text{Peierls}}$-**(P5)**, for each bounded region O on the UV boundary we have a sufficiently local cochain algebra $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$. The bulk operators Q_t, Q_u are defined on local SK functionals by Lie derivatives along $c_t\partial_t$ and $c_u\partial_u$ (Subprogram 1a). Since restriction/pullback to the UV boundary commutes with Lie derivatives along ∂_t and ∂_u, the operators Q_t, Q_u restrict to odd derivations on each $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$. The relations $Q_t^2 = Q_u^2 = \{Q_t, Q_u\} = 0$ are algebraic consequences of Lemma C.1 and therefore continue to hold after restriction to the boundary. Finally, the Q_t Ward identity $\langle Q_t X\rangle = 0$ on SK–admissible states is Proposition C.1. Thus the conditions of Definition B.21 are satisfied on the locality class used later in the construction. □

Remark C.3. This is the point where the BRST/bicomplex operators Q_t, Q_u are introduced; later sections treat them as fixed structure on the chosen locality class.

Appendix D: Subprogram 1b: Local net, GNS reconstruction, and implementation template

This subprogram fixes the Minkowski UV boundary setup in which the odd symmetry constructed in Subprogram 3b will be represented as an operator and classified by HLS. We keep the classification step itself in Subprogram 3c (after the odd derivation is constructed), but we set up the operator-algebraic/Wightman infrastructure here.

1. Local net and vacuum state

Definition D.1 (Dense Borchers–Uhlmann core)**.** Let $O \subset \mathbb{R}^{1,3}$ be a double cone. Define $\mathfrak{A}_{\text{BU}}(O)$ to be the *-algebra generated by symbols $\Phi(F)$ with $F \in \mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$, modulo locality relations for spacelike separated supports and the *-structure $\Phi(F)^* = \Phi(\overline{F})$. Write $\mathfrak{A}_{\text{loc,BU}} := \bigcup_O \mathfrak{A}_{\text{BU}}(O)$.

Remark D.1. The Borchers–Uhlmann net $O \mapsto \mathfrak{A}_{\text{BU}}(O)$ is the dense algebraic core used for GNS reconstruction. A physically relevant completion $\mathfrak{A}(O)$ will be defined in the vacuum representation obtained below.

Definition D.2 (Vacuum state from the SK functional)**.** Assume the SK functional determines a consistent family of vacuum correlators on the Minkowski UV boundary (equal sources on the two SK branches), and that on this equal-source locus the renormalized SK functional is invariant under swapping the two SK branches. Define a linear functional ϖ on $\mathfrak{A}_{\text{loc,BU}}$ by declaring its values on products of generators to be the corresponding SK vacuum correlators, and extending by linearity.

Theorem D.1 (Wightman/GNS net on the Minkowski UV boundary)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$-***(P5)*** *so that the locality presheaf $O \mapsto \mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ is available on the Minkowski UV chart, and assume the energy bounds and covariance interface* $\mathbf{2T}_{\text{EBC}}$ *(Definition B.10). Let $(\mathcal{H}, \pi, \Omega)$ denote the vacuum GNS triple supplied by* $\mathbf{2T}_{\text{EBC}}$-(0i)*. Defining local algebras by completion in this representation,*

$$\mathfrak{A}(O) \;:=\; C^*\big(\pi(\mathfrak{A}_{\text{BU}}(O))\big) \quad (\textit{or equivalently the von Neumann closure}),$$

the assignment $O \mapsto \mathfrak{A}(O)$ is a Haag–Kastler net on $\mathbb{R}^{1,3}$ with vacuum vector Ω and positive-energy Poincaré implementation. Equivalently, the AQFT interface $\mathbf{2T}_{\text{AQFT}}$ *(Definition B.23) holds.*

Proof. We split the argument into the state/representation, the net, locality, and covariance.

a. State and GNS representation. By Definition B.10-(0i), ϖ is a vacuum state on the algebraic inductive limit $*$-algebra $\mathfrak{A}_{\text{loc,BU}} := \bigcup_O \mathfrak{A}_{\text{BU}}(O)$ and $(\mathcal{H}, \pi, \Omega)$ is the corresponding vacuum GNS triple. In particular,

$$\varpi(A) = \langle \Omega, \pi(A)\Omega\rangle \qquad (A \in \mathfrak{A}_{\text{loc,BU}}).$$

b. Local net and isotony. For each double cone O, define the local algebra by completion in the vacuum representation:

$$\mathfrak{A}(O) := C^*\big(\pi(\mathfrak{A}_{\text{BU}}(O))\big).$$

If $O_1 \subset O_2$, then by construction $\mathfrak{A}_{\text{BU}}(O_1) \subset \mathfrak{A}_{\text{BU}}(O_2)$, hence $\pi(\mathfrak{A}_{\text{BU}}(O_1)) \subset \pi(\mathfrak{A}_{\text{BU}}(O_2))$, and taking C*-closures yields isotony: $\mathfrak{A}(O_1) \subset \mathfrak{A}(O_2)$.

c. Locality. Let O_1, O_2 be spacelike separated. Locality holds on the BU core by the defining Borchers–Uhlmann relations, i.e. $[A_1, A_2] = 0$ for all $A_i \in \pi(\mathfrak{A}_{\mathrm{BU}}(O_i))$. To pass to the C*-closures, take arbitrary $A_i \in \mathfrak{A}(O_i)$ and choose sequences $A_{i,n} \in \pi(\mathfrak{A}_{\mathrm{BU}}(O_i))$ with $\|A_{i,n} - A_i\| \to 0$. Then $[A_{1,n}, A_{2,m}] = 0$ for all n, m, and using the operator norm estimate

$$\|[B, C]\| \leq 2\|B\|\,\|C - C'\| + 2\|C'\|\,\|B - B'\|$$

with $B = A_1$, $B' = A_{1,n}$, $C = A_2$, $C' = A_{2,m}$, we obtain $\|[A_1, A_2]\| \to 0$ as $n, m \to \infty$. Hence $[A_1, A_2] = 0$.

d. Poincaré covariance and spectrum. By Definition B.10-(0ii), there exists a strongly continuous positive-energy unitary representation U_g of the Poincaré group implementing the induced action α_g on the locality presheaf $O \mapsto \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ and fixing the vacuum vector, $U_g\Omega = \Omega$. The induced action extends to the BU core by $\alpha_g(\Phi(F)) := \Phi(\alpha_g F)$ and multiplicativity, and the implementation identity $U_g\,\pi(A)\,U_g^{-1} = \pi(\alpha_g(A))$ holds on $\pi(\mathfrak{A}_{\mathrm{BU}}(O))$ by construction. Since conjugation by U_g is isometric, this identity extends by continuity to the C*-closures and yields $U_g\,\mathfrak{A}(O)\,U_g^{-1} = \mathfrak{A}(gO)$ for all double cones O. Finally, the spectrum condition and strong continuity are part of Definition B.10-(0ii), completing the verification that $O \mapsto \mathfrak{A}(O)$ satisfies the Haag–Kastler properties with vacuum and positive-energy Poincaré implementation, i.e. the AQFT interface $\mathbf{2T}_{\mathrm{AQFT}}$ (Definition B.23) holds. □

Remark D.2. One convenient reference mechanism for positive-energy implementation of translations in a vacuum representation is the Borchers–Arveson theorem; in this appendix spine we treat the spectrum condition and strong continuity as part of the energy bounds and covariance interface (Definition B.10).

2. Dense core and inner implementation template

Definition D.3 (Finite polynomial algebra and GNS core)**.** Let $\mathfrak{A}_{\mathrm{fin}} \subset \mathfrak{A}_{\mathrm{loc,BU}}$ denote the *-subalgebra generated by finite polynomials in the local generators $\Phi(F)$ (with F compactly supported in a double cone). Define the common GNS core

$$\mathcal{D}_0 := \pi(\mathfrak{A}_{\mathrm{fin}})\Omega.$$

Proposition D.1 (Inner implementation on the GNS core)**.** *Assume Theorem D.1 and let $\delta : \mathfrak{A}_{\mathrm{fin}} \to \mathfrak{A}_{\mathrm{fin}}$ be a graded *-derivation which is local, Poincaré covariant, and vacuum-invariant in the sense that $\varpi \circ \delta = 0$. Assume moreover that δ is implementable by closable operators on a common invariant core (as will be verified for the constructed δ in Subprogram 3c). Then there exists a densely defined closable operator Q on $\mathcal{H}$ with $\mathcal{D}_0 \subset \mathcal{D}(Q)$ such that, for all $A \in \mathfrak{A}_{\mathrm{fin}}$,*

$$\delta(A) = i[Q, \pi(A)]_{\mathrm{s}} \quad \text{on } \mathcal{D}_0,$$

where $[X, Y]_{\mathrm{s}} := XY - (-1)^{|X||Y|}YX$ is the graded commutator.

Proof. Define an operator Q_0 on the GNS core $\mathcal{D}_0 = \pi(\mathfrak{A}_{\mathrm{fin}})\Omega$ by

$$Q_0\,\pi(B)\Omega \;:=\; i\,\pi(\delta(B))\Omega \qquad (B \in \mathfrak{A}_{\mathrm{fin}}).$$

a. Well-definedness. Suppose $\pi(B)\Omega = 0$. For any $A \in \mathfrak{A}_{\mathrm{fin}}$ one has

$$\langle \pi(A)\Omega, \pi(\delta(B))\Omega \rangle \;=\; \varpi\big(A^*\delta(B)\big).$$

Using the graded Leibniz rule and vacuum invariance $\varpi \circ \delta = 0$,

$$0 = \varpi\big(\delta(A^*B)\big) = \varpi\big(\delta(A^*)B\big) + (-1)^{|A|}\varpi\big(A^*\delta(B)\big),$$

so $\varpi(A^*\delta(B)) = -(-1)^{|A|}\varpi(\delta(A^*)B) = -(-1)^{|A|}\varpi((\delta(A))^*B)$. Since $\pi(B)\Omega = 0$, we have $\varpi(C^*B) = 0$ for all $C \in \mathfrak{A}_{\mathrm{fin}}$, and in particular for $C = \delta(A)$; hence $\varpi(A^*\delta(B)) = 0$ for all A. Therefore $\pi(\delta(B))\Omega$ is orthogonal to the dense set $\pi(\mathfrak{A}_{\mathrm{fin}})\Omega$, so $\pi(\delta(B))\Omega = 0$ and Q_0 is well-defined.

b. Graded commutator identity on the core. For $A, B \in \mathfrak{A}_{\text{fin}}$, using the definition of Q_0 and the graded Leibniz rule for δ,

$$\begin{aligned}[Q_0, \pi(A)]_{\text{s}}\, \pi(B)\Omega &= Q_0\, \pi(AB)\Omega - (-1)^{|A|}\pi(A)\, Q_0\, \pi(B)\Omega \\ &= i\, \pi(\delta(AB))\Omega - (-1)^{|A|} i\, \pi(A\delta(B))\Omega \\ &= i\, \pi(\delta(A)B)\Omega \;=\; i\, \pi(\delta(A))\, \pi(B)\Omega,\end{aligned}$$

so $\delta(A) = i[Q_0, \pi(A)]_{\text{s}}$ on $\mathcal{D}_0$.

c. Closability. By the implementability hypothesis, Q_0 admits a closable extension on a common invariant core. Let Q denote such a densely defined closable extension with $\mathcal{D}_0 \subset \mathcal{D}(Q)$. Then the graded commutator identity continues to hold on $\mathcal{D}_0$, completing the proof. □

Appendix E: Subprogram 2a: Results from the 2T Limited Scope Theorem

Theorem E.1 (Limited Scope import)**.** *Assume* $\mathbf{2T}_{\text{LS}}$*. Then the Limited Scope Theorem provides:*

(LS1) *A presented record-process groupoid* $\mathcal{G}_{\text{proc}}$ *and a parity cocycle* $\omega : \mathcal{G}_{\text{proc}} \to B\mathbb{Z}_2$ *with* $\omega(\text{sqLoop}(t, u, r)) = 1$ *for the canonical square-loop.*

(LS2) *A non-splitting* $\mathbb{Z}_2$*-cover* $\widetilde{\mathcal{G}_{\text{proc}}} \to \mathcal{G}_{\text{proc}}$ *(no global section / no state-label trivialization).*

(LS3) *A nontrivial action on* $\mathsf{Diff} = \text{Cofib(diag)}$ *witnessed by* $y \in \mathsf{Diff}$ *with* $\mathsf{flip}(y) \neq y$ *(detector nontriviality).*

(LS4) *Bicomplex identities on free abelian chains generated by the bisimplicial set* $\mathcal{C}^{\text{SK}}_{m,n}$*, yielding* $(Q^{\text{ch}}_t)^2 = (Q^{\text{ch}}_u)^2 = Q^{\text{ch}}_t Q^{\text{ch}}_u + Q^{\text{ch}}_u Q^{\text{ch}}_t = 0$.

Remark E.1 (Two bicomplexes are not identified)**.** The LS bicomplex $(Q^{\text{ch}}_t, Q^{\text{ch}}_u)$ acts on record-process chains generated by the bisimplicial set $\mathcal{C}^{\text{SK}}_{m,n}$ and is used only to produce the parity cocycle ω_{LS}. The SK/BRST bicomplex (Q_t, Q_u) acts on sufficiently local boundary cochains $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ and is used to formulate Ward identities and descent. These bicomplexes interact only through the tag-action/parity-transport step (Definition G.2 and Lemma G.13); they are not identified.

Appendix F: Subprogram 2b: Cosmology theory Peierls bracket and swap invariance

Theorem F.1 (Cosmology theory Peierls bracket)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$*. Then the Peierls bracket* $\{-,-\}_{\text{P}}$ *is a well-defined graded Poisson bracket on the algebra* $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ *of gauge-invariant sufficiently local boundary functionals, satisfying antisymmetry, Leibniz, and Jacobi. This is the cosmology theory Peierls package (P1–P6), recorded here in abstract interface form.*

Remark F.1. For general background on the Peierls bracket and its covariant phase space formulation, see [32, 33].

Lemma F.1 (Transpose/swap equivariance implies equivariance of $E^{\pm}$ and Δ)**.** *Assume* $\mathbf{2T}_{\text{Peierls}}$*-**(P2)** and **(P6)**, and let* Γ_* *be the involution supplied by **(P6)**. Then*

$$E^{\pm} \circ \Gamma_* = \Gamma_* \circ E^{\pm}, \qquad \Delta \circ \Gamma_* = \Gamma_* \circ \Delta, \quad \textit{where} \quad \Delta := E^- - E^+.$$

Proof. Fix a test source f and consider

$$\psi^{\pm} := E^{\pm}(\Gamma_* f), \qquad \phi^{\pm} := \Gamma_*(E^{\pm} f).$$

By definition, $\psi^{\pm}$ solves $P_{\text{gf,ren}}\psi^{\pm} = \Gamma_* f$ with $B_\Lambda(\psi^{\pm}) = 0$ and advanced/retarded support. Using the commutation assumption from $\mathbf{2T}_{\text{Peierls}}$-**(P6)**,

$$P_{\text{gf,ren}}\phi^{\pm} = P_{\text{gf,ren}}\Gamma_* E^{\pm} f = \Gamma_* P_{\text{gf,ren}} E^{\pm} f = \Gamma_* f, \qquad B_\Lambda(\phi^{\pm}) = B_\Lambda \Gamma_* E^{\pm} f = \Gamma_* B_\Lambda(E^{\pm} f) = 0.$$

Moreover, $\phi^{\pm}$ has the same advanced/retarded support property as $E^{\pm} f$ because Γ_* is pointwise and does not enlarge support. By uniqueness of $E^{\pm}$, one has $\psi^{\pm} = \phi^{\pm}$, i.e. $E^{\pm}\Gamma_* = \Gamma_* E^{\pm}$. Subtracting yields $\Delta\Gamma_* = \Gamma_*\Delta$. □

Lemma F.2 (Swap induces Γ_* on first functional variations)**.** *Let Γ act on functionals by pullback along the induced swap on boundary data, and let Γ_* denote the induced involution on test sources/linearized boundary data (the tangent map of the swap). Then for every $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ one has*

$$(\Gamma F)^{(1)} = \Gamma_*\big(F^{(1)}\big),$$

and similarly for second variations. In the pointer-basis instantiation, Γ_* is the transpose/real structure on linearized boundary data.

Proof. This is the chain rule for pullback: writing $(\Gamma F)(J) := F(\Gamma_* J)$ for boundary data J, differentiation gives $(\Gamma F)^{(1)} = \Gamma_* F^{(1)}$. □

Lemma F.3 (Swap-equivariance and Poisson automorphism)**.** *Assume $\mathbf{2T}_{\mathrm{Peierls}}$, let Γ_* be the involution of **(P6)**, and let Γ act on functionals by pullback along the induced swap on boundary data. Then*

$$\Gamma\{F, G\}_{\mathrm{P}} = \{\Gamma F, \Gamma G\}_{\mathrm{P}} \quad \textit{for all } F, G \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}.$$

Proof. By definition, $\{F, G\}_{\mathrm{P}} = \langle F^{(1)}, \Delta\, G^{(1)}\rangle$. By Lemma F.2, pullback by Γ sends first functional derivatives by Γ_*, and by Lemma F.1 one has $\Delta\Gamma_* = \Gamma_*\Delta$. Using the Γ_*–invariance of the pairing from $\mathbf{2T}_{\mathrm{Peierls}}$-**(P6)** gives the claim. □

Appendix G: Subprogram 2c: Parity transport and physical $\mathbb{Z}_2$ grading

Definition G.1 (SK parity candidate)**.** Let Γ_{SK} be the involutive *-automorphism induced by swapping SK branches (equivalently: sending difference/a-type generators to minus themselves and fixing r-type ones).

Lemma G.1 (Swap invariance of the SK vacuum state)**.** *Assume the vacuum state ϖ is induced from the physical SK functional with equal sources in the sense of Definition D.2. Then ϖ is Γ_{SK}-invariant:*

$$\varpi \circ \Gamma_{\mathrm{SK}} = \varpi.$$

Proof. On products of generators $\Phi(F)$ the state ϖ is, by construction, given by SK vacuum correlators at equal sources. Swap-invariance of the equal-source SK functional implies these correlators are unchanged when every insertion is acted on by Γ_{SK}. By linearity (and continuity on the completion), $\varpi \circ \Gamma_{\mathrm{SK}} = \varpi$ on the net. □

Lemma G.2 (Γ is unitarily implemented in the vacuum GNS representation)**.** *Assume Lemma G.1. Let $(\mathcal{H}, \pi, \Omega)$ be the GNS triple of the vacuum state ϖ. Then there exists a unitary U_Γ on $\mathcal{H}$ with $U_\Gamma\Omega = \Omega$ and*

$$U_\Gamma\, \pi(A)\, U_\Gamma^{-1} = \pi(\Gamma A) \qquad (A \in \mathfrak{A}_{\mathrm{loc,BU}}),$$

and $U_\Gamma^2 = \mathbf{1}$.

Proof. Define U_Γ on the cyclic subspace $\pi(\mathfrak{A}_{\mathrm{loc,BU}})\Omega$ by

$$U_\Gamma\, \pi(A)\Omega := \pi(\Gamma A)\Omega.$$

This is well-defined and isometric because $\varpi \circ \Gamma = \varpi$ implies

$$\|U_\Gamma \pi(A)\Omega\|^2 = \langle \Omega, \pi((\Gamma A)^*(\Gamma A))\Omega\rangle = \varpi(\Gamma(A^*A)) = \varpi(A^*A) = \|\pi(A)\Omega\|^2.$$

Hence U_Γ extends uniquely to a unitary on $\mathcal{H}$. The implementation identity follows by construction, and $U_\Gamma^2 = \mathbf{1}$ because $\Gamma^2 = \mathrm{id}$. □

Lemma G.3 (Γ_{SK} commutes with Q_t and respects the RG decomposition)**.** *On the locality class used in Subprograms 3a and 3b, Γ_{SK} commutes with the two odd differentials:*

$$\Gamma_{\mathrm{SK}} Q_t = Q_t \Gamma_{\mathrm{SK}}, \qquad \Gamma_{\mathrm{SK}} Q_u = Q_u \Gamma_{\mathrm{SK}}.$$

Moreover, for the decomposition $Q_u = Q_u^{\mathrm{phys}} + Q_u^{\mathrm{scheme}}$ used in Definition B.5, we assume that Γ_{SK} commutes with Q_u^{phys} and Q_u^{scheme} on the monotone sector $\mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}$ (Definition B.6):

$$\Gamma_{\mathrm{SK}} Q_u^{\mathrm{phys}} = Q_u^{\mathrm{phys}} \Gamma_{\mathrm{SK}}, \qquad \Gamma_{\mathrm{SK}} Q_u^{\mathrm{scheme}} = Q_u^{\mathrm{scheme}} \Gamma_{\mathrm{SK}} \quad \textit{on } \mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}.$$

Proof. On SK fields and sources, Γ_{SK} exchanges the two SK branches and fixes the base ghosts. The differentials Q_t, Q_u are defined diagonally on the two branches (Subprogram 1a) as Lie derivatives along $c_t\partial_t$ and $c_u\partial_u$. Since these definitions are symmetric under interchanging the branches, Γ_{SK} commutes with Q_t and Q_u on generators; the graded Leibniz rule then extends the commutation to all local cochains in the chosen locality class. The additional equivariance for the decomposition $Q_u = Q_u^{\mathrm{phys}} + Q_u^{\mathrm{scheme}}$ is assumed on the monotone sector $\mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}$. □

Lemma G.4 (Parity involution on observables)**.** *We write $\Gamma := \Gamma_{\mathrm{SK}}$. Let F be any observable functional on $\mathcal{S}_{\mathrm{SK}}$. Define*

$$(\Gamma F)(x) := F(\mathsf{swap}x).$$

Then $\Gamma^2 = \mathrm{id}$ and for all $r \in \mathcal{S}_{\mathrm{rec}}$, $(\Gamma F)(\mathrm{diag}\, r) = F(\mathrm{diag}\, r)$.

Remark G.1. In the pointer basis, swap acts as transposition on states, so for $F(\rho) = \mathrm{Tr}(\rho\,\mathcal{O})$ one has $(\Gamma F)(\rho) = F(\rho^{\mathsf{T}}) = \mathrm{Tr}(\rho\,\mathcal{O}^{\mathsf{T}})$.

Lemma G.5 (Detector witness implies $\Gamma \neq \mathrm{id}$ on observables)**.** *Assume $\mathbf{2T}_{\mathrm{det}}$ in the pointer-basis instantiation, so there exists a swap-odd observable functional F on $\mathcal{S}_{\mathrm{SK}}$ that is constant on diagonal record states and is not identically zero. Then*

$$\Gamma(F) = -F,$$

and in particular $\Gamma \neq \mathrm{id}$ on observables.

Proof. Swap-oddness gives $F(\mathsf{swap}x) = -F(x)$ for some $x \in \mathcal{S}_{\mathrm{SK}}$, hence by definition $(\Gamma F)(x) = F(\mathsf{swap}x) = -F(x) \neq F(x)$. □

Remark G.2. By the non-splitting obstruction from the Limited Scope import, this nontrivial parity cannot be removed by a relabeling of record states; it is a physical grading.

Lemma G.6 (ω-action on additive targets)**.** *Let G be any abelian group. Define*

$$\mathrm{Act}^G_\omega(p)(g) := (-1)^{\omega(p)} g.$$

Then Act^G_ω is a functorial action of $\mathcal{G}_{\mathrm{proc}}$ on G. If Γ acts on an algebra $\mathcal{A}$ of observables, the action $\mathrm{Act}(p) := \Gamma^{\omega(p)}$ is a functorial representation of $\mathcal{G}_{\mathrm{proc}}$ on $\mathcal{A}$.

Definition G.2 (Tag action on observables)**.** Define an action of the two square tags on observables by

$$\mathrm{act}^{\mathrm{obs}}_{\mathrm{fill}}(\mathrm{diag}) = \mathrm{id}, \qquad \mathrm{act}^{\mathrm{obs}}_{\mathrm{fill}}(\mathrm{bridge}) = \Gamma,$$

where Γ is the swap-induced involution of Lemma G.4.

Lemma G.7 (Γ preserves the locality quotient)**.** *For each double cone O, if $F \simeq F'$ in $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ (Definition B.7), then $\Gamma(F) \simeq \Gamma(F')$. Equivalently, Γ induces an involution on $[\,\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)\,]_{\mathrm{loc}}$.*

Proof. Since $\simeq$ is generated by the elementary moves Definition B.7(a)–(d), it suffices to check stability of each move under Γ.

a. Move (a) *(Q_t–exact shifts).* For $X \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$,

$$\Gamma(F + Q_t X) = \Gamma(F) + \Gamma(Q_t X) = \Gamma(F) + Q_t(\Gamma X),$$

using $\Gamma Q_t = Q_t \Gamma$ (Lemma G.3).

b. Move (b) *(scheme move on the monotone sector).* Let $Y \in \mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}(O)$. By Definition B.6, $\Gamma Y \in \mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}(O)$, and by Lemma G.3 (on $\mathcal{F}^{\Lambda}_{\mathrm{s.locmon}}$),

$$\Gamma(Q_u^{\mathrm{scheme}} Y) = Q_u^{\mathrm{scheme}}(\Gamma Y).$$

Moreover, Q_u^{scheme} preserves each $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$ by Definition B.5-**(M2)**, so the shift is well-typed on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$.

c. Move (c) *(total divergences).* Since Γ acts on fields/sources and not on the spacetime coordinates,

$$\Gamma(F + \partial_\mu K^\mu) = \Gamma(F) + \partial_\mu(\Gamma K^\mu).$$

d. Move (d) *(counterterm coboundaries).* Counterterm coboundaries transform covariantly under the SK swap, so $\Gamma(\Delta_{\rm ct})$ is again a counterterm coboundary.

Thus each generating move is preserved by Γ, hence Γ preserves $\simeq$ and induces an involution on $[\,\mathcal{F}^{\Lambda}_{\rm s.loc}(O)\,]_{\rm loc}$. □

Lemma G.8 (Γ–evenness of the anomaly insertion (modulo contact/counterterm ambiguities))**.** *Assume the equal-source SK functional is invariant under SK branch swap, and let $\mathcal{A}^{(1)}(u)$ be the anomaly insertion extracted from the renormalized radial Ward identity in the physical SK sector (Proposition C.2). Assume moreover that any swap-odd remainder in the renormalized Ward insertion is a contact/counterterm ambiguity (i.e. representable by a total divergence or a counterterm coboundary). Then*

$$\big[\,\Gamma(\mathcal{A}^{(1)}(u)) - \mathcal{A}^{(1)}(u)\,\big]_{\rm loc} = 0,$$

where the locality quotient is generated by Q_t–exact shifts (Definition B.7(a)*), total divergences* (c)*, and counterterm coboundaries* (d) *(no use of the scheme move* (b) *is required here).*

Proof. On the physical SK locus (equal sources), swapping the SK branches leaves the renormalized functional invariant by hypothesis. The corresponding radial/Weyl Ward insertion extracted from the renormalized variational principle is therefore Γ–invariant up to the usual local contact and counterterm ambiguities of renormalized Ward identities.

By the additional hypothesis of the lemma, the swap-odd remainder $\Gamma(\mathcal{A}^{(1)}(u)) - \mathcal{A}^{(1)}(u)$ can be represented as a total divergence $\partial_\mu K^\mu$ plus a counterterm coboundary $\Delta_{\rm ct}$ (and, if present, a Q_t–exact shift). Each of these representatives is trivial in the locality quotient by Definition B.7(a), (c), and (d). Hence

$$\big[\,\Gamma(\mathcal{A}^{(1)}(u)) - \mathcal{A}^{(1)}(u)\,\big]_{\rm loc} = 0.$$

No use of the scheme move (b) is required. □

Remark G.3 (Parity of the anomaly insertion)**.** Applying Γ to the descent equation sends the anomaly insertion $\mathcal{A}^{(1)}$ to $\Gamma(\mathcal{A}^{(1)})$. By Lemma G.8, the anomaly insertion is Γ–even in the locality quotient, so the diag and bridge fillers correspond to the same anomaly insertion modulo the standard locality ambiguities.

Remark G.4. In the parity-transport step we work with the physical radial differential $Q_u^{\rm phys}$; scheme terms $Q_u^{\rm scheme}$ are treated as locality-quotient ambiguities (counterterm reparametrisations) and do not affect the transported Γ–odd sector.

Lemma G.9 (Bridge filler acts by Γ on first-descendant potentials)**.** *Assume Lemma G.3. Let $\mathcal{O}^{(0)}$ be Q_t-closed and Γ–even, i.e. $\Gamma(\mathcal{O}^{(0)}) = \mathcal{O}^{(0)}$, and let $\mathcal{J}$ and $\mathcal{A}^{(1)}$ satisfy the (possibly anomalous) first descent equation*

$$\big[\,Q_u^{\rm phys}\mathcal{O}^{(0)} + Q_t\mathcal{J} - \mathcal{A}^{(1)}\,\big]_{\rm loc} = 0, \qquad \big[\,\Gamma(\mathcal{A}^{(1)}) - \mathcal{A}^{(1)}\,\big]_{\rm loc} = 0.$$

Define two tag-associated representatives by

$$\mathcal{J}_{\rm diag} := {\rm act}^{\rm obs}_{\rm fill}({\rm diag})(\mathcal{J}) = \mathcal{J}, \qquad \mathcal{J}_{\rm bridge} := {\rm act}^{\rm obs}_{\rm fill}({\rm bridge})(\mathcal{J}) = \Gamma(\mathcal{J}).$$

Then $\mathcal{J}_{\rm diag}$ and $\mathcal{J}_{\rm bridge}$ are both valid first-descendant potentials of $\mathcal{O}^{(0)}$ (for the same anomaly insertion), and switching the record-level square filler tag from diag *to* bridge *induces*

$$\mathcal{J}_{\rm bridge} = \Gamma(\mathcal{J}_{\rm diag}).$$

Proof. Apply Γ to the descent identity in the locality quotient. By Lemma G.3 and Lemma G.7,

$$\big[\,Q_u^{\rm phys}(\Gamma\mathcal{O}^{(0)}) + Q_t(\Gamma\mathcal{J}) - \Gamma(\mathcal{A}^{(1)})\,\big]_{\rm loc} = 0.$$

Since $\Gamma\mathcal{O}^{(0)} = \mathcal{O}^{(0)}$ by assumption and $[\,\Gamma(\mathcal{A}^{(1)}) - \mathcal{A}^{(1)}\,]_{\rm loc} = 0$, we obtain

$$\big[\,Q_u^{\rm phys}\mathcal{O}^{(0)} + Q_t(\Gamma\mathcal{J}) - \mathcal{A}^{(1)}\,\big]_{\rm loc} = 0,$$

so $\Gamma\mathcal{J}$ is also a valid first-descendant potential (for the same anomaly insertion) in the locality quotient. By Definition G.2 it is exactly the bridge-tag representative. □

Definition G.3 (First-descendant ambiguity group)**.** Let $\mathcal{V}^1$ be the additive group of first-descendant potentials in the chosen locality class, modulo the standard Q_t– and Q_u–exact shifts:

$$\mathcal{V}^1 := \frac{\mathcal{F}^{\Lambda}_{\text{s.loc}}}{Q_t(\mathcal{F}^{\Lambda}_{\text{s.loc}}) + Q_u(\mathcal{F}^{\Lambda}_{\text{s.loc}})}.$$

We write $[\mathcal{J}] \in \mathcal{V}^1$ for the class of a chosen first-descendant potential $\mathcal{J} \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$.

Lemma G.10 (Γ–odd representative choice)**.** *Assume Lemma G.3 and that the locality class is linear over a ring in which 2 is invertible (e.g. over $\mathbb{R}$). Let $\mathcal{J}$ be a representative of a class $[\mathcal{J}] \in \mathcal{V}^1$ satisfying $[\Gamma(\mathcal{J})] = -[\mathcal{J}]$ in $\mathcal{V}^1$ and define*

$$\mathcal{J}_- := \tfrac{1}{2}\big(\mathcal{J} - \Gamma\mathcal{J}\big).$$

Then $\Gamma(\mathcal{J}_-) = -\mathcal{J}_-$ and $[\mathcal{J}_-] = [\mathcal{J}]$ in $\mathcal{V}^1$.

Proof. The parity identity is immediate from $\Gamma^2 = \text{id}$. The class condition implies $\Gamma(\mathcal{J}) + \mathcal{J}$ lies in the ambiguity subgroup $Q_t(\mathcal{F}^{\Lambda}_{\text{s.loc}}) + Q_u(\mathcal{F}^{\Lambda}_{\text{s.loc}})$. Therefore

$$\mathcal{J} - \mathcal{J}_- = \tfrac{1}{2}\big(\Gamma(\mathcal{J}) + \mathcal{J}\big)$$

lies in the ambiguity subgroup, so $[\mathcal{J}_-] = [\mathcal{J}]$ in $\mathcal{V}^1$. □

Lemma G.11 (Odd representative choice preserves the descent identity)**.** *Assume the setup of Lemma G.9 and that the locality class is linear over a ring in which 2 is invertible (e.g. over $\mathbb{R}$). Define*

$$\mathcal{J}_- := \tfrac{1}{2}\big(\mathcal{J} - \Gamma\mathcal{J}\big).$$

Then $\Gamma(\mathcal{J}_-) = -\mathcal{J}_-$ and $\mathcal{J}_-$ is a valid first-descendant potential for the same anomaly insertion, i.e.

$$\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} + Q_t\mathcal{J}_- - \mathcal{A}^{(1)} \,\big]_{\text{loc}} = 0.$$

Moreover, $\mathcal{J}_- \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ and $\mathcal{J}_-$ is EB3-admissible whenever $\mathcal{J}$ is.

Proof. By Lemma G.9, both $\mathcal{J}$ and $\Gamma\mathcal{J}$ are valid first-descendant potentials for the same anomaly insertion, so

$$\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} + Q_t\mathcal{J} - \mathcal{A}^{(1)} \,\big]_{\text{loc}} = 0, \qquad \big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} + Q_t(\Gamma\mathcal{J}) - \mathcal{A}^{(1)} \,\big]_{\text{loc}} = 0.$$

Subtracting gives

$$\big[\, Q_t(\mathcal{J} - \Gamma\mathcal{J}) \,\big]_{\text{loc}} = 0.$$

Define $\mathcal{J}_- := \frac{1}{2}(\mathcal{J} - \Gamma\mathcal{J})$, so $\Gamma(\mathcal{J}_-) = -\mathcal{J}_-$. Now compute the descent residual for $\mathcal{J}_-$ in the locality quotient:

$$\begin{aligned}
\big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} + Q_t\mathcal{J}_- - \mathcal{A}^{(1)} \,\big]_{\text{loc}} &= \Big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} - \mathcal{A}^{(1)} + \tfrac{1}{2}\big(Q_t\mathcal{J} - Q_t(\Gamma\mathcal{J})\big) \,\Big]_{\text{loc}} \\
&= \Big[\, -Q_t\mathcal{J} + \tfrac{1}{2}\big(Q_t\mathcal{J} - Q_t(\Gamma\mathcal{J})\big) \,\Big]_{\text{loc}} \quad \big(\text{using } \big[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} + Q_t\mathcal{J} - \mathcal{A}^{(1)} \,\big]_{\text{loc}} = 0\big) \\
&= \Big[\, -Q_t\big(\tfrac{1}{2}(\mathcal{J} + \Gamma\mathcal{J})\big) \,\Big]_{\text{loc}} \;=\; 0,
\end{aligned}$$

because Q_t–exact shifts are trivial by Definition B.7(a). Finally, Γ preserves $\mathcal{F}^{\Lambda}_{\text{s.loc}}$ and the defining seminorm controls on each O, so $\mathcal{J}_- \in \mathcal{F}^{\Lambda}_{\text{s.loc}}$ and inherits EB3-admissibility from $\mathcal{J}$. □

Remark G.5. Lemma G.11 shows that the Γ–odd projection $\mathcal{J}_- = \frac{1}{2}(\mathcal{J} - \Gamma\mathcal{J})$ preserves the descent data in the locality quotient. The involution Γ preserves the locality class $\mathcal{F}^{\Lambda}_{\text{s.loc}}(O)$ and its defining seminorm controls; hence $\mathcal{J}_- = \frac{1}{2}(\mathcal{J} - \Gamma\mathcal{J})$ remains EB3-admissible whenever $\mathcal{J}$ is.

Because the descent equation is imposed only in the locality quotient, it fixes the cohomology class $[\, Q_u^{\text{phys}}\mathcal{O}^{(0)} \,] \in H^1(\mathcal{F}^{\Lambda}_{\text{s.loc}}, Q_t)$ but does not uniquely fix the potential. The LS parity transport identifies the physically relevant Γ–odd descendant class, and Theorem H.1 then selects its canonical representative by calibration minimization.

Lemma G.12 (Filler–descent dictionary)**.** *Assume the setup of Lemma G.9 so that $\mathcal{J}_{\mathrm{diag}}$ and $\mathcal{J}_{\mathrm{bridge}}$ are defined from a first-descendant potential $\mathcal{J}$. Let $[\mathcal{J}] \in \mathcal{V}^1$ denote the class of a first-descendant modulo the standard Q_t/Q_u ambiguities. By the Limited Scope import, the canonical square-loop* $\mathrm{sqLoop}(t,u,r)$ *satisfies* $\omega(\mathrm{sqLoop}(t,u,r)) = 1$*. In the sign transport induced by ω on the additive group $\mathcal{V}^1$ (Lemma G.6), changing the filler tag from* diag *to* bridge *multiplies the descendant class by -1:*

$$[\,\mathcal{J}_{\mathrm{bridge}}\,] = -\,[\,\mathcal{J}_{\mathrm{diag}}\,] \quad \textit{in } \mathcal{V}^1.$$

Equivalently, the descendant sector is a torsor for the ω-twisted local system.

Proof. Apply Lemma G.6 to the abelian group $G = \mathcal{V}^1$. This defines a functorial action of the record-process groupoid $\mathcal{G}_{\mathrm{proc}}$ on $\mathcal{V}^1$ by

$$\mathrm{Act}_{\omega}^{\mathcal{V}^1}(p)([\mathcal{J}]) = (-1)^{\omega(p)}[\mathcal{J}].$$

By the Limited Scope import (Theorem E.1), the canonical square-loop $p := \mathrm{sqLoop}(t,u,r)$ satisfies $\omega(p) = 1$, hence

$$\mathrm{Act}_{\omega}^{\mathcal{V}^1}(p)([\mathcal{J}]) = -[\mathcal{J}].$$

Changing the record-level square filler tag from diag to bridge corresponds to transporting the first-descendant class along this canonical square-loop, so $[\mathcal{J}_{\mathrm{bridge}}] = \mathrm{Act}_{\omega}^{\mathcal{V}^1}(p)([\mathcal{J}_{\mathrm{diag}}])$. Therefore $[\mathcal{J}_{\mathrm{bridge}}] = -[\mathcal{J}_{\mathrm{diag}}]$ in $\mathcal{V}^1$. □

Lemma G.13 (Parity transport: LS square-loop acts as Γ)**.** *Assume $\mathbf{2T}_{\mathrm{LS}}$ and the tag action Definition G.2. On the first-descendant ambiguity group $\mathcal{V}^1$ (Definition G.3), changing the square filler from* diag *to* bridge *acts by Γ on representatives and by multiplication by -1 on classes:*

$$[\Gamma(\mathcal{J})] = -[\mathcal{J}] \quad \textit{in } \mathcal{V}^1.$$

In particular, if $[\mathcal{J}] \neq 0$ then it lies in the odd sector of the Γ–grading.

Proof. By Lemma G.9, $\mathcal{J}_{\mathrm{bridge}} = \Gamma(\mathcal{J}_{\mathrm{diag}})$, while Lemma G.12 gives $[\mathcal{J}_{\mathrm{bridge}}] = -[\mathcal{J}_{\mathrm{diag}}]$ in $\mathcal{V}^1$. Thus $[\Gamma(\mathcal{J})] = -[\mathcal{J}]$. □

Appendix H: Subprogram 3: Selected-line nonvanishing, Hamiltonized odd derivation, AQFT and HLS

1. Subprogram 3a: Selected-line nonvanishing, canonicalization, and characteristic activity

In this subsection we work directly with the two-time bicomplex $(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t, Q_u)$ supplied by Subprogram 1a and allow an explicit anomaly insertion as in (B1). This keeps the mixed-anomaly bookkeeping in the full two-ghost complex.

Theorem H.1 (Finite-cutoff canonical odd representative)**.** *Assume the calibration characterization package $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$, namely Definition B.12 together with Definition B.13. For each ledger step k, let $[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k}$ be the admissible odd descendant class. Then there exists a unique representative*

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$$

such that

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_k} = \arg\min_{J \in [\Xi^{(1)}_{\Lambda_k}]} \mathsf{B}_{\Lambda_k}(J,J). \tag{H1}$$

Proof. By Definition B.13, the form B_{Λ_k} Hilbertizes $E^{-}_{\Lambda_k}$; write H_{Λ_k} for the resulting Hilbert space with inner product still denoted B_{Λ_k}. Since $\mathcal{A}^{-}_{\Lambda_k} \subset E^{-}_{\Lambda_k}$ is closed, the affine set $[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathcal{A}^{-}_{\Lambda_k}$ is a closed affine subspace of H_{Λ_k}. The Hilbert projection theorem therefore gives a unique element of this affine subspace of minimal B_{Λ_k}-norm, which is exactly $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$. □

Proposition H.1 (Orthogonality characterization of $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$)**.** *Under the hypotheses of Theorem H.1, the unique minimizer $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ is characterized equivalently by*

$$\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A) = 0 \qquad \forall\, A \in \mathcal{A}^-_{\Lambda_k}. \tag{H2}$$

Proof. Write $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} = J_{0,\Lambda_k} + A_*$ with $A_* \in \mathcal{A}^-_{\Lambda_k}$. For any $A \in \mathcal{A}^-_{\Lambda_k}$ and $s \in \mathbb{R}$, the point $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} + sA$ lies in the same affine class. Minimality of $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ implies

$$0 = \frac{d}{ds}\Big|_{s=0} \mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k} + sA,\, \mathcal{J}^{\mathrm{can}}_{\Lambda_k} + sA) = 2\,\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A),$$

which is (H2). Conversely, if $J \in [\Xi^{(1)}_{\Lambda_k}]$ satisfies (H2), then for every $A \in \mathcal{A}^-_{\Lambda_k}$,

$$\mathsf{B}_{\Lambda_k}(J + A, J + A) = \mathsf{B}_{\Lambda_k}(J, J) + \mathsf{B}_{\Lambda_k}(A, A) \geq \mathsf{B}_{\Lambda_k}(J, J),$$

so J is the unique minimizer on the affine class. □

Remark H.1 (Finite-dimensional projection formula)**.** In any finite-dimensional realization of the affine odd descendant class

$$[\Xi^{(1)}_{\Lambda_k}] = J_{0,\Lambda_k} + \mathrm{im}(U_{\Lambda_k}),$$

where U_{Λ_k} is any full-column-rank basis matrix for $\mathcal{A}^-_{\Lambda_k}$, let B_{Λ_k} be the matrix of the calibration form B_{Λ_k}. Then the B_{Λ_k}-orthogonal projector onto the ambiguity subspace is

$$P^{\mathsf{B}_{\Lambda_k}}_{\mathcal{A}^-_{\Lambda_k}} := U_{\Lambda_k}\big(U^\top_{\Lambda_k} B_{\Lambda_k} U_{\Lambda_k}\big)^{-1} U^\top_{\Lambda_k} B_{\Lambda_k},$$

and the canonical representative is

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_k} = \big(I - P^{\mathsf{B}_{\Lambda_k}}_{\mathcal{A}^-_{\Lambda_k}}\big) J_{0,\Lambda_k}.$$

Equivalently, if

$$a_* := -\big(U^\top_{\Lambda_k} B_{\Lambda_k} U_{\Lambda_k}\big)^{-1} U^\top_{\Lambda_k} B_{\Lambda_k} J_{0,\Lambda_k},$$

then

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_k} = J_{0,\Lambda_k} + U_{\Lambda_k} a_*.$$

This formula is independent of the chosen basis U_{Λ_k} of $\mathcal{A}^-_{\Lambda_k}$: replacing U_{Λ_k} by $U_{\Lambda_k}T$ with $T \in \mathrm{GL}(r, \mathbb{R})$ leaves the projector $P^{\mathsf{B}_{\Lambda_k}}_{\mathcal{A}^-_{\Lambda_k}}$ unchanged. It is also independent of the chosen representative $J_{0,\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$: replacing J_{0,Λ_k} by $J_{0,\Lambda_k} + U_{\Lambda_k} h$ leaves $\big(I - P^{\mathsf{B}_{\Lambda_k}}_{\mathcal{A}^-_{\Lambda_k}}\big) J_{0,\Lambda_k}$ unchanged.

Remark H.2 (Proof strategy for same-tick obstruction nonvanishing)**.** The nonvanishing step is purely quotient-theoretic for the same-tick obstruction lane once one has a faithful LS-to-descendant intertwiner. One first isolates a selected odd descendant line inside the raw first-descendant ambiguity group $\mathcal{V}^1$ and uses Lemma G.13 to see that the LS square-loop acts there by sign. One then maps that line into the characterization quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$. If this realization map is faithful on the selected line, then the image of a nonzero generator is itself nonzero; equivariance therefore upgrades the sign transport on the raw line to a nontrivial sign transport on the selected quotient point. At that stage the conclusion $0 \notin [\Xi^{(1)}_{\Lambda_k}]$ is immediate, because 0 is the only fixed point of the sign involution on a real vector space. This is the abstract same-tick obstruction route; the physically correct delayed route is the lift recorded below.

Remark H.3 (Proof skeleton for the same-tick LS-to-descendant faithfulness step)**.** Fix a ledger step k. The structured proof has four steps:

(i) choose a selected odd descendant line $L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k} \subset \mathcal{V}^1$ with $\ell_{\Lambda_k} \neq 0$;

(ii) use Lemma G.13 to identify the LS square-loop action on that line with multiplication by -1;

(iii) realize the selected line on the characterization quotient by a linear map $\mathfrak{R}_{\Lambda_k} : L^{\mathrm{sel}}_{\Lambda_k} \to E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$ whose value on ℓ_{Λ_k} is the selected quotient point $\overline{\Xi}^{(1)}_{\Lambda_k}$;

(iv) prove faithfulness on the selected line, i.e. $\ker \mathfrak{R}_{\Lambda_k} = 0$.

Then $\overline{\Xi}^{(1)}_{\Lambda_k} = \mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) \neq 0$, and equivariance gives

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}(\overline{\Xi}^{(1)}_{\Lambda_k}) = -\,\overline{\Xi}^{(1)}_{\Lambda_k} \neq \overline{\Xi}^{(1)}_{\Lambda_k}.$$

The theorem below expands this same-tick obstruction skeleton into a full proof.

Proposition H.2 (Selected-line compatibility suffices for the same-tick faithfulness hypotheses)**.** *Assume* $\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$*. Fix a ledger step* k*, and let*

$$q_{\Lambda_k} : E^-_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$$

be the quotient map. Assume the selected-line supplement package $\mathbf{2T}_{\mathrm{sel}}$ *of Definition B.17, and write*

$$\ell_{\Lambda_k} := [J^{\mathrm{sel}}_{\Lambda_k}]_{\mathcal{V}^1} \neq 0.$$

Define

$$L^{\mathrm{sel}}_{\Lambda_k} := \mathbb{R}\,\ell_{\Lambda_k}, \qquad \mathfrak{R}_{\Lambda_k}(a\,\ell_{\Lambda_k}) := a\,q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}).$$

Then hypotheses (a)–(e) *of Theorem H.3 hold.*

Proof. Condition (a) of Theorem H.3 holds by construction: $L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k}$ is one-dimensional and nonzero because $\ell_{\Lambda_k} \neq 0$.

Condition (b) holds because $\mathfrak{R}_{\Lambda_k}$ is defined on a real line by scalar multiplication of the fixed quotient element $q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k})$.

Condition (c) holds by direct substitution:

$$\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) = q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}).$$

For condition (d), linearity on the one-dimensional real line gives

$$\mathfrak{R}_{\Lambda_k}(-\ell_{\Lambda_k}) = -\,\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}).$$

It remains to prove faithfulness (e). Let $a\,\ell_{\Lambda_k} \in \ker \mathfrak{R}_{\Lambda_k}$. Then

$$a\,q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}) = 0.$$

If $a \neq 0$, then $q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}) = 0$, so the selected-line compatibility implication (B2) yields

$$[J^{\mathrm{sel}}_{\Lambda_k}]_{\mathcal{V}^1} = 0,$$

contradicting the definition of $\ell_{\Lambda_k} \neq 0$. Hence $a = 0$, and therefore

$$\ker \mathfrak{R}_{\Lambda_k} = \{0\}.$$

So condition (e) holds as well. □

Definition H.1 (Scheme-refined raw descendant quotient)**.** Fix a ledger step k. Define the *scheme-refined raw descendant quotient* by

$$\widehat{\mathcal{V}}^1_{\Lambda_k} := \frac{\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}}{Q_t(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}) + Q^{\mathrm{scheme}}_u(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}})}.$$

The coarse ambiguity group $\mathcal{V}^1$ of Definition G.3 is the further quotient obtained by killing the remaining Q^{phys}_u-exact shifts. We write

$$\pi_{\Lambda_k} : \widehat{\mathcal{V}}^1_{\Lambda_k} \twoheadrightarrow \mathcal{V}^1$$

for that quotient map.

Remark H.4 (What the refined quotient isolates)**.** The quotient $\widehat{\mathcal{V}}^1_{\Lambda_k}$ is introduced only to separate the genuinely physical Q_u^{phys}-collapse from the already isolated scheme sector. It does not attempt to incorporate boundary-admissible divergences or local counterterm coboundaries into the raw descendant group itself. Those local improvement directions are already handled separately: they appear in the locality quotient relation (Definition B.7) and, on the characterization side, are absorbed into $\mathcal{A}^-_{\Lambda_k}$ by the characterization supplement.

Theorem H.2 (Scheme-refined same-tick no-extra-collapse yields the selected-line supplement)**.** *Assume* $\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$. *Fix a ledger step k, and let*

$$q_{\Lambda_k} : E^-_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$$

be the quotient map. Assume there exist:

(a) *a one-dimensional selected odd line*

$$\widehat{L}^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\widehat{\ell}_{\Lambda_k} \subset \widehat{\mathcal{V}}^1_{\Lambda_k};$$

(b) *noncollapse under the coarse quotient,*

$$\pi_{\Lambda_k}(\widehat{\ell}_{\Lambda_k}) \neq 0;$$

(c) *a selected representative*

$$J^{\mathrm{sel}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}] \cap E^-_{\Lambda_k};$$

(d) *a linear realization map*

$$\widehat{\mathfrak{R}}_{\Lambda_k} : \widehat{L}^{\mathrm{sel}}_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$$

satisfying

$$\widehat{\mathfrak{R}}_{\Lambda_k}(\widehat{\ell}_{\Lambda_k}) = q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k});$$

(e) *exact no-extra-collapse on the selected line,*

$$\ker\big(\widehat{\mathfrak{R}}_{\Lambda_k}|_{\widehat{L}^{\mathrm{sel}}_{\Lambda_k}}\big) = \ker\big(\pi_{\Lambda_k}|_{\widehat{L}^{\mathrm{sel}}_{\Lambda_k}}\big).$$

Then the selected-line Main supplement package $\mathbf{2T}_{\mathrm{sel}}$ *of Definition B.17 holds with*

$$\ell_{\Lambda_k} := \pi_{\Lambda_k}(\widehat{\ell}_{\Lambda_k}) \in \mathcal{V}^1.$$

Consequently, Proposition H.2 applies.

Proof. By assumption (b), the class $\ell_{\Lambda_k} := \pi_{\Lambda_k}(\widehat{\ell}_{\Lambda_k})$ is nonzero in $\mathcal{V}^1$, so the first part of Definition B.17 holds with $L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k}$. The second part holds by the chosen representative $J^{\mathrm{sel}}_{\Lambda_k}$.

It remains to verify the selected-line compatibility implication (B2). Assume

$$q_{\Lambda_k}(J^{\mathrm{sel}}_{\Lambda_k}) = 0.$$

Using assumption (d), this means

$$\widehat{\mathfrak{R}}_{\Lambda_k}(\widehat{\ell}_{\Lambda_k}) = 0,$$

so $\widehat{\ell}_{\Lambda_k} \in \ker(\widehat{\mathfrak{R}}_{\Lambda_k}|_{\widehat{L}^{\mathrm{sel}}_{\Lambda_k}})$. By assumption (e),

$$\widehat{\ell}_{\Lambda_k} \in \ker(\pi_{\Lambda_k}|_{\widehat{L}^{\mathrm{sel}}_{\Lambda_k}}),$$

hence

$$\pi_{\Lambda_k}(\widehat{\ell}_{\Lambda_k}) = 0.$$

By definition of ℓ_{Λ_k}, this is exactly

$$[J^{\mathrm{sel}}_{\Lambda_k}]_{\mathcal{V}^1} = 0.$$

So the selected-line compatibility implication holds, completing the selected-line supplement contract.

The final sentence follows by Proposition H.2. □

Theorem H.3 (Same-tick faithful LS-to-descendant intertwiner yields a nontrivial square-loop witness)**.** *Assume* $\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$. *Fix a ledger step* k, *and let*

$$q_{\Lambda_k} : E^-_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$$

be the quotient map. Assume there exist:

(a) *a one-dimensional selected odd descendant line*

$$L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k} \subset \mathcal{V}^1 \qquad \textit{with} \qquad \ell_{\Lambda_k} \neq 0;$$

(b) *a linear realization map*

$$\mathfrak{R}_{\Lambda_k} : L^{\mathrm{sel}}_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k};$$

(c) *a representative* $J_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ *such that*

$$\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) = q_{\Lambda_k}(J_{\Lambda_k});$$

(d) *equivariance of the realization map with respect to the LS square-loop:*

$$\mathfrak{R}_{\Lambda_k}(-\ell_{\Lambda_k}) = -\,\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k});$$

(e) *faithfulness on the selected line:*

$$\ker \mathfrak{R}_{\Lambda_k} = \{0\}.$$

Define

$$\overline{\Xi}^{(1)}_{\Lambda_k} := q_{\Lambda_k}(J) \qquad \big(J \in [\Xi^{(1)}_{\Lambda_k}]\big),$$

which is well defined because $[\Xi^{(1)}_{\Lambda_k}]$ *is an affine class modulo* $\mathcal{A}^-_{\Lambda_k}$. *Then*

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}(\overline{\Xi}^{(1)}_{\Lambda_k}) = -\,\overline{\Xi}^{(1)}_{\Lambda_k} \neq \overline{\Xi}^{(1)}_{\Lambda_k}. \tag{H3}$$

Consequently the hypotheses of Theorem H.4 hold at ledger step k.

Proof. By Lemma G.13, the LS square-loop acts by sign on raw first-descendant classes in $\mathcal{V}^1$:

$$[\Gamma(\mathcal{J})] = -[\mathcal{J}] \qquad \text{in } \mathcal{V}^1.$$

Since $L^{\mathrm{sel}}_{\Lambda_k} = \mathbb{R}\,\ell_{\Lambda_k} \subset \mathcal{V}^1$ is assumed to be a selected odd descendant line, this sign transport restricts to

$$\mathrm{Act}^{\mathcal{V}^1}_{\mathrm{sqLoop}}(\ell_{\Lambda_k}) = -\,\ell_{\Lambda_k}.$$

Next, identify the selected quotient point with the image of the chosen generator. By assumption (c),

$$\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) = q_{\Lambda_k}(J_{\Lambda_k})$$

for some $J_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$. But by definition of $\overline{\Xi}^{(1)}_{\Lambda_k}$, the quotient value $q_{\Lambda_k}(J)$ is independent of the chosen representative $J \in [\Xi^{(1)}_{\Lambda_k}]$. Therefore

$$\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) = \overline{\Xi}^{(1)}_{\Lambda_k}.$$

We now prove that this quotient point is nonzero. Because $\ell_{\Lambda_k} \neq 0$ by assumption (a) and $\ker \mathfrak{R}_{\Lambda_k} = \{0\}$ by assumption (e), one has

$$\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) \neq 0.$$

Using the identification just proved, this gives

$$\overline{\Xi}^{(1)}_{\Lambda_k} \neq 0 \qquad \text{in} \qquad E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}.$$

Next, transport the square-loop action to the quotient. By assumption (d),

$$\mathfrak{R}_{\Lambda_k}(-\ell_{\Lambda_k}) = -\,\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}).$$

Substituting $\mathfrak{R}_{\Lambda_k}(\ell_{\Lambda_k}) = \overline{\Xi}^{(1)}_{\Lambda_k}$, we obtain

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}\big(\overline{\Xi}^{(1)}_{\Lambda_k}\big) = -\,\overline{\Xi}^{(1)}_{\Lambda_k}.$$

Since $\overline{\Xi}^{(1)}_{\Lambda_k} \neq 0$ and the quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$ is a real vector space, the only fixed point of $x \mapsto -x$ is $x = 0$. Hence

$$-\,\overline{\Xi}^{(1)}_{\Lambda_k} \neq \overline{\Xi}^{(1)}_{\Lambda_k}.$$

This proves (H3).

Finally, compare with Theorem H.4. That theorem requires exactly two inputs on the selected quotient point: sign transport and nontriviality of the square-loop action. These are precisely what was established above, namely

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}\big(\overline{\Xi}^{(1)}_{\Lambda_k}\big) = -\,\overline{\Xi}^{(1)}_{\Lambda_k} \qquad \text{and} \qquad \mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}\big(\overline{\Xi}^{(1)}_{\Lambda_k}\big) \neq \overline{\Xi}^{(1)}_{\Lambda_k}.$$

Therefore the hypotheses of Theorem H.4 hold at ledger step k. □

Theorem H.4 (Nonvanishing from a same-tick square-loop witness on the transported descendant quotient)**.** *Assume the weak polarized characterization interface* $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ (Definition B.12)*. Fix a ledger step k, and let*

$$q_{\Lambda_k} : E^-_{\Lambda_k} \longrightarrow E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$$

be the quotient map. Define the transported descendant quotient point by

$$\overline{\Xi}^{(1)}_{\Lambda_k} := q_{\Lambda_k}(J) \qquad \big(J \in [\Xi^{(1)}_{\Lambda_k}]\big),$$

which is well defined because $[\Xi^{(1)}_{\Lambda_k}]$ is an affine class modulo $\mathcal{A}^-_{\Lambda_k}$. Assume moreover that the abstract LS square-loop transports this selected quotient point by sign:

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}\big(\overline{\Xi}^{(1)}_{\Lambda_k}\big) = -\,\overline{\Xi}^{(1)}_{\Lambda_k}, \tag{H4}$$

and that this transport is nontrivial on the selected descendant point:

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}\big(\overline{\Xi}^{(1)}_{\Lambda_k}\big) \neq \overline{\Xi}^{(1)}_{\Lambda_k}. \tag{H5}$$

Then

$$0 \notin [\Xi^{(1)}_{\Lambda_k}].$$

Proof. First note that $\overline{\Xi}^{(1)}_{\Lambda_k}$ is well defined. Indeed, if $J_1, J_2 \in [\Xi^{(1)}_{\Lambda_k}]$, then $J_2 - J_1 \in \mathcal{A}^-_{\Lambda_k}$, so $q_{\Lambda_k}(J_1) = q_{\Lambda_k}(J_2)$ in the quotient $E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}$.

Assume for contradiction that

$$0 \in [\Xi^{(1)}_{\Lambda_k}].$$

Then, since $[\Xi^{(1)}_{\Lambda_k}]$ is an affine class modulo $\mathcal{A}^-_{\Lambda_k}$, its quotient point is the zero class:

$$\overline{\Xi}^{(1)}_{\Lambda_k} = q_{\Lambda_k}(0) = 0 \qquad \text{in} \qquad E^-_{\Lambda_k}/\mathcal{A}^-_{\Lambda_k}.$$

Applying the square-loop transport law (H4) then gives

$$\mathrm{Act}^{\mathrm{desc}}_{\mathrm{sqLoop}}(\overline{\Xi}^{(1)}_{\Lambda_k}) = -\,\overline{\Xi}^{(1)}_{\Lambda_k} = -0 = 0 = \overline{\Xi}^{(1)}_{\Lambda_k},$$

which contradicts the nontriviality assumption (H5). Therefore the assumption $0 \in [\Xi^{(1)}_{\Lambda_k}]$ is false, proving

$$0 \notin [\Xi^{(1)}_{\Lambda_k}].$$

□

Theorem H.5 (Delayed selected line yields a nonzero quotient point)**.** *Assume* $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$*. Fix consecutive ledger steps* k *and* $k+1$*. Assume there are delayed-lift data* $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}, J^{\mathrm{split}}_{\Lambda_{k+1}}, \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}$*, and a scalar* $c_{\Lambda_{k+1}}$ *as displayed below. In particular, let*

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} := \mathfrak{L}^{\mathrm{split}}_{k\to k+1}(\widehat{\ell}^{\mathrm{delay}}_k) = q_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) \in \mathfrak{D}^-_{\Lambda_{k+1}},$$

with

$$J^{\mathrm{split}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}.$$

Assume moreover that

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0 \qquad \textit{and} \qquad \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}} \neq 0.$$

Then:

(i) $\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$*;*

(ii)

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}} \neq 0;$$

(iii) *the map* $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}$ *is faithful on* $\widehat{L}^{\mathrm{sel,delay}}_k$*;*

(iv) *the delayed selected-line supplement package* $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$ *of Definition B.18 holds.*

Proof. Items (i) and (ii) are assumed. Since $\widehat{L}^{\mathrm{sel,delay}}_k$ is one-dimensional and $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}(\widehat{\ell}^{\mathrm{delay}}_k) = \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$, the map $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}$ is faithful on $\widehat{L}^{\mathrm{sel,delay}}_k$, proving (iii). The representative $J^{\mathrm{split}}_{\Lambda_{k+1}}$ there satisfies

$$q_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}.$$

So every clause of Definition B.18 is satisfied. □

Theorem H.6 (Nonvanishing from the delayed lift)**.** *Assume* $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$*. Fix consecutive ledger steps* k *and* $k+1$*. Assume the delayed selected-line supplement package* $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$ *from Definition B.18. Then*

$$0 \notin [\Xi^{(1)}_{\Lambda_{k+1}}].$$

Proof. By Definition B.18, there exists $J^{\mathrm{split}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^{-}_{\Lambda_{k+1}}$ with

$$q_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}.$$

The same definition gives $\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$. Assume for contradiction that $0 \in [\Xi^{(1)}_{\Lambda_{k+1}}]$. Then every representative of the affine class has quotient image zero, hence in particular

$$q_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = 0,$$

contradicting $\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$. Therefore $0 \notin [\Xi^{(1)}_{\Lambda_{k+1}}]$. □

Corollary H.1 (Canonical representative is nonzero on the delayed route)**.** *Assume* $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ *and the hypotheses of Theorem H.6. Then*

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}} \neq 0.$$

Proof. By Theorem H.6, $0 \notin [\Xi^{(1)}_{\Lambda_{k+1}}]$. If $\mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}} = 0$, then since $\mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}]$ by Theorem H.1, one would have $0 \in [\Xi^{(1)}_{\Lambda_{k+1}}]$, contradiction. □

Corollary H.2 (Canonical dominant-kernel reentry on the delayed selected quotient line)**.** *Fix consecutive ledger steps k and $k+1$. Assume the hypotheses of Theorem H.5. Define*

$$N^{\mathrm{dom-null}}_{\Lambda_{k+1}} := \ker \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

Assume furthermore that $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is admissible as the mod-null datum and satisfies the required transversality condition. In particular, the mod-null selected line and its faithful realization map are canonically defined at Λ_{k+1}.

Proof. By the additional hypothesis, the dominant-kernel datum $N^{\mathrm{dom-null}}_{\Lambda_{k+1}}$ is admissible as the mod-null datum, the required transversality holds, and the conclusions of the canonical mod-null reentry theorem hold at Λ_{k+1}. □

Remark H.5 (Exact boundary of the delayed selected-line lane)**.** The delayed selected-line lane establishes a quotient-first delayed route: there is a delayed selected carrier at step k, a faithful delayed lift to a nonzero dominant-active quotient point in $\mathfrak{D}^{-}_{\Lambda_{k+1}}$, and a canonical dominant-kernel reentry lane. What is not yet derived from the delayed quotient data alone is the existence of one subordinate unit descendant realization refining that quotient point. Accordingly, this lane should not be presented as a fully closed microscopic reconstruction theorem.

Corollary H.3 (Canonical representative is nonzero once the class is nonzero)**.** *Assume* $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ *and the hypotheses of Theorem H.4. Then*

$$\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \neq 0.$$

Proof. By Theorem H.4, $0 \notin [\Xi^{(1)}_{\Lambda_k}]$. If $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} = 0$, then since $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ by Theorem H.1, one would have $0 \in [\Xi^{(1)}_{\Lambda_k}]$, contradiction. □

Theorem H.7 (The characterization algebra is derived from the canonical representative)**.** *Assume* $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$*. For each ledger step k, let*

$$\mathcal{O}^{\mathrm{char}}_{\Lambda_k} := \mathrm{alg}\Big\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \big(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}\big)^2\Big\} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}.$$

Then the characterization-algebra supplement $\mathbf{2T}_{\mathrm{alg}}$ (Definition B.15) *holds.*

Proof. By Theorem H.1, the canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in E^-_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ is defined for each ledger step k. Since $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ is an algebra, the algebra generated by $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ and its even square $\left(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}\right)^2$ is a distinguished subalgebra of $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. This is exactly the datum required in Definition B.15.

□

The nonzero-class input used below may now be supplied either by the same-tick obstruction/square-loop route of Theorem H.4 or by the delayed lift route of Theorem H.6; the latter passes through the delayed $k \to k+1$ step.

Proposition H.3 (Canonical activity functional from a nonzero odd descendant class)**.** *Assume* $\mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$. *Fix a ledger step k. If $0 \notin [\Xi^{(1)}_{\Lambda_k}]$, equivalently $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \neq 0$, define*

$$\chi^{\mathrm{can}}_{\Lambda_k}(F) := \frac{\mathsf{B}_{\Lambda_k}(F, \mathcal{J}^{\mathrm{can}}_{\Lambda_k})}{\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \mathcal{J}^{\mathrm{can}}_{\Lambda_k})} \qquad (F \in E^-_{\Lambda_k}).$$

Then $\chi^{\mathrm{can}}_{\Lambda_k}$ is a continuous nonzero linear functional on $E^-_{\Lambda_k}$ satisfying

$$\chi^{\mathrm{can}}_{\Lambda_k}(A) = 0 \qquad \forall\, A \in \mathcal{A}^-_{\Lambda_k},$$

and

$$\chi^{\mathrm{can}}_{\Lambda_k}(J) = 1 \qquad \forall\, J \in [\Xi^{(1)}_{\Lambda_k}].$$

In particular, once the class is known to be nonzero, activity/orientation are derived canonically and require no extra weak-interface input.

Proof. Continuity and linearity are immediate from the bilinearity of B_{Λ_k}, and the denominator is strictly positive because $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \neq 0$ and B_{Λ_k} is positive definite. For $A \in \mathcal{A}^-_{\Lambda_k}$, Proposition H.1 gives

$$\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A) = 0,$$

hence $\chi^{\mathrm{can}}_{\Lambda_k}(A) = 0$. Now let $J \in [\Xi^{(1)}_{\Lambda_k}]$. Since $J - \mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in \mathcal{A}^-_{\Lambda_k}$, write $J = \mathcal{J}^{\mathrm{can}}_{\Lambda_k} + A$ with $A \in \mathcal{A}^-_{\Lambda_k}$. Then

$$\chi^{\mathrm{can}}_{\Lambda_k}(J) = \frac{\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \mathcal{J}^{\mathrm{can}}_{\Lambda_k} + A)}{\mathsf{B}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \mathcal{J}^{\mathrm{can}}_{\Lambda_k})} = 1$$

by the orthogonality just proved. In particular $\chi^{\mathrm{can}}_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = 1$, so the functional is nonzero. □

Remark H.6 (Canonicalization lives on the descendant class)**.** Theorem H.1 is intentionally a theorem about the affine odd descendant class $[\Xi^{(1)}_{\Lambda_k}]$, not about all of $E^-_{\Lambda_k}$. Once the class is known to be nonzero, Proposition H.3 supplies a canonical orientation functional, but that functional still lives on the already identified affine class and does not by itself pick a unique element of the full odd generator space.

Remark H.7 (Selected-line lanes)**.** There are two selected-line sublanes. The same-tick package $\mathbf{2T}_{\mathrm{sel}}$ is the abstract obstruction lane used to transport the square-loop sign and rule out extra collapse on a same-tick selected carrier. The delayed package $\mathbf{2T}^{\mathrm{delay}}_{\mathrm{sel}}$ is the quotient-first delayed lane: it yields a nonzero dominant-active quotient point at Λ_{k+1}, feeds the canonical dominant-kernel reentry theorem, and supplies the delayed odd-to-even route. Its nondegeneracy clause $c_{\Lambda_{k+1}} \neq 0$ is built into the delayed package as an imported hypothesis; deriving that clause from stronger upstream transport/overlap input remains open, and the bridge-closing existence of one subordinate realization remains open.

2. Subprogram 3b: RG monotone and Hamiltonized odd derivation

In this subsection we return to the RG monotone package and combine it with the canonicalized odd data obtained above to produce the Hamiltonized local odd derivation and its physical consequences on the sufficiently local boundary algebra.

Definition H.2 (RG monotone)**.** Assume $\mathbf{2T}_{\mathrm{RG}}$. We call $\mathcal{O}^{(0)}(u_0) \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ a *good observable* if it is Q_t–closed, represents a nontrivial class in $H^0(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t)$, and defines the monotone via $M(u) = \langle \mathcal{O}^{(0)}(u) \rangle_{\rho_u}$.

Lemma H.1 (RG slope equals the anomaly insertion)**.** *Assume* $\mathbf{2T}_{\mathrm{RG}}$ *and the anomalous descent* (B1)*. Then at the UV ceiling,*

$$\partial_u M(u)\big|_{u_0} \;=\; -\big\langle \mathcal{A}^{(1)}(u_0) \big\rangle_{\rho_{u_0}}.$$

In particular, if $\partial_u M(u)|_{u_0} < 0$*, then* $\big\langle \mathcal{A}^{(1)}(u_0) \big\rangle_{\rho_{u_0}} > 0$*.*

Proof. By $\mathbf{2T}_{\mathrm{RG}}$-**(M2)**, $\partial_u M(u_0) = -\langle (Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0) \rangle_{\rho_{u_0}}$. Taking the expectation value of (B1) and using $\langle Q_t X \rangle_{\rho_{u_0}} = 0$ gives $\langle (Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0) \rangle_{\rho_{u_0}} = \langle \mathcal{A}^{(1)}(u_0) \rangle_{\rho_{u_0}}$. □

Proposition H.4 (Nontrivial descendant sector detected by the RG slope)**.** *Assume* $\mathbf{2T}_{\mathrm{BRST}} + \mathbf{2T}_{\mathrm{RG}}$*. If the monotone has nontrivial UV slope* $\partial_u M(u)|_{u_0} < 0$*, then the degree-one cohomology class*

$$\big[\, (Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0) \,\big] \;\neq\; 0 \qquad \text{in} \qquad H^1(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t).$$

Proof. By $\mathbf{2T}_{\mathrm{RG}}$-**(M2)**, $(Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0)$ is Q_t–closed (so it defines a class in $H^1(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t)$). Suppose for contradiction that it is Q_t–exact: $(Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0) = Q_t Y$ for some $Y \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. Then $\mathbf{2T}_{\mathrm{BRST}}$-**(B2)** gives $\langle (Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0) \rangle_{\rho_{u_0}} = \langle Q_t Y \rangle_{\rho_{u_0}} = 0$. This contradicts Lemma H.1 under $\partial_u M(u)|_{u_0} < 0$. □

Theorem H.8 (Good observable from RG monotone)**.** *Under* $\mathbf{2T}_{\mathrm{BRST}} + \mathbf{2T}_{\mathrm{RG}}$*, there exists a nontrivial* Q_t*–closed representative* $\mathcal{O}^{(0)}$ *of the RG monotone and, by* (B1)*, a raw first-descendant potential* $\Xi^{(1)}$ *together with an anomaly insertion* $\mathcal{A}^{(1)}$ *implementing the anomalous first descent. Moreover, every sufficiently local representative* $J \in [\Xi^{(1)}]$ *is EB3-admissible in the sense of Definition B.11 (Lemma B.1). Moreover, if the UV slope is nontrivial (*$\partial_u M(u)|_{u_0} < 0$*), then* $(Q_u^{\mathrm{phys}} \mathcal{O}^{(0)})(u_0)$ *defines a nonzero class in* $H^1(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}, Q_t)$ *(Proposition H.4).*

Remark H.8 (Cohomological vs. physical parity)**.** Recall Definition B.4. In particular, the first-descendant data $\Xi^{(1)}$ have cohomological degree 0 at representative level, whereas the canonical odd representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ is chosen in the physically odd sector $\Gamma(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = -\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$.

Definition H.3 (Hamiltonized odd derivation)**.** Assume $\mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ so the Peierls bracket is defined on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ and the canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ is available on the ledger-step skeleton. For a fixed ledger step k, define

$$\delta_{\Lambda_k}(A) := \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k}$$

for $A \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$.

Lemma H.2 (Hamiltonian parity ⇒ derivation parity)**.** *Assume* Γ *is a Poisson automorphism on* $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ *(Lemma F.3). If* $\Gamma(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = -\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$*, then the Hamiltonian derivation* $\delta_{\Lambda_k} := \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}$ *is* Γ*–odd:*

$$\Gamma \delta_{\Lambda_k} \Gamma^{-1} = -\delta_{\Lambda_k}.$$

Proof. For $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$, $\Gamma \delta_{\Lambda_k}(F) = \Gamma\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, F\}_{\mathrm{P},\Lambda_k} = \{\Gamma \mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \Gamma F\}_{\mathrm{P},\Lambda_k} = -\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \Gamma F\}_{\mathrm{P},\Lambda_k} = -\delta_{\Lambda_k}(\Gamma F)$. □

Remark H.9 (Optional Peierls closure identity)**.** For the chosen first-descendant potential $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$, one may prove the existence of a translation-direction Hamiltonian $\mathcal{H}_\xi \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ (for some constant Minkowski vector ξ^μ) such that, modulo Peierls-central terms and the standard Q_t-exact/contact ambiguities in the locality quotient,

$$\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \mathcal{J}^{\mathrm{can}}_{\Lambda_k}\}_{\mathrm{P},\Lambda_k} \;=\; 2\,\mathcal{H}_\xi. \tag{H6}$$

If (H6) holds, then for all $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$,

$$\delta^2_{\Lambda_k}(F) \;=\; \tfrac{1}{2}\,\big\{\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \mathcal{J}^{\mathrm{can}}_{\Lambda_k}\}_{\mathrm{P},\Lambda_k}, F\big\}_{\mathrm{P},\Lambda_k} \;=\; \{\mathcal{H}_\xi, F\}_{\mathrm{P}},$$

so $\delta^2_{\Lambda_k}$ acts as the (inner) translation derivation generated by $\mathcal{H}_\xi$.

Proposition H.5 (Characterization activity implies nontriviality of δ on the BU/AQFT core)**.** *Assume* $\mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}} + \mathbf{2T}^{\mathrm{act}}_{\mathrm{char}}$. *Fix a ledger step k, and let $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$ be the canonical representative of Theorem H.1. Then there exists an even sufficiently local probe*

$$A \in \mathcal{P}^{+}_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$$

such that

$$\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k} \neq 0.$$

Equivalently, the Hamiltonized derivation $\delta_{\Lambda_k} = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}$ is nontrivial on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. After passage to the Borchers–Uhlmann core (Definition D.1 and Lemma H.6)*, this gives a nontrivial graded *-derivation on $\mathfrak{A}_{\mathrm{fin}}$.*

Proof. Because $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in [\Xi^{(1)}_{\Lambda_k}]$, Definition B.12**(C5)** gives

$$\chi_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = \varepsilon_{\Lambda_k} \neq 0.$$

If instead

$$\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k} = 0 \qquad \forall\, A \in \mathcal{P}^{+}_{\Lambda_k},$$

then Definition B.14 would force

$$\chi_{\Lambda_k}(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = 0,$$

contradiction. Therefore some $A \in \mathcal{P}^{+}_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$ satisfies

$$\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k} \neq 0.$$

For that same A, the BU-core extension from Lemma H.6 satisfies

$$\delta_{\Lambda_k}(\Phi(A)) = \Phi\big(\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k}\big).$$

Since the Borchers–Uhlmann core imposes only the linearity, *-structure, and locality relations on generator symbols, a nonzero local functional gives a nonzero generator symbol. Hence δ_{Λ_k} is nontrivial on $\mathfrak{A}_{\mathrm{fin}}$ as well. □

Lemma H.3 (Derivation and locality of the canonical Hamiltonian action)**.** *Assume* $\mathbf{2T}_{\mathrm{Peierls}} + \mathbf{2T}_{\mathrm{EBC}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$*, and fix the EB3-admissible canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in E^{-}_{\Lambda_k}$* (Theorem H.1 and Lemma B.1)*. Then*

$$\delta_{\Lambda_k}(F) := \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, F\}_{\mathrm{P},\Lambda_k}$$

is a graded derivation on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$. Moreover, for every double cone O,

$$\delta_{\Lambda_k}(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)) \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O),$$

so the BU-core assignment $\delta_{\Lambda_k}(\Phi(F)) := \Phi(\delta_{\Lambda_k} F)$ from Definition B.10(EB3) *is local on generators.*

Proof. By Theorem H.1, the canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ belongs to the odd sufficiently local class $E^{-}_{\Lambda_k} \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$, so it is an admissible input for the Peierls bracket. Theorem F.1 identifies the Peierls bracket as a graded Poisson bracket on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$; for fixed $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$, the Leibniz and Jacobi identities therefore imply that the Hamiltonian adjoint action $F \mapsto \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, F\}_{\mathrm{P},\Lambda_k}$ is a graded derivation on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$.

For locality, let $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$. The sufficiently local class is closed under the Peierls construction in the same locality regime used in $\mathbf{2T}_{\mathrm{Peierls}}$-**(P5)** and the EB3 estimate, so the bracket with the fixed local generator $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ remains supported in the same double cone O. Hence $\delta_{\Lambda_k}(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)) \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$. The BU-core extension from Definition B.10(EB3) is therefore local on generators. □

Lemma H.4 (Γ-oddness of the Hamiltonized derivation)**.** *Assume* $\mathbf{2T}_{\mathrm{LS}} + \mathbf{2T}^{\mathrm{cal}}_{\mathrm{char}}$ *and Lemma F.3. For the canonical representative* $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in E^-_{\Lambda_k}$*, one has*

$$\Gamma \delta_{\Lambda_k} \Gamma^{-1} = -\delta_{\Lambda_k}.$$

Proof. With $\Gamma(\mathcal{J}^{\mathrm{can}}_{\Lambda_k}) = -\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$, Lemma F.3 gives for all $A \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$:

$$\Gamma \delta_{\Lambda_k}(A) = \Gamma\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, A\}_{\mathrm{P},\Lambda_k} = \{\Gamma \mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \Gamma A\}_{\mathrm{P},\Lambda_k} = -\{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, \Gamma A\}_{\mathrm{P},\Lambda_k} = -\delta_{\Lambda_k}(\Gamma A),$$

so $\Gamma \delta_{\Lambda_k} \Gamma^{-1} = -\delta_{\Lambda_k}$. □

Theorem H.9 (Peierls Hamiltonization in $\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}}$)**.** *Assume* $\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}}$ (Definition B.22)*. Then for each ledger step* k*, the canonical odd representative* $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$ *Hamiltonizes to a well-defined local odd*

$$\delta_{\Lambda_k} := \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}$$

on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$*, and* δ_{Λ_k} *is physically* Γ*-odd.*

Proof. By Theorem H.1, each ledger step k carries a canonical odd representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k} \in E^-_{\Lambda_k}$, and Lemma B.1 shows that this representative is EB3-admissible. The Hamiltonian action $\delta_{\Lambda_k} = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, -\}_{\mathrm{P},\Lambda_k}$ is therefore well-defined on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$, and Lemma H.3 gives both the graded derivation property and locality. Lemma H.4 supplies the parity statement $\Gamma \delta_{\Lambda_k} \Gamma^{-1} = -\delta_{\Lambda_k}$. Hence the canonical representative Hamiltonizes to a well-defined local physically odd derivation, as claimed. □

Proposition H.6 (Physical invariance under ambiguity-triviality)**.** *Fix a ledger step* k*, and let* $J_1, J_2 \in [\Xi^{(1)}_{\Lambda_k}]$*, so* $J_2 - J_1 \in \mathcal{A}^-_{\Lambda_k}$*. Define*

$$\delta^{(i)}_{\Lambda_k} := \{J_i, -\}_{\mathrm{P},\Lambda_k} \qquad (i = 1, 2).$$

If every ambiguity direction $A \in \mathcal{A}^-_{\Lambda_k}$ *acts trivially on the physical observable sector, then* $\delta^{(1)}_{\Lambda_k}$ *and* $\delta^{(2)}_{\Lambda_k}$ *agree there. More generally, if for every* $A \in \mathcal{A}^-_{\Lambda_k}$ *and every physical local observable* F *the Peierls bracket* $\{A, F\}_{\mathrm{P},\Lambda_k}$ *vanishes in the physical quotient* (equivalently: belongs to the physical null ideal)*, then the induced physical Peierls derivation is independent of the chosen representative in* $[\Xi^{(1)}_{\Lambda_k}]$*.*

Proof. Since $J_2 - J_1 \in \mathcal{A}^-_{\Lambda_k}$, bilinearity of the Peierls bracket gives

$$\delta^{(2)}_{\Lambda_k}(F) - \delta^{(1)}_{\Lambda_k}(F) = \{J_2 - J_1, F\}_{\mathrm{P},\Lambda_k}.$$

If every ambiguity direction acts trivially on the physical observable sector, the right-hand side vanishes there, so the two derivations agree on physical observables. Under the weaker null-ideal hypothesis, the same identity shows that the difference vanishes after passage to the physical quotient, hence the induced physical derivation depends only on the class $[\Xi^{(1)}_{\Lambda_k}]$. □

Remark H.10 (Need for the ambiguity-triviality hypothesis)**.** The weak package $\mathbf{2T}^{\mathrm{weak}}_{\mathrm{char}}$ does not by itself imply ambiguity-triviality on physical observables. A separate ambiguity supplement may impose ambiguity-triviality on the characterization algebra and representative-independence of the induced action of $[\Xi^{(1)}_{\Lambda_k}]$ on the relevant reduced sector. Proposition H.6 is the corresponding statement after passage to the AQFT/HLS physical sector.

3. Subprogram 3c: AQFT net and HLS on the Minkowski boundary

a. Minkowski boundary chart. Cosmology theory provides the boundary local algebraic data and the Peierls bracket. For the AQFT/HLS analysis below we work on a chosen Minkowski boundary chart on which $\mathbf{2T}_{\mathrm{EBC}}$ is assumed. No further details of the cosmology construction enter in this part of the argument.

Definition H.4 (Borchers–Uhlmann local net)**.** Recall Definition D.1. The assignment $O \mapsto \mathfrak{A}_{\mathrm{BU}}(O)$ defines a net of local *-algebras on $\mathbb{R}^{1,3}$. We write $\mathfrak{A}(O)$ for the completion in the vacuum representation obtained in Subprogram 1b (Theorem D.1).

Proposition H.7 (Haag–Kastler properties)**.** *Assume Theorem D.1. Then the net $O \mapsto \mathfrak{A}(O)$ satisfies isotony and locality and carries a Poincaré covariant action $\alpha_{(a,\Lambda)}$ with positive-energy representation in the vacuum GNS space.*

Lemma H.5 (Parity on the net)**.** *The involution Γ extends to a *-automorphism of the net, $\Gamma : \mathfrak{A}(O) \to \mathfrak{A}(O)$, and defines a $\mathbb{Z}_2$ grading $\mathfrak{A} = \mathfrak{A}^+ \oplus \mathfrak{A}^-$.*

Lemma H.6 (Local graded derivation)**.** *Assume Theorem D.1 and Theorem H.9. Define δ_{Λ_k} on BU generators by*

$$\delta_{\Lambda_k}\big(\Phi(F)\big) \;:=\; \Phi\big(\delta_{\Lambda_k} F\big), \qquad \delta_{\Lambda_k} F := \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, F\}_{\mathrm{P},\Lambda_k}.$$

*Then δ_{Λ_k} extends uniquely to a graded *-derivation on the polynomial core $\mathfrak{A}_{\mathrm{fin}}$ (Definition D.3) and is local in the sense that $\delta_{\Lambda_k}(\mathfrak{A}_{\mathrm{fin}} \cap \mathfrak{A}_{\mathrm{BU}}(O)) \subset \mathfrak{A}_{\mathrm{BU}}(O)$.*

Proof. The Borchers–Uhlmann algebra $\mathfrak{A}_{\mathrm{BU}}(O)$ is generated by the symbols $\Phi(F)$ with $F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$, modulo the defining *-structure $\Phi(F)^* = \Phi(\overline{F})$ and the locality relations for spacelike separated supports. We must check that the assignment on generators is compatible with these relations.

*b. Compatibility with the *-structure.* Using $\Phi(F)^* = \Phi(\overline{F})$ and the definition of δ on generators,

$$\delta_{\Lambda_k}(\Phi(F)^*) = \delta_{\Lambda_k}(\Phi(\overline{F})) = \Phi(\delta_{\Lambda_k}(\overline{F})).$$

Since $\delta_{\Lambda_k} F = \{\mathcal{J}^{\mathrm{can}}_{\Lambda_k}, F\}_{\mathrm{P},\Lambda_k}$ is defined by the (real) Peierls bracket, it respects complex conjugation on sufficiently local functionals: $\delta_{\Lambda_k}(\overline{F}) = \overline{\delta_{\Lambda_k} F}$. Therefore $\delta_{\Lambda_k}(\Phi(F)^*) = \Phi(\overline{\delta_{\Lambda_k} F}) = \delta_{\Lambda_k}(\Phi(F))^*$, so δ_{Λ_k} is a *-derivation on generators.

c. Compatibility with locality relations. Let $F_i \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O_i)$ with O_1 spacelike separated from O_2, so the defining relations impose $[\Phi(F_1), \Phi(F_2)] = 0$ in the BU algebra. By Lemma H.3, δ_{Λ_k} preserves locality on $\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}$, so $\delta_{\Lambda_k} F_i \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O_i)$. Hence $[\Phi(\delta_{\Lambda_k} F_1), \Phi(F_2)] = [\Phi(F_1), \Phi(\delta_{\Lambda_k} F_2)] = 0$ by the same BU locality relations, and applying the graded Leibniz rule to the commutator shows $\delta_{\Lambda_k}([\Phi(F_1), \Phi(F_2)]) = 0$. Thus δ_{Λ_k} descends to the quotient defining $\mathfrak{A}_{\mathrm{BU}}(O)$.

d. Extension and uniqueness. The polynomial core $\mathfrak{A}_{\mathrm{fin}}$ is generated by finite polynomials in the generators $\Phi(F)$. Since δ_{Λ_k} is fixed on generators and required to satisfy the graded Leibniz rule, it extends uniquely to a graded *-derivation on $\mathfrak{A}_{\mathrm{fin}}$. Locality of the extension follows from $\delta_{\Lambda_k}(\mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)) \subset \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O)$. □

Lemma H.7 (Vacuum invariance on the even subalgebra)**.** *Assume Theorem D.1 and Lemma G.1 (so $\varpi \circ \Gamma = \varpi$). Assume moreover that δ_{Λ_k} is Γ-odd in the sense that $\Gamma \delta_{\Lambda_k} \Gamma^{-1} = -\delta_{\Lambda_k}$ (Lemma H.4). Then*

$$\varpi(\delta_{\Lambda_k}(A)) = 0 \qquad \textit{for all } A \in \mathfrak{A}^+.$$

Proof. If $A \in \mathfrak{A}^+$, then $\Gamma(A) = A$, hence

$$\varpi(\delta_{\Lambda_k}(A)) = \varpi(\Gamma(\delta_{\Lambda_k}(A))) = \varpi\big((\Gamma\delta_{\Lambda_k}\Gamma^{-1})(\Gamma A)\big) = \varpi\big(-\delta_{\Lambda_k}(A)\big) = -\varpi(\delta_{\Lambda_k}(A)).$$

□

Theorem H.10 (Uniform graph bound for the canonical implementer candidate)**.** *Assume Theorem D.1, Lemma H.6, Definition B.10* (EB3)*, and the fixed-generator EB3-admissibility package of Definition B.11 for the canonical representative $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$. Define Q_0 on the vacuum core $\mathcal{D}_0$ by*

$$Q_0\, \pi(B)\Omega := i\, \pi(\delta_{\Lambda_k}(B))\Omega \qquad (B \in \mathfrak{A}_{\mathrm{fin}}).$$

Then there exist an integer $m \geq 0$ and a constant $C_* > 0$, depending only on the fixed canonical representative and the vacuum representation, such that

$$\|Q_0\psi\| \leq C_*\, \|(1+H)^m \psi\| \qquad (\psi \in \mathcal{D}_0).$$

Equivalently, Q_0 is graph-bounded on $\mathcal{D}_0$ by one fixed power of $(1+H)$.

Proof. Definition B.11 fixes one Sobolev order $\sigma > \frac{5}{2}$ for the chosen canonical generator, and Lemma B.1 shows that the corresponding finitely many locality/Sobolev seminorms are cutoff-uniform for that fixed $\mathcal{J}^{\mathrm{can}}_{\Lambda_k}$. In the cosmology proof of EB3, these seminorms are the only source of the graph-bound constants, while the power of $(1+H)$ depends only on the fixed Sobolev order. Hence the regionwise EB3 bounds for δ_{Λ_k} may be restated, for this fixed generator, in the form

$$\|\pi(\delta_{\Lambda_k}(B))\Omega\| \le C_* \,\|(1+H)^m \pi(B)\Omega\| \qquad (B \in \mathfrak{A}_{\mathrm{fin}})$$

with one common exponent m and one common constant C_*. Writing $\psi = \pi(B)\Omega$ gives the displayed graph bound for Q_0. □

Proposition H.8 (Implementability from an energy bound)**.** *Assume the hypotheses of Theorem H.10. Then there exists a densely defined closable operator Q on $\mathcal{H}$ with $\Omega \in \mathcal{D}(Q)$, $Q\Omega = 0$, such that on the core $\mathcal{D}_0$ one has*

$$\delta_{\Lambda_k}(A) = i[Q, \pi(A)]_{\mathrm{s}} \qquad (A \in \mathfrak{A}_{\mathrm{fin}}).$$

Proof. Define an operator Q_0 on the GNS core $\mathcal{D}_0 = \pi(\mathfrak{A}_{\mathrm{fin}})\Omega$ by

$$Q_0\, \pi(B)\Omega \;:=\; i\,\pi(\delta_{\Lambda_k}(B))\Omega \qquad (B \in \mathfrak{A}_{\mathrm{fin}}).$$

e. Well-definedness. If $\pi(B)\Omega = 0$, then Theorem H.10 gives

$$\|\pi(\delta_{\Lambda_k}(B))\Omega\| \le C_* \,\|(1+H)^m\, \pi(B)\Omega\| = 0,$$

so $\pi(\delta_{\Lambda_k}(B))\Omega = 0$ and Q_0 is well-defined.

f. Implementation identity on $\mathcal{D}_0$. Let $A, B \in \mathfrak{A}_{\mathrm{fin}}$. Using the definition of Q_0 and the graded Leibniz rule for δ,

$$\begin{aligned}
[Q_0, \pi(A)]_{\mathrm{s}}\, \pi(B)\Omega &= Q_0\, \pi(AB)\Omega - (-1)^{|A|}\pi(A)\, Q_0\, \pi(B)\Omega \\
&= i\,\pi(\delta_{\Lambda_k}(AB))\Omega - (-1)^{|A|} i\,\pi(A\delta_{\Lambda_k}(B))\Omega \\
&= i\,\pi(\delta_{\Lambda_k}(A)B)\Omega \;=\; i\,\pi(\delta_{\Lambda_k}(A))\,\pi(B)\Omega,
\end{aligned}$$

so $\delta_{\Lambda_k}(A) = i[Q_0, \pi(A)]_{\mathrm{s}}$ on $\mathcal{D}_0$.

g. Closability and extension. By Theorem H.10,

$$\|Q_0\psi\| \le C_* \,\|(1+H)^m \psi\| \qquad (\psi \in \mathcal{D}_0).$$

Thus Q_0 is bounded relative to the closed positive selfadjoint operator $(1+H)^m$ on the invariant core $\mathcal{D}_0$. Relative graph-boundedness therefore implies that Q_0 is closable; let Q be its closure. Then Q is densely defined and closable, contains Ω in its domain, and the implementation identity remains valid on $\mathcal{D}_0$.

Finally, since $1 \in \mathfrak{A}_{\mathrm{fin}}$ and every derivation annihilates the unit,

$$Q_0\Omega = Q_0\, \pi(1)\Omega = i\,\pi(\delta_{\Lambda_k}(1))\Omega = 0.$$

Because Q is the closure of Q_0, it follows that $\Omega \in \mathcal{D}(Q)$ and $Q\Omega = 0$. □

Lemma H.8 (Noether current datum for a Peierls-inner odd derivation)**.** *Assume Theorem D.1, Definition B.10, and* $\mathbf{2T}_{\mathrm{Peierls}}$-(P2)–(P5)*. Let $\mathcal{J} \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}(O_{\mathcal{J}})$ be physically odd and EB3-admissible, and define the local graded *-derivation*

$$\delta_{\mathcal{J}}(F) := \{\mathcal{J}, F\}_{\mathrm{P}} \qquad (F \in \mathcal{F}^{\Lambda}_{\mathrm{s.loc}}),$$

extended to $\mathfrak{A}_{\mathrm{fin}}$ by $\delta_{\mathcal{J}}(\Phi(F)) := \Phi(\delta_{\mathcal{J}} F)$ and the graded Leibniz rule. Then there exists a local vector-valued operator-valued distribution

$$j^{\mu}_{\delta_{\mathcal{J}}}(x)$$

on the vacuum core $\mathcal{D}_0$ with the following properties.

(i) ***Locality, *-compatibility, and temperedness.*** *For every compactly supported smooth test vector h_μ, the smeared operator*

$$j_{\delta_{\mathcal{J}}}(h) := \int d^4x\, h_\mu(x)\, j^\mu_{\delta_{\mathcal{J}}}(x)$$

is localized in $\operatorname{supp} h$*, acts on $\mathcal{D}_0$, and satisfies*

$$j_{\delta_{\mathcal{J}}}(h)^* = j_{\delta_{\mathcal{J}}}(\overline{h})$$

on $\mathcal{D}_0$. For every $\psi, \phi \in \mathcal{D}_0$, the map $h \mapsto \langle \psi, j_{\delta_{\mathcal{J}}}(h)\phi\rangle$ is a tempered distribution of finite order.

(ii) ***Conservation.***

$$\partial_\mu j^\mu_{\delta_{\mathcal{J}}} = 0$$

in the distributional/Wightman sense.

(iii) ***Poincaré covariance.*** *Writing $g = (a, \Lambda)$ and $(g \cdot h)_\mu(x) := \Lambda_\mu{}^\nu h_\nu(g^{-1}x)$, one has on $\mathcal{D}_0$*

$$U(g)\, j_{\delta_{\mathcal{J}}}(h)\, U(g)^{-1} = j_{\delta_{\mathcal{J}}}(g \cdot h).$$

(iv) ***Charge-implementation on local vectors.*** *Fix a standard time mollifier $\chi_\varepsilon \in C_c^\infty(\mathbb{R})$ with $\int \chi_\varepsilon(s)\, ds = 1$, and let $f_R \in C_c^\infty(\mathbb{R}^3)$ be an equal-time spatial cutoff family with $0 \le f_R \le 1$ and $f_R \to 1$ pointwise. Define*

$$h^\mu_{t,R,\varepsilon}(x) := \delta^\mu_0\, \chi_\varepsilon(x^0 - t)\, f_R(\vec{x}), \qquad Q_{R,\varepsilon}(t) := j_{\delta_{\mathcal{J}}}(h_{t,R,\varepsilon}).$$

After the vacuum normalization $Q_{R,\varepsilon}(t)\Omega = 0$, the following holds: for every double cone O and every $\psi = \pi(B)\Omega \in \mathcal{D}_0(O)$, there exists $R_0 = R_0(O, t, \varepsilon)$ such that for all $R \ge R_0$,

$$Q_{R,\varepsilon}(t)\psi = i\, \pi(\delta_{\mathcal{J}}(B))\Omega.$$

Equivalently, for every local $A \in \mathfrak{A}_{\mathrm{fin}}(O)$ and every $\psi \in \mathcal{D}_0(O)$,

$$i\, [Q_{R,\varepsilon}(t), \pi(A)]_{\mathrm{s}}\psi = \pi(\delta_{\mathcal{J}}(A))\psi \qquad (R \gg 1).$$

(v) ***Uniqueness of the datum.*** *Any other current datum with the same properties differs from $j^\mu_{\delta_{\mathcal{J}}}$ by a local improvement $\partial_\nu B^{[\nu\mu]}$ and a conserved current whose cutoff charges act trivially on the induced derivation.*

Proof. The proof has three steps.

h. Step 1: classical Noether current from the symplectic current. By $\mathbf{2T}_{\mathrm{Peierls}}$-(P2)–(P4), the causal propagator Δ is defined on first functional derivatives and is compatible with the renormalized symplectic current $\omega^\mu_{\mathrm{ren}}$ of $\mathbf{2T}_{\mathrm{Peierls}}$-(P3). Set

$$X_{\mathcal{J}} := \Delta\, \mathcal{J}^{(1)}.$$

This is the Hamiltonian linearized solution associated with $\mathcal{J}$. For every compactly supported linearized solution η, define the classical current density

$$c^\mu_{\mathcal{J}}[\eta](x) := \omega^\mu_{\mathrm{ren}}(X_{\mathcal{J}}, \eta)(x),$$

with the argument order chosen so that the associated Cauchy-surface charge equals $\{\mathcal{J}, -\}_{\mathrm{P}}$ rather than $\{-, \mathcal{J}\}_{\mathrm{P}}$.

Because both entries satisfy the linearized equations, conservation of the symplectic current gives

$$\partial_\mu c^\mu_{\mathcal{J}}[\eta] = 0.$$

For every Cauchy surface Σ,

$$\int_\Sigma d\Sigma_\mu\, c^\mu_{\mathcal{J}}[\eta] = \Omega_{\mathrm{ren},\Sigma}(X_{\mathcal{J}}, \eta),$$

and by the compatibility of Ω_{ren} with Δ,

$$\Omega_{\mathrm{ren},\Sigma}(X_{\mathcal{J}}, \eta_F) = \{\mathcal{J}, F\}_{\mathrm{P}} \qquad (\eta_F := \Delta F^{(1)}).$$

Thus the classical Cauchy-surface charge of $c^\mu_{\mathcal{J}}$ implements exactly the Peierls derivation $\delta_{\mathcal{J}}$.

The locality of $\omega^\mu_{\mathrm{ren}}$, the retarded/advanced support of Δ, and $\mathcal{J} \in \mathcal{F}^\Lambda_{\mathrm{s.loc}}(O_{\mathcal{J}})$ imply that $c^\mu_{\mathcal{J}}$ is sufficiently local. By $\mathbf{2T}_{\mathrm{Peierls}}$-(P5), compactly smearing $c^\mu_{\mathcal{J}}$ against a test vector h_μ again produces an element of $\mathcal{F}^\Lambda_{\mathrm{s.loc}}$.

i. Step 2: Wightman/GNS realization on the vacuum core. For every test vector h_μ, let $C_{\mathcal{J}}(h) \in \mathcal{F}^\Lambda_{\mathrm{s.loc}}$ denote the smeared current functional constructed in Step 1. Theorem D.1 realizes every element of $\mathcal{F}^\Lambda_{\mathrm{s.loc}}$ as a local operator on the vacuum GNS core, so set

$$j_{\delta_{\mathcal{J}}}(h) := \pi(\Phi(C_{\mathcal{J}}(h))).$$

By Definition B.10-(EB1), every such smeared local field obeys a polynomial H-bound on $\mathcal{D}_0$; hence it is closable and its matrix elements against vectors in $\mathcal{D}_0$ are tempered distributions of finite order. This gives item (i).

The distributional conservation in item (ii) is the quantum realization of the classical identity $\partial_\mu c^\mu_{\mathcal{J}} = 0$: for every test scalar $g \in C_c^\infty(\mathbb{R}^{1,3})$,

$$j_{\delta_{\mathcal{J}}}(\partial^\mu g) = 0$$

on $\mathcal{D}_0$. Likewise, the covariance clause of Definition B.10 and the covariance of $\omega^\mu_{\mathrm{ren}}$ imply

$$U(g)\, j_{\delta_{\mathcal{J}}}(h)\, U(g)^{-1} = j_{\delta_{\mathcal{J}}}(g \cdot h),$$

which is item (iii).

j. Step 3: charge identity on local vectors. Let O be a double cone and $\psi = \pi(B)\Omega \in \mathcal{D}_0(O)$. Fix t and ε. Choose R so large that the shell on which ∇f_R is supported is spacelike to O after thickening by the time slab $\operatorname{supp} \chi_\varepsilon(\cdot - t)$. Conservation of the current then allows one to deform the cutoff charge $Q_{R,\varepsilon}(t)$ to any other Cauchy-surface representative with the same interior region but no shell contribution against vectors localized in O. Replacing the cutoff surface by a full Cauchy surface and using the Step 1 surface-charge identity shows that, in vacuum matrix elements against local vectors,

$$\langle \pi(C)\Omega, Q_{R,\varepsilon}(t)\pi(B)\Omega\rangle = \langle \pi(C)\Omega,\, i\,\pi(\delta_{\mathcal{J}}(B))\Omega\rangle$$

for every $\pi(C)\Omega \in \mathcal{D}_0$. Because $\mathcal{D}_0$ is dense, this proves

$$Q_{R,\varepsilon}(t)\pi(B)\Omega = i\,\pi(\delta_{\mathcal{J}}(B))\Omega \qquad (R \gg 1),$$

after the vacuum normalization $Q_{R,\varepsilon}(t)\Omega = 0$. Applying the same argument to graded commutators with a local observable A gives

$$i\,[Q_{R,\varepsilon}(t), \pi(A)]_{\mathrm{s}}\psi = \pi(\delta_{\mathcal{J}}(A))\psi \qquad (R \gg 1),$$

which is item (iv).

Finally, if $\widetilde{j}^\mu$ is another current datum with the same charge action, their difference $k^\mu := \widetilde{j}^\mu - j^\mu_{\delta_{\mathcal{J}}}$ is conserved and has vanishing cutoff charges on all local vectors. On each double cone, the local Poincaré lemma writes a conserved compactly supported current with vanishing Cauchy charge as a divergence $\partial_\nu B^{[\nu\mu]}$, up to a conserved remainder whose charges act trivially on the induced derivation. This proves item (v). □

Theorem H.11 (Conserved-current implementation theorem on the vacuum core)**.** *Assume Theorem D.1. Let δ be a local graded *-derivation on $\mathfrak{A}_{\mathrm{fin}}$. Assume there exists a current datum $j^\mu_\delta(x)$ on $\mathcal{D}_0$ with the properties listed in Lemma H.8*(i)–(v) *for δ. Then:*

(i) *for every $\psi \in \mathcal{D}_0$ and every time t, the equal-time cutoff charges $Q_{R,\varepsilon}(t)$ stabilize for $R \gg 1$ to a value $Q_\delta(t)\psi$ independent of R and ε;*

(ii) *the operator $Q_\delta(t)$ on $\mathcal{D}_0$ is densely defined, odd, and closable;*

(iii) *$Q_\delta(t)$ is independent of t; we write the common operator simply as Q_δ;*

(iv) *on $\mathcal{D}_0$,*

$$\delta(A) = i[Q_\delta, \pi(A)]_{\mathrm{s}} \qquad (A \in \mathfrak{A}_{\mathrm{fin}});$$

(v) *j^μ_δ is unique up to local improvements and currents acting trivially on the induced derivation.*

Proof. Step 1: stabilization on local vectors. Let $\psi = \pi(B)\Omega \in \mathcal{D}_0(O)$. By the charge-implementation property of the current datum, there exists R_0 such that for all $R \geq R_0$,

$$Q_{R,\varepsilon}(t)\psi = i\,\pi(\delta(B))\Omega.$$

Thus the family stabilizes on every vector of the local core $\mathcal{D}_0(O)$, hence on all of $\mathcal{D}_0 = \bigcup_O \mathcal{D}_0(O)$. This proves item (i) and defines $Q_\delta(t)$ on $\mathcal{D}_0$.

Step 2: implementation identity. Let $A, B \in \mathfrak{A}_{\mathrm{fin}}$. Choose R so large that the charge datum applies simultaneously to B and to AB. Using the graded Leibniz rule and the vacuum normalization $Q_{R,\varepsilon}(t)\Omega = 0$,

$$\begin{aligned}
[Q_\delta(t), \pi(A)]_{\mathrm{s}}\,\pi(B)\Omega &= Q_\delta(t)\pi(AB)\Omega - (-1)^{|A|}\pi(A)Q_\delta(t)\pi(B)\Omega \\
&= i\,\pi(\delta(AB))\Omega - (-1)^{|A|} i\,\pi(A\,\delta(B))\Omega \\
&= i\,\pi(\delta(A)B)\Omega \\
&= i\,\pi(\delta(A))\,\pi(B)\Omega.
\end{aligned}$$

Since such vectors span $\mathcal{D}_0$, this gives

$$\delta(A) = i[Q_\delta(t), \pi(A)]_{\mathrm{s}} \quad \text{on } \mathcal{D}_0.$$

This is item (iv).

Step 3: independence of the time slice. Let $t_1, t_2 \in \mathbb{R}$ and $\psi \in \mathcal{D}_0(O)$. Choose R so large that the shell region of the two cutoffs between the slices t_1 and t_2 is spacelike to O. Conservation of the current datum implies that the two cutoff charges have the same action on ψ. Hence

$$Q_\delta(t_1)\psi = Q_\delta(t_2)\psi \qquad (\psi \in \mathcal{D}_0),$$

which proves item (iii).

Step 4: closability. Fix t and write $Q_\delta := Q_\delta(t)$. Suppose $\psi_n \in \mathcal{D}_0$, $\psi_n \to 0$, and $Q_\delta\psi_n \to \eta$ in $\mathcal{H}$. Take any local vector $\phi = \pi(C)\Omega \in \mathcal{D}_0(O)$. For $R \gg 1$, the same stabilization argument applied to the adjoint current datum gives a vector $Q^\sharp_\delta\phi \in \mathcal{H}$ such that

$$\langle \phi, Q_\delta\psi_n \rangle = \langle Q^\sharp_\delta\phi, \psi_n \rangle$$

for all sufficiently large n. Because $\psi_n \to 0$, the right-hand side tends to 0, so

$$\langle \phi, \eta \rangle = 0 \qquad \forall\, \phi \in \mathcal{D}_0.$$

Since $\mathcal{D}_0$ is dense, $\eta = 0$. Therefore Q_δ is closable. This proves item (ii).

Step 5: uniqueness. If $\widetilde{j}^\mu_\delta$ is another current datum for the same derivation, then $k^\mu := \widetilde{j}^\mu_\delta - j^\mu_\delta$ is conserved and its cutoff charges vanish on $\mathcal{D}_0$. By the local Poincaré lemma on Minkowski space, k^μ is a local improvement $\partial_\nu B^{[\nu\mu]}$ plus a conserved current whose charges act trivially on the induced derivation. This is item (v). □

Corollary H.4 (Canonical current reconstruction for the Main Hamiltonized derivation)**.** *Assume* $\mathbf{2T}^{\text{spine}}_{\text{Main}}$*, Theorem D.1, Theorem H.9, and Lemma B.1. Then the Hamiltonized derivation* $\delta_{\Lambda_k} = \{\mathcal{J}^{\text{can}}_{\Lambda_k}, -\}_{\text{P},\Lambda_k}$ *admits a Poincaré-covariant conserved local current*

$$\mathcal{J}^{\mu}_{\Lambda_k}(x)$$

on the vacuum core $\mathcal{D}_0$*, unique up to local improvements and derivation-trivial terms, whose equal-time cutoff charges stabilize on* $\mathcal{D}_0$ *to a densely defined closable odd operator* Q_{Λ_k} *satisfying*

$$\delta_{\Lambda_k}(A) = i[Q_{\Lambda_k}, \pi(A)]_{\text{s}} \qquad (A \in \mathfrak{A}_{\text{fin}})$$

on $\mathcal{D}_0$*.*

Proof. By Theorem H.9, the canonical representative $\mathcal{J}^{\text{can}}_{\Lambda_k}$ defines a local odd Peierls-inner derivation

$$\delta_{\Lambda_k} = \{\mathcal{J}^{\text{can}}_{\Lambda_k}, -\}_{\text{P},\Lambda_k}.$$

Lemma B.1 gives the required EB3-admissibility of the fixed canonical generator, and $\mathbf{2T}^{\text{spine}}_{\text{Main}}$ contains both $\mathbf{2T}_{\text{Peierls}}$ and $\mathbf{2T}_{\text{EBC}}$. Therefore Lemma H.8 applies and supplies a current datum $j^{\mu}_{\delta_{\Lambda_k}}$ for δ_{Λ_k}. Applying Theorem H.11 to that datum yields the current $\mathcal{J}^{\mu}_{\Lambda_k}$ and the closable operator Q_{Λ_k} with the stated implementation identity. □

Theorem H.12 (Translation covariance of the Hamiltonized odd derivation)**.** *Assume Corollary H.4 and Definition B.10* (Covariance)*. Then* δ_{Λ_k} *commutes with translations on* $\mathfrak{A}_{\text{fin}}$*:*

$$\alpha_a \circ \delta_{\Lambda_k} \circ \alpha_a^{-1} = \delta_{\Lambda_k}.$$

Proof. Let $Q_{\Lambda_k,R,\varepsilon}(t)$ be the cutoff charges built from the current $\mathcal{J}^{\mu}_{\Lambda_k}$ of Corollary H.4. By the Poincaré covariance of that current,

$$U(a)\, Q_{\Lambda_k,R,\varepsilon}(t)\, U(a)^{-1} = Q^{(a)}_{\Lambda_k,R,\varepsilon}(t + a^0),$$

where the superscript $^{(a)}$ means that the spatial cutoff is translated by $\vec{a}$.

Fix $\psi \in \mathcal{D}_0(O)$. Choose R so large that both the original and translated cutoff charges have stabilized on ψ. Since the stabilized charge is independent of the precise cutoff representative and of the time slice,

$$U(a)\, Q_{\Lambda_k}\, U(a)^{-1}\psi = Q_{\Lambda_k}\psi \qquad (\psi \in \mathcal{D}_0).$$

Hence $U(a)Q_{\Lambda_k}U(a)^{-1} = Q_{\Lambda_k}$ on $\mathcal{D}_0$. Therefore, for every $A \in \mathfrak{A}_{\text{fin}}$,

$$\alpha_a(\delta_{\Lambda_k}(A)) = i[U(a)Q_{\Lambda_k}U(a)^{-1}, \alpha_a(\pi(A))]_{\text{s}} = i[Q_{\Lambda_k}, \alpha_a(\pi(A))]_{\text{s}} = \delta_{\Lambda_k}(\alpha_a A).$$

This is exactly $\alpha_a \circ \delta_{\Lambda_k} \circ \alpha_a^{-1} = \delta_{\Lambda_k}$. □

Definition H.5 (Lorentz-covariant family of odd derivations)**.** Let δ be a local derivation on $\mathfrak{A}_{\text{fin}}$. For each proper orthochronous Lorentz transformation $\Lambda \in SO^+(1,3)$ define

$$\delta_\Lambda := \alpha_\Lambda \circ \delta \circ \alpha_\Lambda^{-1}, \qquad \mathscr{D} := \text{span}_{\mathbb{R}}\{\delta_\Lambda : \Lambda \in SO^+(1,3)\}.$$

We say δ is *Lorentz-covariant of finite type* if $\mathscr{D}$ is finite-dimensional.

Lemma H.9 (Hyperplane realization of the reconstructed charge)**.** *Assume Corollary H.4. Let* $\Sigma_0 := \{x^0 = 0\}$*, let* Q_{Λ_k} *be the equal-time reconstructed global charge, and let* $\Sigma = g\Sigma_0$ *be any oriented affine Cauchy hyperplane obtained from* Σ_0 *by a proper orthochronous Poincaré transformation* $g = (a, \Lambda)$*. For a transported cutoff function on* Σ*, let* $Q_{R,\Sigma}$ *denote the cutoff flux of the conserved current* $\mathcal{J}^{\mu}_{\Lambda_k}$ *through* Σ*. Then:*

(i) $Q_{R,\Sigma}$ *stabilizes on* $\mathcal{D}_0$ *for* $R \gg 1$ *to a closable odd operator* $Q_{\Lambda_k}(\Sigma)$*;*

(ii) $Q_{\Lambda_k}(\Sigma)$ *is independent of* Σ*;*

(iii) $U(g)Q_{\Lambda_k}(\Sigma_0)U(g)^{-1} = Q_{\Lambda_k}(g\Sigma_0)$ *on* $\mathcal{D}_0$.

Proof. Because $\mathcal{J}^{\mu}_{\Lambda_k}$ is Poincaré-covariant, the transported cutoff flux is related to the standard equal-time cutoff charge by

$$Q_{R,g\Sigma_0} = U(g)\, Q_{R,\Sigma_0}\, U(g)^{-1}$$

on $\mathcal{D}_0$. Since the equal-time charges stabilize on $\mathcal{D}_0$ by Theorem H.11, the same is true for the transported family, proving (i) and (iii).

For (ii), fix a local vector $\psi = \pi(B)\Omega \in \mathcal{D}_0(O)$. Choose R so large that the cutoff shell between Σ_0 and Σ lies spacelike to O. Exactly the shell argument used in Step 3 of the proof of Theorem H.11 then gives

$$Q_{R,\Sigma}\psi = Q_{R,\Sigma_0}\psi.$$

Passing to the stabilized values yields

$$Q_{\Lambda_k}(\Sigma)\psi = Q_{\Lambda_k}(\Sigma_0)\psi.$$

Since the local vectors span $\mathcal{D}_0$, the two operators agree on $\mathcal{D}_0$. As Σ was arbitrary, the reconstructed global charge is hyperplane-independent. □

Theorem H.13 (Lorentz finite type from the canonical conserved current)**.** *Assume Corollary H.4. Then the Hamiltonized odd derivation* δ_{Λ_k} *is Lorentz-covariant of finite type in the sense of Definition H.5. In fact,*

$$\alpha_\Lambda \circ \delta_{\Lambda_k} \circ \alpha_\Lambda^{-1} = \delta_{\Lambda_k} \qquad (\Lambda \in SO^+(1,3)).$$

Proof. Let $\Sigma_0 = \{x^0 = 0\}$. By Lemma H.9,

$$U(\Lambda)\, Q_{\Lambda_k}\, U(\Lambda)^{-1} = Q_{\Lambda_k}(\Lambda\Sigma_0) = Q_{\Lambda_k}(\Sigma_0) = Q_{\Lambda_k}$$

on $\mathcal{D}_0$. Therefore for every local observable A,

$$\begin{aligned}(\alpha_\Lambda \delta_{\Lambda_k} \alpha_\Lambda^{-1})(A) &= i\,[U(\Lambda)Q_{\Lambda_k}U(\Lambda)^{-1}, \pi(A)]_{\mathrm{s}} \\ &= i\,[Q_{\Lambda_k}, \pi(A)]_{\mathrm{s}} = \delta_{\Lambda_k}(A).\end{aligned}$$

Hence every Lorentz conjugate derivation equals δ_{Λ_k} itself, so

$$\mathscr{D} = \mathrm{span}_{\mathbb{R}}\{\alpha_\Lambda \delta_{\Lambda_k} \alpha_\Lambda^{-1}\} = \mathbb{R}\,\delta_{\Lambda_k}$$

is one-dimensional. This is stronger than finite type. □

Lemma H.10 (Covariance of the Lorentz orbit)**.** *Assume Theorem D.1 so the net carries a Poincaré action* α_g*. Let* δ *be a local derivation on* $\mathfrak{A}_{\mathrm{fin}}$ *and define* δ_Λ *as in Definition H.5. Then each* δ_Λ *is again a local derivation on* $\mathfrak{A}_{\mathrm{fin}}$*, and* $\Lambda \mapsto \delta_\Lambda$ *defines a representation of* $SO^+(1,3)$ *on* $\mathscr{D}$ *by conjugation.*

Proof. Since each α_Λ is a *-automorphism of the net, conjugation preserves the graded Leibniz rule and the *-structure, so δ_Λ is a graded *-derivation. If A is localized in a double cone O, then $\alpha_\Lambda^{-1}(A)$ is localized in $\Lambda^{-1}O$ and locality of δ gives $\delta(\alpha_\Lambda^{-1}(A))$ localized in $\Lambda^{-1}O$; applying α_Λ shows $\delta_\Lambda(A)$ is localized in O. Thus δ_Λ is local. Finally,

$$\delta_{\Lambda_2\Lambda_1} = \alpha_{\Lambda_2\Lambda_1}\delta\alpha^{-1}_{\Lambda_2\Lambda_1} = \alpha_{\Lambda_2}\big(\alpha_{\Lambda_1}\delta\alpha^{-1}_{\Lambda_1}\big)\alpha^{-1}_{\Lambda_2} = \alpha_{\Lambda_2}\delta_{\Lambda_1}\alpha^{-1}_{\Lambda_2},$$

so $\Lambda \mapsto \delta_\Lambda$ defines a representation by conjugation on the span $\mathscr{D}$. □

Theorem H.14 (HLS admissibility of δ)**.** *Assume the Main spine package* $\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}}$ (Definition B.22)*. Fix a ledger step k for which Subprogram 1b, the Main Hamiltonization chain, the current-reconstruction theorem, and the UV conformal/common-sector route have been established, namely Theorem D.1, Theorem H.9, Lemma G.1, Corollary H.4, Theorem H.12, and Theorem H.13. Then the derivation* δ_{Λ_k} *satisfies the HLS interface* $\mathbf{2T}_{\mathrm{HLS}}$ (Definition B.24)*.*

Proof. Lemma H.6 gives a densely defined local graded *-derivation on $\mathfrak{A}_{\mathrm{fin}}$. Proposition H.5 gives nontriviality of the induced derivation. Lemma H.4 gives the required physical Γ-oddness. Lemma H.7 gives vacuum invariance on $\mathfrak{A}^+$. Corollary H.4 gives implementability by a densely defined closable operator on the common invariant core $\mathcal{D}_0$. Theorem H.12 gives translation covariance. Theorem H.13 gives the finite-dimensional Lorentz covariance datum. These are exactly the clauses of the HLS interface $\mathbf{2T}_{\mathrm{HLS}}$ of Definition B.24. □

Theorem H.15 (Haag–Łopuszański–Sohnius (invoked under $\mathbf{2T}_{\mathrm{HLS}}$) [16])**.** *Assume the HLS classification theorem applies under $\mathbf{2T}_{\mathrm{HLS}}$. Let $(\mathcal{H}, \pi, U, \Omega)$ satisfy the Wightman/AQFT axioms and the spectrum condition. Suppose there exists a nontrivial Γ–odd local derivation δ obeying $\mathbf{2T}_{\mathrm{HLS}}$. Then there exist:*

- *a finite-dimensional complex odd charge module $\mathcal{V}_Q$,*
- *a nonzero vector $v_0 \in \mathcal{V}_Q$,*
- *and densely defined operators $Q_\alpha(v)$, complex-linear in $v \in \mathcal{V}_Q$, acting on a common invariant core $\mathcal{D}_Q$,*

such that

(i) *the original derivation is implemented by one nonzero element of this module:*

$$\delta = i[Q(v_0), \cdot]_{\mathrm{s}} \quad \text{on } \mathcal{D}_Q, \qquad Q(v_0) := (Q_1(v_0), Q_2(v_0));$$

(ii) *for each $v \in \mathcal{V}_Q$, the pair $Q(v) = (Q_1(v), Q_2(v))$ acts on the common invariant core $\mathcal{D}_Q$ as a local odd symmetry generator and transforms as a Weyl spinor doublet under the Lorentz generators;*

(iii) *after choosing any orthonormal basis $\{e_I\}_{I=1}^N$ of $\mathcal{V}_Q$ and writing*

$$Q^I_\alpha := Q_\alpha(e_I), \qquad \bar{Q}_{\dot\alpha I} := \big(Q^I_\alpha\big)^\dagger,$$

the graded anticommutators close on translations in indexed form:

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2\,\delta^I{}_J\,\sigma^\mu_{\alpha\dot\beta} P_\mu, \qquad \{Q^I_\alpha, Q^J_\beta\} = \{\bar{Q}_{\dot\alpha I}, \bar{Q}_{\dot\beta J}\} = 0;$$

(iv) *the internal HLS symmetry sector, when present, acts on the odd generators through endomorphisms of $\mathcal{V}_Q$; equivalently, in any chosen basis it is represented by matrices acting on the index I.*

Theorem H.16 (Indexed super-Poincaré seed at the UV boundary)**.** *Assume $\mathbf{2T}^{\mathrm{spine}}_{\mathrm{Main}}$, Theorem D.1, and Theorem H.14. Then, by Theorem H.15, the Hamiltonized odd derivation δ_{Λ_k} extends to a finite-dimensional odd charge module $\mathcal{V}_Q$ on the Minkowski UV boundary. Choosing a basis $\{e_I\}_{I=1}^N$ of $\mathcal{V}_Q$ and writing $Q^I_\alpha := Q_\alpha(e_I)$, the UV boundary carries densely defined odd generators*

$$Q^I_\alpha, \qquad \bar{Q}_{\dot\alpha I},$$

on a common invariant core $\mathcal{D}_Q$ such that

$$\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2\,\delta^I{}_J\,\sigma^\mu_{\alpha\dot\beta} P_\mu, \qquad \{Q^I_\alpha, Q^J_\beta\} = \{\bar{Q}_{\dot\alpha I}, \bar{Q}_{\dot\beta J}\} = 0, \qquad [P_\mu, Q^I_\alpha] = [P_\mu, \bar{Q}_{\dot\alpha I}] = 0,$$

with Q^I_α and $\bar{Q}_{\dot\alpha I}$ transforming in the two Weyl spinor representations under $M_{\mu\nu}$. The internal HLS symmetry sector acts on this odd charge multiplet through endomorphisms of $\mathcal{V}_Q$. Every module element $Q(v)$ acts on the same invariant core $\mathcal{D}_Q$ as a local odd symmetry generator.

Proof. Theorem H.14 proves that the Hamiltonized derivation δ_{Λ_k} satisfies the interface $\mathbf{2T}_{\mathrm{HLS}}$. Hence the invoked HLS classification theorem (Theorem H.15) applies. Choosing an orthonormal basis $\{e_I\}_{I=1}^N$ of the resulting odd charge module $\mathcal{V}_Q$ and writing $Q^I_\alpha := Q_\alpha(e_I)$ gives the indexed generators on the common invariant core $\mathcal{D}_Q$, together with the displayed translation anticommutators, vanishing same-chirality anticommutators, and Weyl-spinor Lorentz transformation law. The final sentence is the corresponding module statement from Theorem H.15. □

Lemma H.11 (Injectivity of the HLS module realization)**.** *Assume Theorem H.16. If $Q(v) = 0$ as a Weyl pair, then $v = 0$.*

Proof. By the invoked HLS theorem used in Theorem H.16, the original nonzero odd derivation is implemented by some nonzero $v_0 \in \mathcal{V}_Q$; in particular $Q(v_0) \neq 0$ on $\mathcal{D}_Q$.

We first show that the translation representation is nontrivial. If all translation generators vanished, then $P_\mu = 0$ for all μ, hence also $P_{\alpha\dot\beta} = 0$ for all spinor indices. The HLS anticommutator would then give

$$0 = \{Q_\alpha(v_0), Q_\alpha(v_0)^\dagger\}$$

on $\mathcal{D}_Q$. Therefore, for every $\psi \in \mathcal{D}_Q$,

$$0 = \langle \psi, \{Q_\alpha(v_0), Q_\alpha(v_0)^\dagger\}\psi\rangle = \|Q_\alpha(v_0)\psi\|^2 + \|Q_\alpha(v_0)^\dagger\psi\|^2.$$

Hence $Q_\alpha(v_0) = 0$ for both $\alpha = 1, 2$, contradicting that $Q(v_0)$ implements a nontrivial derivation. So not all P_μ vanish. Since the matrices $\sigma^\mu_{\alpha\dot\beta}$ span the Hermitian 2×2 matrices, some $P_{\alpha\dot\beta}$ is nonzero.

Now let $Q(v) = 0$. For every $w \in \mathcal{V}_Q$, the indexed HLS anticommutator gives

$$0 = \{Q_\alpha(v), \bar{Q}_{\dot\beta}(w)\} = 2\,\langle v, w\rangle_Q\, P_{\alpha\dot\beta}$$

on $\mathcal{D}_Q$. Choose $\alpha, \dot\beta$ such that $P_{\alpha\dot\beta} \neq 0$. Then $\langle v, w\rangle_Q = 0$ for every w, and therefore $v = 0$. □

Theorem H.17 (HLS spinor completeness on the UV vacuum representation)**.** *Assume Theorem H.16, and assume the invoked HLS classification step identifies the full translation-invariant local odd Weyl-spinor sector of the given chirality with the HLS charge module $\mathcal{V}_Q$, in the standard indexed supercharge-module presentation and classification picture recorded in [17, 34]. Let $\mathscr{Q}_+$ be the space of all translation-invariant local odd Weyl-spinor symmetry generators of the given chirality on the vacuum representation $(\mathcal{H}, \pi, U, \Omega)$, and define $\mathscr{Q}_-$ similarly for the conjugate chirality. Then the HLS charge module $\mathcal{V}_Q$ is complete in the sense that for every $X_\alpha \in \mathscr{Q}_+$ there exists a unique $v \in \mathcal{V}_Q$ with*

$$X_\alpha = Q_\alpha(v),$$

and similarly every barred generator in $\mathscr{Q}_-$ is a unique linear combination of the $\bar{Q}_{\dot\alpha}(v)$.

Proof. Theorem H.16 already identifies a finite-dimensional HLS odd charge module $\mathcal{V}_Q$ whose elements act as translation-invariant local odd Weyl spinor symmetry generators on the same vacuum representation. The invoked HLS classification input states that this finite-dimensional odd module exhausts the corresponding translation-invariant local odd spinor sector. Hence every such generator X_α lies in the span of the $Q_\alpha(v)$. Uniqueness is not merely basis independence: if $Q(v) = Q(v')$, then $Q(v - v') = 0$, so Lemma H.11 gives $v = v'$. The barred statement is identical. □

Theorem H.18 (Optional indexed current reconstruction under separate current data)**.** *Assume Theorem H.16. Choose a basis $\{e_I\}_{I=1}^N$ of $\mathcal{V}_Q$ and write $Q^I_\alpha := Q_\alpha(e_I)$, $\bar{Q}_{\dot\alpha I} := (Q^I_\alpha)^\dagger$. Assume moreover that for every I and α there is given a conserved local current datum*

$$\mathcal{J}^\mu{}_\alpha{}^I(x)$$

whose equal-time cutoff charges stabilize to Q^I_α on a dense invariant core $\mathcal{D}^{I,\alpha}_{\mathrm{curr}} \subset \mathcal{D}_Q$, and whose ambiguity is only by local improvements and derivation-trivial terms. Then, after replacing $\mathcal{D}^{I,\alpha}_{\mathrm{curr}}$ by the finite intersection over I, α, the whole indexed family may be realized on one dense common current domain.

Proof. This is exactly the additional datum assumed in the statement. Because the family $\{(I, \alpha)\}$ is finite, intersecting the dense invariant cores $\mathcal{D}^{I,\alpha}_{\mathrm{curr}}$ again yields a dense invariant domain, and the stated ambiguity for each current remains the standard one by improvements and derivation-trivial terms. □

Remark H.11 (Optional indexed current reconstruction)**.** Theorem H.18 is an optional AQFT supplement. If separate current data are chosen for the HLS generators, it yields a common dense current domain for the indexed family. A separate indexed current theorem therefore requires its own current data for those generators.

Corollary H.5 (Same-$\mathcal{H}$ superconformal closure)**.** *Assume Theorem H.16, Theorem H.17, the UV conformal common-sector interface* $\mathbf{2T}_{\mathrm{conf}}$*, and the same-*$\mathcal{H}$ *conformal-closure results of Subprograms 4a/4b. Then on the same UV Hilbert space* $\mathcal{H}$ *the generators*

$$P_\mu,\ M_{\mu\nu},\ D,\ K_\mu,\ Q^I_\alpha,\ \bar{Q}_{\dot{\alpha}I},\ S^\alpha{}_I,\ \bar{S}^{\dot{\alpha}I},\ R^I{}_J$$

close on a dense common invariant core to the full $4D$ *superconformal algebra. More precisely, the resulting operator Lie superalgebra is* $\mathfrak{su}(2,2|N)$ *for* $N \neq 4$*, while in the* $N = 4$ *case the standard simple form is obtained after quotienting the central* $\mathfrak{u}(1)$*, giving* $\mathfrak{psu}(2,2|4)$*.*

Proof. Combine Theorem H.16, Theorem H.17, Theorem J.4, and Theorem J.10. □

Appendix I: Subprogram 4a: Dilatation weight $\frac{1}{2}$ of the odd charges on the common UV sector

In this section we determine the scaling weight of the odd charges that appear on the Minkowski UV vacuum sector after the Subprogram 1b reconstruction and the Subprogram 3c HLS step. The argument uses:

- the indexed HLS super-Poincare seed on the UV vacuum representation (Theorem H.16),
- the HLS completeness statement (Theorem H.17),
- and the conformal common-sector supplement $\mathbf{2T}_{\mathrm{conf}}$ (Definition B.19).

The logic is:

(1) place the conformal even generators $P_\mu, M_{\mu\nu}, D, K_\mu$ on the same UV vacuum representation as the HLS odd module;

(2) use HLS completeness to prove that the dilation subgroup acts on the finite-dimensional HLS charge module itself;

(3) run the Jacobi argument against $\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} \propto P_\mu$ to force $\Delta(Q) = \frac{1}{2}$.

1. Input from Subprograms 1b and 3c

We work on the Minkowski UV chart used in Subprogram 3c. The inputs are:

- the Wightman net and vacuum representation $(\mathcal{H}, \pi, U, \Omega)$ with positive-energy Poincare covariance from Theorem D.1;
- the indexed HLS odd charge module $\mathcal{V}_Q$ and the operators $Q^I_\alpha, \bar{Q}_{\dot\alpha I}$ from Theorem H.16;
- the completeness statement that every translation-invariant local odd Weyl-spinor symmetry generator of the given chirality is already of the form $Q_\alpha(v)$ for a unique $v \in \mathcal{V}_Q$, from Theorem H.17;
- the same-representation conformal package $\mathbf{2T}_{\mathrm{conf}}$, which places $P_\mu, M_{\mu\nu}, D, K_\mu$ on the same UV Hilbert space.

In this subsection we also write

$$P_{\alpha\dot\beta} := (\sigma^\mu)_{\alpha\dot\beta} P_\mu.$$

Lemma I.1 (Dilatation and translations on the common UV conformal sector)**.** *Assume the UV conformal common-sector interface* $\mathbf{2T}_{\mathrm{conf}}$ (Definition B.19) *on the same vacuum representation* $(\mathcal{H}, \pi, \Omega)$ *used in Theorem H.16. Then the selfadjoint generators* D *and* P_μ *act on the common Stone core supplied by* $\mathbf{2T}_{\mathrm{conf}}$ *and satisfy*

$$[D, P_\mu] = iP_\mu. \tag{I1}$$

In particular, the conformal even sector and the HLS odd charge module act on one common UV vacuum representation with a dense common invariant core.

Proof. Definition B.19 places the conformal generators $P_\mu, M_{\mu\nu}, D, K_\mu$ on the common UV vacuum representation and on a common Stone core on which (I1) holds. Theorem H.16 places the indexed HLS odd charges on that same vacuum representation. Intersecting the HLS invariant core with the Stone core gives the claimed common UV vacuum representation and common invariant core. □

Lemma I.2 (Nontriviality of the HLS translation block)**.** *Assume Theorem H.16. Then some* $P_{\alpha\dot\beta}$ *is nonzero on the common HLS core.*

Proof. Let $E := \mathcal{V}_Q$. By the HLS theorem invoked in Theorem H.16, the original nonzero odd derivation is implemented by some nonzero $v_0 \in E$, so $Q(v_0) \neq 0$.

Suppose for contradiction that $P_{\alpha\dot{\beta}} = 0$ for all $\alpha, \dot{\beta}$. Then the HLS anticommutator gives

$$0 = \{Q_\alpha(v_0), Q_\alpha(v_0)^\dagger\}$$

on the common core. Hence for every core vector ψ,

$$0 = \langle \psi, \{Q_\alpha(v_0), Q_\alpha(v_0)^\dagger\}\psi\rangle = \|Q_\alpha(v_0)\psi\|^2 + \|Q_\alpha(v_0)^\dagger\psi\|^2.$$

Therefore $Q_\alpha(v_0) = 0$ for both $\alpha = 1, 2$, contradicting the nontriviality of the implementing HLS charge. Thus some $P_{\alpha\dot{\beta}} \neq 0$. □

Lemma I.3 (Finite-dimensional odd charge module is stable under D)**.** *Assume Theorem H.16, Definition B.19, and Theorem H.17. Let $\mathcal{V}_Q$ be the finite-dimensional odd charge module furnished by Theorem H.16, and write $Q_\alpha(v)$ for the corresponding odd generators. Then:*

(a) the dilation one-parameter subgroup acts on $\mathcal{V}_Q$ through a finite-dimensional representation e^{tA_Q} such that on the common UV core

$$e^{itD} Q_\alpha(v) e^{-itD} = Q_\alpha(e^{tA_Q} v);$$

(b) differentiating at $t = 0$ gives

$$[D, Q_\alpha(v)] = i\, Q_\alpha(A_Q v);$$

(c) with respect to the Hermitian form determined by the indexed HLS anticommutator, $A_Q + A_Q^\dagger = \mathbf{1}_{\mathcal{V}_Q}$. In particular, $A_Q - \frac{1}{2}\mathbf{1}$ is skew-Hermitian.

Proof. Set $E := \mathcal{V}_Q$, and let

$$\mathscr{Q}_+ := \{\text{translation-invariant local odd Weyl-spinor symmetry generators of the fixed chirality}\}.$$

By Theorem H.17, the map

$$J : E \longrightarrow \mathscr{Q}_+, \qquad v \longmapsto Q(v) := (Q_1(v), Q_2(v))$$

is a linear isomorphism. In particular, $\mathscr{Q}_+$ is finite-dimensional, and the full Weyl pair records both spinor components in a single map.

For the dilation subgroup from Definition B.19, define

$$\Theta_t(X) := e^{itD} X e^{-itD} \qquad (X \in \mathscr{Q}_+).$$

Because dilations preserve locality, translation-invariance, and Weyl-spinor type, Θ_t maps $\mathscr{Q}_+$ to itself and satisfies

$$\Theta_{t+s} = \Theta_t \Theta_s, \qquad \Theta_0 = \mathbf{1}_{\mathscr{Q}_+}.$$

Transfer this action to E by

$$T_t := J^{-1} \Theta_t J \in \mathrm{GL}(E).$$

Then, by construction,

$$e^{itD} Q_\alpha(v) e^{-itD} = Q_\alpha(T_t v)$$

for both $\alpha = 1, 2$.

It remains to prove continuity of $t \mapsto T_t$. Because $\mathscr{Q}_+$ is finite-dimensional, it is enough to prove continuity of a separating family of linear functionals. Choose a basis $X^{(1)}, \ldots, X^{(m)}$ of $\mathscr{Q}_+$. By finite-dimensionality, one can choose vectors ϕ_j, ψ_j in the common UV core and local observables A_j so that the linear functionals

$$\ell_j(X) := \langle \phi_j, X\, \pi(A_j)\Omega\rangle \qquad (1 \leq j \leq m)$$

separate $\mathscr{Q}_+$. For fixed $X \in \mathscr{Q}_+$,

$$\ell_j(\Theta_t X) = \big\langle e^{-itD}\phi_j,\, X\, e^{-itD}\pi(A_j)\Omega\big\rangle.$$

The conformal representation is strongly continuous, and the conformal local core is stable under the dilation subgroup by Definition B.19; hence each $t \mapsto \ell_j(\Theta_t X)$ is continuous. Therefore $t \mapsto \Theta_t$ is continuous on the finite-dimensional space $\mathscr{Q}_+$, and hence $t \mapsto T_t$ is continuous on E.

The standard finite-dimensional one-parameter-group theorem therefore yields a unique $A_Q \in \mathrm{End}(E)$ such that

$$T_t = e^{tA_Q}.$$

This proves item (a). Differentiating the defining identity at $t = 0$ on the common UV core gives

$$[D, Q_\alpha(v)] = i\, Q_\alpha(A_Q v),$$

which is item (b).

For item (c), define the HLS Hermitian form on $E = \mathcal{V}_Q$ by

$$\{Q_\alpha(v), \bar{Q}_{\dot\beta}(w)\} = 2\,\langle v, w\rangle_Q\, P_{\alpha\dot\beta}.$$

By item (b) and selfadjointness of D, the barred generators obey

$$[D, \bar{Q}_{\dot\beta}(w)] = i\,\bar{Q}_{\dot\beta}(A_Q^\dagger w)$$

under the HLS Hermitian identification. Applying D to the HLS anticommutator and using $[D, P_{\alpha\dot\beta}] = iP_{\alpha\dot\beta}$ gives

$$\begin{aligned} i\,2\,\langle v, w\rangle_Q\, P_{\alpha\dot\beta} &= [D, \{Q_\alpha(v), \bar{Q}_{\dot\beta}(w)\}] \\ &= \{[D, Q_\alpha(v)], \bar{Q}_{\dot\beta}(w)\} + \{Q_\alpha(v), [D, \bar{Q}_{\dot\beta}(w)]\} \\ &= i\,2\,\big(\langle A_Q v, w\rangle_Q + \langle v, A_Q^\dagger w\rangle_Q\big) P_{\alpha\dot\beta}. \end{aligned}$$

Hence

$$2i\,\big\langle (A_Q + A_Q^\dagger - \mathbf{1}_E)v, w\big\rangle_Q\, P_{\alpha\dot\beta} = 0 \qquad (v, w \in E).$$

By Lemma I.2, choose $(\alpha_0, \dot\beta_0)$ with $P_{\alpha_0\dot\beta_0} \neq 0$. Specializing to that pair, the scalar coefficient must vanish, so

$$\big\langle (A_Q + A_Q^\dagger - \mathbf{1}_E)v, w\big\rangle_Q = 0 \qquad (v, w \in E).$$

Therefore

$$A_Q + A_Q^\dagger = \mathbf{1}_E.$$

Equivalently, $A_Q - \frac{1}{2}\mathbf{1}_E$ is skew-Hermitian. □

Theorem I.1 (Odd-charge dilatation normal form)**.** *Assume Lemma I.3. Set*

$$K_Q := A_Q - \frac{1}{2}\mathbf{1}_{\mathcal{V}_Q}.$$

Then $K_Q^\dagger = -K_Q$, and on the common UV core one has

$$[D, Q_\alpha(v)] = \frac{i}{2}\, Q_\alpha(v) + i\, Q_\alpha(K_Q v).$$

If the barred generators are viewed through the Hermitian identification $\mathcal{V}_Q \cong \overline{\mathcal{V}_Q}^{\,*}$, then likewise

$$[D, \bar{Q}_{\dot\alpha}(v)] = \frac{i}{2}\,\bar{Q}_{\dot\alpha}(v) - i\,\bar{Q}_{\dot\alpha}(K_Q v).$$

In particular, the only possible obstruction to exact weight $\frac{1}{2}$ is a commuting skew-Hermitian endomorphism of the internal multiplicity space.

Proof. Lemma I.3 gives

$$A_Q + A_Q^\dagger = \mathbf{1}_{\mathcal{V}_Q},$$

hence $K_Q^\dagger = -K_Q$. Substituting $A_Q = \frac{1}{2}\mathbf{1}_{\mathcal{V}_Q} + K_Q$ into the commutator formula of Lemma I.3 immediately gives

$$[D, Q_\alpha(v)] = i\, Q_\alpha(A_Q v) = \frac{i}{2}\, Q_\alpha(v) + i\, Q_\alpha(K_Q v).$$

For the barred sector, take Hilbert adjoints and use the selfadjointness of D:

$$[D, \bar{Q}_{\dot\alpha}(v)] = -[D, Q_\alpha(v)]^\dagger = \frac{i}{2}\, \bar{Q}_{\dot\alpha}(v) - i\, \bar{Q}_{\dot\alpha}(K_Q v).$$

No further conclusion about K_Q follows from the stated interfaces alone. □

Remark I.1 (Meaning of the K_Q term)**.** Theorem I.1 isolates the most general dilatation normal form allowed by the stated interfaces. The endomorphism K_Q measures the internal skew-Hermitian part of the dilatation action on the multiplicity space. In Subprogram 4b that term is absorbed into the internal degree-zero sector once that sector is realized on the same Hilbert space.

Corollary I.1 (Standard half-weight after central internal normalization)**.** *Assume Theorem I.1, Theorem J.5, Theorem J.6, Lemma J.9, Lemma J.10, Lemma J.11, and Corollary J.1. Let*

$$K_Q = \kappa\, \mathbf{1}_E, \qquad R_{K_Q} := T_{K_Q}, \qquad D_{\mathrm{std}} := D - iR_{K_Q}.$$

Then

$$[D_{\mathrm{std}}, Q_\alpha(v)] = \frac{i}{2}\, Q_\alpha(v), \qquad [D_{\mathrm{std}}, \bar{Q}_{\dot\alpha}(v)] = \frac{i}{2}\, \bar{Q}_{\dot\alpha}(v).$$

$$[D_{\mathrm{std}}, S^\alpha(\lambda)] = -\frac{i}{2}\, S^\alpha(\lambda), \qquad [D_{\mathrm{std}}, \bar{S}^{\dot\alpha}(\lambda)] = -\frac{i}{2}\, \bar{S}^{\dot\alpha}(\lambda).$$

Moreover, for every $A \in \mathrm{End}(E)$,

$$[D_{\mathrm{std}}, T_A] = 0.$$

Finally,

$$[D_{\mathrm{std}}, P_\mu] = iP_\mu, \qquad [D_{\mathrm{std}}, M_{\mu\nu}] = [D, M_{\mu\nu}],$$

Proof. Choose an orthonormal basis $\{e_I\}_{I=1}^N$ of E with dual basis $\{e^I\}_{I=1}^N$, and write

$$R^I{}_J := T_{e_I, e^J}, \qquad r := \sum_I R^I{}_I.$$

Since $K_Q = \kappa\, \mathbf{1}_E$, one has

$$R_{K_Q} = \kappa r.$$

By Corollary J.1,

$$\left[r, Q_\alpha^I\right] = Q_\alpha^I, \qquad \left[r, \bar{Q}_{\dot\alpha I}\right] = -\bar{Q}_{\dot\alpha I},$$

$$\left[r, S^\alpha{}_I\right] = -S^\alpha{}_I, \qquad \left[r, \bar{S}^{\dot\alpha I}\right] = \bar{S}^{\dot\alpha I}.$$

Hence

$$\left[iR_{K_Q}, Q_\alpha^I\right] = i\kappa\, Q_\alpha^I, \qquad \left[iR_{K_Q}, \bar{Q}_{\dot\alpha I}\right] = -i\kappa\, \bar{Q}_{\dot\alpha I},$$

$$\big[iR_{K_Q}, S^{\alpha}{}_{I}\big] = -i\kappa\, S^{\alpha}{}_{I}, \qquad \big[iR_{K_Q}, \bar{S}^{\dot\alpha I}\big] = i\kappa\, \bar{S}^{\dot\alpha I}.$$

On the other hand, Theorem I.1, Lemma J.10, and Lemma J.11 give

$$[D, Q^I_\alpha] = \Big(\frac{i}{2} + i\kappa\Big) Q^I_\alpha, \qquad [D, \bar{Q}_{\dot\alpha I}] = \Big(\frac{i}{2} - i\kappa\Big) \bar{Q}_{\dot\alpha I},$$

$$[D, S^{\alpha}{}_{I}] = \Big(-\frac{i}{2} - i\kappa\Big) S^{\alpha}{}_{I}, \qquad [D, \bar{S}^{\dot\alpha I}] = \Big(-\frac{i}{2} + i\kappa\Big) \bar{S}^{\dot\alpha I}.$$

Subtracting the iR_{K_Q}-commutators therefore yields the exact half-weight relations for all four odd families.

For the internal block, let $A \in \mathrm{End}(E)$ and set

$$U_A := [R_{K_Q}, T_A].$$

Because R_{K_Q} and T_A are degree-zero internal operators, U_A is again degree-zero internal. By linearity from Theorem J.5,

$$[D, T_A] = 0, \qquad [T_A, P_\mu] = [T_A, M_{\mu\nu}] = 0.$$

Moreover, for every $w \in E$,

$$\begin{aligned}[U_A, Q_\alpha(w)] &= [R_{K_Q}, Q_\alpha(Aw)] - [T_A, Q_\alpha(K_Q w)] \\ &= Q_\alpha(K_Q A w) - Q_\alpha(A K_Q w) \\ &= 0\end{aligned}$$

by Lemma J.10. Hence U_A is a degree-zero internal operator acting trivially on the lower odd layer. By Lemma J.9, it acts trivially on the entire odd closure, so $U_A = 0$. Therefore

$$[R_{K_Q}, T_A] = 0 \qquad (A \in \mathrm{End}(E)),$$

and thus

$$[D_{\mathrm{std}}, T_A] = [D, T_A] - i[R_{K_Q}, T_A] = 0.$$

Since R_{K_Q} is internal, it already commutes with P_μ and $M_{\mu\nu}$, so

$$[D_{\mathrm{std}}, P_\mu] = [D, P_\mu] = iP_\mu, \qquad [D_{\mathrm{std}}, M_{\mu\nu}] = [D, M_{\mu\nu}].$$

□

Remark I.2 (Current-theoretic consequence of D_{std})**.** For any conserved local current representative of an HLS charge that is conformal-primary with respect to D_{std}, the relations of Corollary I.1 force engineering dimension $\frac{7}{2}$.

Appendix J: Subprogram 4b: Same-$\mathcal{H}$ conformal closure of the HLS odd sector

In this section the HLS odd sector is closed conformally on the same UV Hilbert space. The argument proceeds as follows:

(1) strengthen the HLS import once, by using Theorem H.17;

(2) take the finite conformal closure of the HLS odd seed;

(3) prove same-$\mathcal{H}$ implementability of that finite closure on a common invariant core;

(4) recover the internal degree-zero algebra from mixed Q-S brackets and prove saturation to $\mathfrak{u}(N)$;

(5) identify the resulting operator superalgebra with the standard 4D superconformal algebra.

Theorem H.18 from the main spine gives a separate AQFT result for choosing indexed current representatives, while the closure argument below is formulated directly at the operator and module level.

1. Representation-theoretic input and odd conformal closure

Theorem J.1 (Generated conformal prolongation of the HLS seed [35, 36])**.** *Assume Theorem H.16, Definition B.19, Theorem H.17, and Theorem I.1. Let $E := \mathcal{V}_Q$, and let W_Q be the translation-invariant HLS odd seed, viewed through the normal form of Theorem I.1 as the tensor product of the standard weight-$\frac{1}{2}$ Weyl module with the internal multiplicity space E. Let*

$$W_{\mathrm{sc}} := U\big(\mathfrak{so}(4,2)_{\mathbb{C}}\big) \cdot W_Q$$

inside the odd local-derivation space on the UV vacuum representation. Then:

(i) W_{sc} is finite-dimensional.

(ii) W_{sc} has exactly two odd weight layers,

$$W_{\mathrm{sc}} = W_Q \oplus W_S.$$

(iii) with respect to that normalized conformal part, W_Q has weight $+\frac{1}{2}$ and W_S has weight $-\frac{1}{2}$.

(iv) P_μ annihilates W_Q, and K_μ annihilates W_S.

(v) No further odd layers occur.

Proof. Let

$$\mathfrak{g} := \mathfrak{so}(4,2)_{\mathbb{C}} = \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}$$

be the standard conformal $|1|$-grading, with

$$\mathfrak{g}_{+1} \cong \mathrm{span}\{P_\mu\}, \qquad \mathfrak{g}_{-1} \cong \mathrm{span}\{K_\mu\}, \qquad \mathfrak{g}_0 \cong \mathfrak{so}(1,3)_{\mathbb{C}} \oplus \mathbb{C}D.$$

By Theorem H.16 and Theorem I.1, the HLS seed W_Q carries the Weyl-spinor factor with standard weight $\frac{1}{2}$, while the skew-Hermitian part K_Q acts only on the multiplicity space E and commutes with the Lorentz-spinor factor. Hence W_Q may be viewed as a finite-dimensional $\mathfrak{g}_0$-module whose conformal part is the standard lowest-weight Weyl module, annihilated by $\mathfrak{g}_{+1}$ because the HLS generators are translation-invariant. Thus W_Q is naturally a module for the parabolic subalgebra

$$\mathfrak{p}^+ := \mathfrak{g}_0 \oplus \mathfrak{g}_{+1}.$$

Form the generalized Verma module

$$M(W_Q) := U(\mathfrak{g}) \otimes_{U(\mathfrak{p}^+)} W_Q.$$

The concrete conformal closure in Definition J.1 is exactly the image of $M(W_Q)$ in the odd local-derivation space on the UV vacuum representation.

Chirality by chirality, the conformal part of W_Q is E tensored with the lowest-weight Weyl spinor of weight $\frac{1}{2}$. The imported conformal-spinor prolongation classification cited in the appendix ([35, 36]) says that the generated module for such a seed has the standard two-layer conformal-spinor quotient and no higher odd prolongation. Therefore the concrete image generated by the actual HLS seed factors through a finite-dimensional module with exactly the two odd normalized weights $\pm\frac{1}{2}$. The lowest layer is the seed module W_Q, the upper layer is its conformal prolongation W_S, P_μ kills the lower layer, K_μ kills the upper layer, and no further odd layers occur. □

Definition J.1 (Odd conformal closure)**.** Let W_Q be the odd derivation module generated by the HLS charges of Theorem H.16; equivalently,

$$W_Q := \{A \mapsto i[Q(v), \pi(A)]_{\mathrm{s}} \ : \ v \in \mathcal{V}_Q\}.$$

Define W_{sc} to be the smallest linear space of odd local derivations on the UV net that:

- contains W_Q, and

- is stable under commutator with the even conformal generators $P_\mu, M_{\mu\nu}, D, K_\mu$.

We call $W_{\rm sc}$ the post-HLS odd conformal closure of the HLS seed.

Theorem J.2 (Finite-dimensional conformal closure of the HLS odd sector)**.** *Assume Theorem J.1. Then $W_{\rm sc}$ is finite-dimensional and splits as*

$$W_{\rm sc} = W_Q \oplus W_S,$$

where:

- *with respect to the normalized conformal part, W_Q has weight $+\frac{1}{2}$,*
- *with respect to the normalized conformal part, W_S has weight $-\frac{1}{2}$,*
- *P_μ annihilates W_Q,*
- *K_μ annihilates W_S,*
- *and W_S has exactly the left/right Weyl-spinor chirality pattern of the standard $(S, \bar{S})$ layer.*

Proof. This is the content of Theorem J.1, restated with the notation of Definition J.1. □

2. Same-$\mathcal{H}$ implementability

Lemma J.1 (Conformal Nelson smooth domain)**.** *Let*

$$N_{\rm conf} := 1 + \sum_\mu P_\mu^2 + \sum_{\mu<\nu} M_{\mu\nu}^2 + D^2 + \sum_\mu K_\mu^2$$

with closure taken on the common Stone core from Definition B.19. Then

$$\mathcal{D}_\infty(N_{\rm conf}) := \bigcap_{r\geq 0} \operatorname{Dom} N_{\rm conf}^r$$

is dense, invariant under the connected conformal group, and coincides with the smooth-vector space for the conformal representation.

Proof. This is the standard Nelson theorem for a finite family of essentially selfadjoint generators of a strongly continuous unitary representation of a finite-dimensional Lie group. Definition B.19 supplies exactly those hypotheses for the conformal generator family $\{P_\mu, M_{\mu\nu}, D, K_\mu\}$. □

Lemma J.2 (The indexed HLS seed has a common linear H-bound)**.** *Assume Theorem H.16. Then, for any chosen norm on the two-component Weyl pair $Q(v) = (Q_1(v), Q_2(v))$, there exists $C_0 > 0$ such that*

$$\|Q(v)\psi\| \leq C_0 \, \|v\|_Q \, \|(1+H)\psi\| \qquad (v \in \mathcal{V}_Q,\ \psi \in \mathcal{D}_Q),$$

where $H := P^0$. In particular, the seed polynomial energy bound holds with $m_0 = 1$.

Proof. Choose an orthonormal basis $\{e_I\}_{I=1}^N$ of $E := \mathcal{V}_Q$, write

$$v = \sum_{I=1}^N v_I e_I,$$

and define

$$Q_\alpha(v) := \sum_{I=1}^N v_I Q_\alpha^I, \qquad \bar{Q}_{\dot\beta}(v) := \sum_{I=1}^N \overline{v_I}\, \bar{Q}_{\dot\beta I}.$$

By bilinearity and the indexed HLS anticommutator from Theorem H.16,

$$\{Q_\alpha(v), \bar{Q}_{\dot\beta}(v)\} = 2\,\|v\|_Q^2\,\sigma^\mu_{\alpha\dot\beta} P_\mu$$

on the common invariant core $\mathcal{D}_Q$. Summing the diagonal entries $\alpha = \dot\alpha = 1, 2$ and using $\sigma^0 = \mathbf{1}_2$ together with the tracelessness of the spatial Pauli matrices gives

$$\sum_{\alpha=1}^{2} \{Q_\alpha(v), \bar{Q}_{\dot\alpha}(v)\} = 4\,\|v\|_Q^2\, H.$$

Therefore, for every $\psi \in \mathcal{D}_Q$,

$$\sum_{\alpha=1}^{2} \Big(\|Q_\alpha(v)\psi\|^2 + \|\bar{Q}_{\dot\alpha}(v)\psi\|^2 \Big) = 4\,\|v\|_Q^2\,\langle \psi, H\psi\rangle.$$

Because Theorem H.16 is built on the positive-energy vacuum representation from Theorem D.1, the spectrum condition gives $H \geq 0$. Hence the spectral theorem yields

$$\langle \psi, H\psi\rangle \leq \|(1+H)\psi\|^2.$$

It follows that

$$\sum_{\alpha=1}^{2} \|Q_\alpha(v)\psi\|^2 \leq 4\,\|v\|_Q^2\,\|(1+H)\psi\|^2.$$

If $\|\cdot\|_{\mathrm{W}}$ denotes the chosen norm on the two-component Weyl pair, finite-dimensional norm equivalence gives $c_{\mathrm{W}} > 0$ such that for every $(\xi_1, \xi_2) \in \mathcal{H} \oplus \mathcal{H}$,

$$\|(\xi_1, \xi_2)\|_{\mathrm{W}} \leq c_{\mathrm{W}}\,(\|\xi_1\|^2 + \|\xi_2\|^2)^{1/2}$$

Applying this pointwise to $(Q_1(v)\psi, Q_2(v)\psi)$ yields

$$\|Q(v)\psi\| \leq 2c_{\mathrm{W}}\,\|v\|_Q\,\|(1+H)\psi\|.$$

Thus the stated bound holds with $C_0 := 2c_{\mathrm{W}}$, and in particular one may take $m_0 = 1$. □

Lemma J.3 (Seed bounds transfer from H to N_{conf})**.** *Assume Theorem H.16 and Definition B.19. Then, after restricting to $\mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q$, there exist $m_1 \geq 0$ and $C_1 > 0$ such that*

$$\|Q(v)\psi\| \leq C_1\,\|v\|_Q\,\|(1+N_{\mathrm{conf}})^{m_1}\psi\| \qquad (v \in \mathcal{V}_Q,\ \psi \in \mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q).$$

Proof. By Lemma J.2, there exists $C_0 > 0$ such that

$$\|Q(v)\psi\| \leq C_0\,\|v\|_Q\,\|(1+H)\psi\| \qquad (v \in \mathcal{V}_Q,\ \psi \in \mathcal{D}_Q).$$

Since $H = P^0$ is one of the conformal generators entering N_{conf}, the conformal Nelson domain carries a polynomial comparison

$$\|(1+H)\psi\| \leq C\,\|(1+N_{\mathrm{conf}})^{m_1}\psi\|$$

for suitable m_1 and C. Combining the two inequalities gives the claim with $C_1 := CC_0$. □

Lemma J.4 (One conformal derivative costs one Sobolev order)**.** *Let X be a closable operator satisfying*

$$\|X\psi\| \leq C\,\|(1+N_{\mathrm{conf}})^m\psi\| \qquad (\psi \in \mathcal{D}_\infty(N_{\mathrm{conf}})).$$

For any one-parameter conformal subgroup with selfadjoint generator G, the orbit map

$$t \longmapsto e^{itG} X e^{-itG}$$

is differentiable as a map

$$\mathrm{Dom}(1+N_{\mathrm{conf}})^{(m+1)/2} \longrightarrow \mathcal{H},$$

and its derivative at $t = 0$ is the commutator $[iG, X]$ on $\mathcal{D}_\infty(N_{\mathrm{conf}})$.

Proof. The conformal group acts continuously on each Nelson-Sobolev space built from N_{conf}. Differentiating the conjugation orbit loses one group derivative and hence one Sobolev order. This is the standard smooth-vector calculus for finite-dimensional unitary Lie-group representations. □

Theorem J.3 (Uniform orbit energy bounds for the odd conformal closure)**.** *Assume Theorem H.16, Definition B.19, and Theorem J.2. Then there exist an integer $M \geq 0$, a constant $C > 0$, and a norm $\|\cdot\|_{\mathrm{sc}}$ on W_{sc} such that every odd derivation in W_{sc} is implemented on $\mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q$ by a closable operator X satisfying*

$$\|X\psi\| \leq C\,\|X\|_{\mathrm{sc}}\,\|(1+N_{\mathrm{conf}})^M\psi\| \qquad (\psi \in \mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q).$$

In particular, the entire odd conformal closure obeys one common polynomial graph bound on the same UV Hilbert space.

Proof. By Lemma J.3, the HLS seed implementers satisfy a common N_{conf}-Sobolev bound on $\mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q$. By Theorem J.2, the whole odd conformal closure has depth one:

$$W_{\mathrm{sc}} = W_Q \oplus W_S, \qquad W_S = [K, W_Q].$$

The W_Q-layer already satisfies the transferred Nelson bound. Every element of W_S is obtained by one conformal derivative of a seed implementer, so Lemma J.4 yields the same kind of bound with one additional Sobolev order. Because W_{sc} is finite-dimensional, one may choose a basis $\{X_a\}$ of $W_Q \oplus W_S$, let M be the maximum of the finitely many required Sobolev exponents, and define

$$\|X\|_{\mathrm{sc}} := \sum_a |c_a| \qquad \text{for } X = \sum_a c_a X_a.$$

Then one constant C works for every $X \in W_{\mathrm{sc}}$, giving the displayed bound. □

Lemma J.5 (Gårding vectors from a dense Hilbert subspace are smooth-dense)**.** *Let G be a finite-dimensional Lie group with strongly continuous unitary representation U on $\mathcal{H}$, and let $\mathcal{H}^\infty$ be the smooth-vector Fréchet space. If $E \subset \mathcal{H}$ is dense, then*

$$\mathcal{G}(E) := \operatorname{span}\{U(f)\eta : f \in C_c^\infty(G),\ \eta \in E\}$$

is Fréchet-dense in $\mathcal{H}^\infty$.

Proof. Fix $\psi \in \mathcal{H}^\infty$ and a finite family of Fréchet seminorms $p_{Y_1,\dots,Y_m}(\xi) := \|dU(Y_1)\cdots dU(Y_m)\xi\|$. Let (f_n) be a standard approximate identity in $C_c^\infty(G)$. Then

$$U(f_n)\psi \longrightarrow \psi$$

in every smooth-vector seminorm. For each fixed n, the operators $U(L_{Y_1}\cdots L_{Y_m} f_n)$ are bounded on $\mathcal{H}$. Since E is dense in $\mathcal{H}$, one may choose $\eta_n \in E$ such that

$$\|U(L_{Y_1}\cdots L_{Y_m} f_n)(\eta_n - \psi)\|$$

is arbitrarily small for the finitely many seminorms under consideration. Then

$$p_{Y_1,\dots,Y_m}\big(U(f_n)\eta_n - \psi\big) \leq p_{Y_1,\dots,Y_m}\big(U(f_n)(\eta_n - \psi)\big) + p_{Y_1,\dots,Y_m}\big(U(f_n)\psi - \psi\big)$$

can be made arbitrarily small. A diagonal argument over a countable defining family of seminorms shows that $\mathcal{G}(E)$ is Fréchet-dense in $\mathcal{H}^\infty$. □

Lemma J.6 (Finite conformal operator families preserve conformal smooth vectors)**.** *Assume Definition B.19, Theorem J.3, and that W_{sc} is finite-dimensional and stable under the even conformal action. Set*

$$\mathcal{D}_{\mathrm{seed}} := \mathcal{D}_\infty(N_{\mathrm{conf}}) \cap \mathcal{D}_Q.$$

Then every $X \in W_{\mathrm{sc}}$ extends uniquely from $\mathcal{D}_{\mathrm{seed}}$ to a continuous endomorphism of the Fréchet space $\mathcal{D}_\infty(N_{\mathrm{conf}})$, and the whole family W_{sc} preserves $\mathcal{D}_\infty(N_{\mathrm{conf}})$.

Proof. By Lemma J.1, $\mathcal{D}_\infty(N_{\rm conf})$ is dense in $\mathcal{H}$, and by Theorem H.16 the HLS core $\mathcal{D}_Q$ is dense in $\mathcal{H}$. Hence their intersection $\mathcal{D}_{\rm seed}$ is dense in $\mathcal{H}$.

Let $G := {\rm Conf}_0(\mathbb{R}^{1,3})$, let $U := U_{\rm conf}$, and for $X \in W_{\rm sc}$ write

$$\Theta_g(X) := U(g)XU(g)^{-1}.$$

Choose a basis $X_1, \dots, X_r$ of the finite-dimensional space $W_{\rm sc}$ and write

$$\Theta_g(X) = \sum_{a=1}^{r} c_a(g; X)X_a$$

with smooth coefficient functions $c_a(g; X)$.

For $f \in C_c^\infty(G)$ and $\psi \in \mathcal{D}_{\rm seed}$, define

$$X\,U(f)\psi := \int_G f(g)\,U(g)\,\Theta_{g^{-1}}(X)\psi\,dg.$$

Because $\Theta_{g^{-1}}(X)$ stays in the finite-dimensional family $W_{\rm sc}$, the uniform Sobolev bound from Theorem J.3 implies that the integrand is norm-continuous and compactly supported in g; thus the integral is well defined. Expanding $\Theta_{g^{-1}}(X)$ in the basis gives

$$XU(f)\psi = \sum_{a=1}^{r} \int_G f(g)c_a(g^{-1}; X)U(g)X_a\psi\,dg,$$

so $XU(f)\psi$ is a finite sum of conformal smooth vectors.

Now apply any Lie derivative $dU(Y_1)\cdots dU(Y_m)$ to the above formula. Differentiation under the integral is allowed because orbit derivatives of $\Theta_g(X)$ still lie in $W_{\rm sc}$, and each such derivative obeys the same kind of polynomial Sobolev bound by repeated use of Lemma J.4 and finite-dimensionality. Hence, for each Fréchet seminorm p_m on $\mathcal{D}_\infty(N_{\rm conf})$, there exist seminorms $q_{m,X}$ on $\mathcal{D}_{\rm seed}$ and constants $C_{m,X,f}$ such that

$$p_m\big(XU(f)\psi\big) \le C_{m,X,f}\,q_{m,X}(\psi).$$

Therefore X acts continuously on the Gårding subspace generated from $\mathcal{D}_{\rm seed}$.

By Lemma J.5, the conformal Gårding vectors generated from the dense set $\mathcal{D}_{\rm seed}$ are Fréchet-dense in $\mathcal{D}_\infty(N_{\rm conf})$. The seminorm estimates above therefore extend X uniquely and continuously to all conformal smooth vectors. This extension preserves $\mathcal{D}_\infty(N_{\rm conf})$. □

Theorem J.4 (Same-UV-Hilbert-space odd conformal closure package)**.** *Assume Theorem H.16, Definition B.19, Theorem I.1, Theorem J.2, and Theorem J.3. Then every derivation in $W_{\rm sc}$ is implemented by a closable operator on the same UV Hilbert space $\mathcal{H}$, and the dense domain*

$$\mathcal{D}_{\rm sc} := \mathcal{D}_\infty(N_{\rm conf}) \subset \mathcal{H}$$

is a common invariant core for

$$P_\mu,\ M_{\mu\nu},\ D,\ K_\mu$$

and for all odd implementers belonging to $W_{\rm sc}$. After choosing bases of the two odd layers furnished by Theorem J.1, one obtains operators

$$Q^I_\alpha,\ \bar{Q}_{\dot\alpha I},\ S^\alpha{}_I,\ \bar{S}^{\dot\alpha I}$$

on that same common core.

Proof. Lemma J.1 gives a dense conformal smooth-vector space $\mathcal{D}_{\rm sc}$ preserved by the even conformal generators. Theorem J.3 gives one common polynomial $N_{\rm conf}$-Sobolev bound for all $X \in W_{\rm sc}$ on the seed domain

$$\mathcal{D}_{\rm seed} := \mathcal{D}_\infty(N_{\rm conf}) \cap \mathcal{D}_Q.$$

Lemma J.6 upgrades each $X \in W_{\rm sc}$ to a continuous endomorphism of $\mathcal{D}_\infty(N_{\rm conf})$. In particular, every odd derivation in $W_{\rm sc}$ is implemented by a closable operator on the same UV Hilbert space and preserves $\mathcal{D}_\infty(N_{\rm conf})$.

Therefore $\mathcal{D}_{\rm sc} = \mathcal{D}_\infty(N_{\rm conf})$ is a dense common invariant core for the full even conformal sector and the entire odd conformal closure. Naming the lower and upper odd layers by $Q^I_\alpha, \bar{Q}_{\dot\alpha I}$ and $S^\alpha{}_I, \bar{S}^{\dot\alpha I}$ gives the stated same-$\mathcal{H}$ realization. □

3. Mixed brackets and the internal algebra

Write $E := \mathcal{V}_Q$, with $\dim_{\mathbb{C}} E = N$. The HLS Hermitian form from Theorem H.16 identifies E anti-linearly with E^*; write

$$\flat : E \to E^*, \quad v \mapsto v^\flat, \qquad \sharp : E^* \to E, \quad \lambda \mapsto \lambda^\sharp,$$

so that $\lambda(v) = \langle v, \lambda^\sharp \rangle_Q$. For $A \in \mathrm{End}(E)$, write

$$A^\vee : E^* \to E^*, \qquad (A^\vee \lambda)(v) := \lambda(Av),$$

and let $A^\star \in \mathrm{End}(E)$ be defined by

$$\langle Av, w \rangle_Q = \langle v, A^\star w \rangle_Q.$$

For rank-one operators one then has

$$(v \otimes \lambda)^\star = \lambda^\sharp \otimes v^\flat.$$

We write the upper odd layer as $S^\alpha(\lambda)$ and $\bar{S}^{\dot\alpha}(\lambda)$ for $\lambda \in E^*$, and we set

$$P_{\alpha\dot\beta} := (\sigma^\mu)_{\alpha\dot\beta} P_\mu, \qquad K^{\alpha\dot\beta} := (\bar\sigma^\mu)^{\alpha\dot\beta} K_\mu.$$

We regard $Q_\alpha(\cdot)$ and $S^\alpha(\cdot)$ as linear in their labels. For the barred families we state all internal-action formulas in the invariant $A^\star$-/$\flat, \sharp$-form; the component formulas in an orthonormal basis are the unambiguous concrete version.

Lemma J.7 (Normalization of the upper odd layer)**.** *After choosing bases of W_S, one may and shall normalize the S-operators so that on the common core $\mathcal{D}_{\mathrm{sc}}$ the unique nonzero $\mathfrak{so}(1,3)$-intertwiners are*

$$[P_{\alpha\dot\beta}, S^\gamma(\lambda)] = 2\,\delta^\gamma_\alpha\, \bar{Q}_{\dot\beta}(\lambda^\sharp), \qquad [P_{\alpha\dot\beta}, \bar{S}^{\dot\gamma}(\lambda)] = 2\,\delta^{\dot\gamma}_{\dot\beta}\, Q_\alpha(\lambda^\sharp),$$

$$[K^{\alpha\dot\beta}, Q_\gamma(v)] = 2\,\delta^\alpha_\gamma\, \bar{S}^{\dot\beta}(v^\flat), \qquad [K^{\alpha\dot\beta}, \bar{Q}_{\dot\gamma}(v)] = 2\,\delta^{\dot\beta}_{\dot\gamma}\, S^\alpha(v^\flat),$$

with

$$[K, S] = [K, \bar{S}] = 0, \qquad [P, Q] = [P, \bar{Q}] = 0.$$

Proof. By Theorem J.1, W_S is the unique normalized weight-$-\frac{1}{2}$ layer in the conformal prolongation of W_Q. For each chirality, the tensor product of the vector representation with W_S contains the lower odd layer with multiplicity one, and likewise $V \otimes W_Q$ contains W_S with multiplicity one. Therefore the corresponding conformal intertwiners are unique up to an overall nonzero scalar, which we fix once and for all by the displayed normalization. The annihilation relations by P and K are already part of Theorem J.1. □

Lemma J.8 (Upper-layer basis may be chosen adjointly)**.** *Assume Theorem J.4. The bases of the two chiral pieces of the upper layer W_S may be chosen so that*

$$\big(S^\alpha(\lambda)\big)^\dagger = \bar{S}^{\dot\alpha}(\lambda)$$

under the $\flat/\sharp$ identification, and simultaneously the normalization conventions of Lemma J.7 hold.

Proof. Because W_S is realized on the same Hilbert space $\mathcal{H}$, the Hilbert adjoint sends one chiral piece of W_S to the opposite chiral piece. Choose a basis of one chirality and define the basis of the opposite chirality by taking adjoints. By Theorem J.2, both chiral pieces occur with the standard multiplicity-one Lorentz type. Therefore the bridge intertwiners in Lemma J.7 are still unique up to a common nonzero scalar on each chirality, and that scalar can be fixed once and for all so that the displayed normalization continues to hold. This preserves the adjoint pairing. □

Definition J.2 ($\mathfrak{gl}(2) \oplus \mathfrak{gl}(2)$ degree-zero generators)**.** Let

$$L_\alpha{}^\beta, \qquad \bar{L}^{\dot\alpha}{}_{\dot\beta}$$

denote the standard complex-linear combinations of $M_{\mu\nu}$ and D that realize the left and right $\mathfrak{gl}(2)$ actions inside $\mathfrak{so}(4,2)_{\mathbb{C}}$. They are characterized by

$$[L_\alpha{}^\beta, Q_\gamma(v)] = \delta^\beta_\gamma Q_\alpha(v), \qquad [L_\alpha{}^\beta, S^\gamma(\lambda)] = -\delta^\gamma_\alpha S^\beta(\lambda),$$

$$[\bar{L}^{\dot\alpha}{}_{\dot\beta}, \bar{Q}_{\dot\gamma}(v)] = \delta^{\dot\alpha}_{\dot\gamma} \bar{Q}_{\dot\beta}(v), \qquad [\bar{L}^{\dot\alpha}{}_{\dot\beta}, \bar{S}^{\dot\gamma}(\lambda)] = -\delta^{\dot\gamma}_{\dot\beta} \bar{S}^{\dot\alpha}(\lambda),$$

and by the analogous tautological action on $P_{\alpha\dot\beta}$ and $K^{\alpha\dot\beta}$. This is only a repackaging of the already available $M_{\mu\nu}$ and D.

Theorem J.5 (Unbarred degree-zero mixed-bracket decomposition)**.** *Assume Theorem J.4. Then for every $v \in E$ and $\lambda \in E^*$ there exists a unique even operator $T_{v,\lambda}$ on $\mathcal{D}_{\mathrm{sc}}$ such that*

$$\{Q_\alpha(v), S^\beta(\lambda)\} = \lambda(v)\, L_\alpha{}^\beta + \delta^\beta_\alpha\, T_{v,\lambda},$$

$$\{Q_\alpha(v), \bar{S}^{\dot\beta}(\lambda)\} = 0, \qquad \{\bar{Q}_{\dot\alpha}(v), S^\beta(\lambda)\} = 0.$$

Moreover,

$$[T_{v,\lambda}, P_\mu] = [T_{v,\lambda}, M_{\mu\nu}] = [T_{v,\lambda}, D] = 0.$$

Proof. The anticommutator $\{Q_\alpha(v), S^\beta(\lambda)\}$ is even, has total D-weight zero, and transforms in

$$S_+ \otimes S_+^* \cong \mathrm{End}(S_+) \cong \mathfrak{gl}(S_+).$$

Hence it decomposes uniquely into the tautological $\mathfrak{gl}(S_+)$ piece and a scalar-on-spinors remainder; this defines $T_{v,\lambda}$ and yields the first displayed formula. The cross-chiral mixed anticommutators have weight zero and vector Lorentz type, but the degree-zero even conformal algebra contains no vector piece, so they vanish.

By construction, the scalar remainder commutes with the degree-zero conformal algebra generated by L and $\bar{L}$. To see that it also commutes with translations, apply $[P_{\gamma\dot\delta}, -]$ to the first mixed-bracket formula. Lemma J.7 and the HLS anticommutator give

$$[P_{\gamma\dot\delta}, \{Q_\alpha(v), S^\beta(\lambda)\}] = 4\,\delta^\beta_\gamma\, \lambda(v)\, P_{\alpha\dot\delta}.$$

With the normalization of Definition J.2, this is exactly the translation commutator with $\lambda(v) L_\alpha{}^\beta$, so comparison yields

$$[P_{\gamma\dot\delta}, T_{v,\lambda}] = 0.$$

□

Theorem J.6 (Bilinear factorization and faithful lower-layer internal action from mixed brackets)**.** *Assume Theorem J.5 and Lemma H.11. Then:*

(i) the assignment $(v,\lambda) \mapsto T_{v,\lambda}$ is bilinear and therefore factors uniquely through a linear map

$$\Theta : E \otimes E^* \cong \mathrm{End}(E) \longrightarrow \mathrm{End}(\mathcal{D}_{\mathrm{sc}}), \qquad A \longmapsto T_A,$$

with $T_{v\otimes\lambda} = T_{v,\lambda}$;

(ii) for every $A \in \mathrm{End}(E)$ and every $w \in E$,

$$[T_A, Q_\alpha(w)] = Q_\alpha(Aw);$$

(iii) the map $A \mapsto T_A$ is injective.

Proof. The bilinearity of $(v, \lambda) \mapsto T_{v,\lambda}$ is immediate from the mixed-bracket formula

$$\{Q_\alpha(v), S^\beta(\lambda)\} = \lambda(v) L_\alpha{}^\beta + \delta_\alpha^\beta T_{v,\lambda},$$

since both the anticommutator and the scalar coefficient $\lambda(v)$ are bilinear in (v, λ). This gives the unique linear factorization through $E \otimes E^* \cong \mathrm{End}(E)$.

For rank-one operators, the same Schur-lemma argument as before shows that $T_{v,\lambda}$ acts on the lower odd layer

$$W_Q \cong S_+ \otimes E.$$

as the identity on the spinor factor tensored with an endomorphism of E. Evaluating the super-Jacobi identity on the triple $(Q_\alpha(v), S^\beta(\lambda), Q_\gamma(w))$ gives

$$\lambda(v)\, \delta_\gamma^\beta Q_\alpha(w) + \delta_\alpha^\beta [T_{v,\lambda}, Q_\gamma(w)] = \lambda(w)\, \delta_\alpha^\beta Q_\gamma(v) + \delta_\gamma^\beta [T_{w,\lambda}, Q_\alpha(v)].$$

Comparing the δ_α^β and δ_γ^β components yields

$$[T_{v,\lambda}, Q_\alpha(w)] = \lambda(w)\, Q_\alpha(v) = Q_\alpha\big((v \otimes \lambda)w\big).$$

By linearity, this extends to

$$[T_A, Q_\alpha(w)] = Q_\alpha(Aw)$$

for every $A \in \mathrm{End}(E)$.

Finally, if $T_A = 0$, then

$$0 = [T_A, Q_\alpha(w)] = Q_\alpha(Aw)$$

for every $w \in E$. Lemma H.11 therefore gives $Aw = 0$ for all w, hence $A = 0$. □

Write

$$R_{\mathrm{cl},\mathbb{C}} := \Theta(\mathrm{End}(E)) = \{T_A : A \in \mathrm{End}(E)\}.$$

Lemma J.9 (Faithfulness of the degree-zero internal block on the odd closure)**.** *Assume Theorem H.17, Theorem J.1, and Theorem J.4, Lemma J.10, and Lemma J.11. Let U be an even operator on $\mathcal{D}_{\mathrm{sc}}$ such that:*

(i) U preserves the odd closure $W_{\mathrm{sc}} = W_Q \oplus W_S$ under commutator;

(ii) U commutes with P_μ, $M_{\mu\nu}$, and D;

(iii) U acts trivially on the lower odd layer, i.e.

$$[U, Q_\alpha(v)] = [U, \bar{Q}_{\dot\alpha}(v)] = 0 \qquad (v \in E).$$

Then

$$[U, S^\alpha(\lambda)] = [U, \bar{S}^{\dot\alpha}(\lambda)] = 0 \qquad (\lambda \in E^*).$$

In particular, inside the generated degree-zero block, an operator is determined by its action on W_Q.

Proof. Let X belong to one chiral component of W_S. Since U commutes with translations,

$$[P_\mu, [U, X]] = [[P_\mu, U], X] + [U, [P_\mu, X]] = [U, [P_\mu, X]].$$

By Theorem J.1, translations map the upper layer into the lower layer, and U annihilates the lower layer by hypothesis. Hence $[P_\mu, [U, X]] = 0$ for every μ, so $[U, X]$ is a translation-invariant odd generator.

By Theorem H.17, every translation-invariant odd generator lies in the HLS lower layer W_Q. By Lemma J.10, one has $K_Q = \kappa\,\mathbf{1}_E$, so Theorem I.1 and Lemma J.11 give exact scalar D-weights:

$$[D, Q_\alpha(v)] = \Big(\frac{i}{2} + i\kappa\Big) Q_\alpha(v), \qquad [D, S^\alpha(\lambda)] = \Big(-\frac{i}{2} - i\kappa\Big) S^\alpha(\lambda),$$

$$[D, \bar{Q}_{\dot\alpha}(v)] = \Big(\frac{i}{2} - i\kappa\Big) \bar{Q}_{\dot\alpha}(v), \qquad [D, \bar{S}^{\dot\alpha}(\lambda)] = \Big(-\frac{i}{2} + i\kappa\Big) \bar{S}^{\dot\alpha}(\lambda).$$

Because U commutes with D, the commutator $[U, X]$ has the same D-weight as X. But the lower and upper weights are distinct, so a translation-invariant element of W_Q cannot have the D-weight of an element of W_S. Hence $[U, X] = 0$. The barred case is identical. □

Lemma J.10 (The skew part of the lower-layer D-action is scalar)**.** *Assume Theorem I.1, Theorem J.5, and Theorem J.6. Then*

$$[K_Q, A] = 0 \qquad (A \in \mathrm{End}(E)).$$

Consequently,

$$K_Q = \kappa\,\mathbf{1}_E \qquad \textit{for a unique } \kappa \in i\mathbb{R}.$$

Proof. Fix $A \in \mathrm{End}(E)$. By linearity, Theorem J.5 gives

$$[D, T_A] = 0.$$

Apply this commutator identity to $Q_\alpha(w)$ for arbitrary $w \in E$:

$$0 = [D, [T_A, Q_\alpha(w)]] - [T_A, [D, Q_\alpha(w)]].$$

Using $[T_A, Q_\alpha(w)] = Q_\alpha(Aw)$ and the lower-layer normal form from Theorem I.1, one obtains

$$0 = Q_\alpha(K_Q A w) - Q_\alpha(A K_Q w).$$

Lemma H.11 therefore gives

$$(K_Q A - A K_Q) w = 0 \qquad (w \in E),$$

hence $[K_Q, A] = 0$ for every $A \in \mathrm{End}(E)$. The only endomorphisms commuting with all of $\mathrm{End}(E)$ are the scalars, so $K_Q = \kappa\,\mathbf{1}_E$. The skew-Hermiticity of K_Q from Theorem I.1 forces $\kappa \in i\mathbb{R}$. □

Lemma J.11 (Upper-layer dilatation normal form)**.** *Assume Theorem J.4, Lemma J.7, Theorem I.1, and Lemma H.11. Then there exist endomorphisms $B_S, C_S \in \mathrm{End}(E^*)$ such that*

$$[D, S^\alpha(\lambda)] = i\, S^\alpha(B_S \lambda), \qquad [D, \bar{S}^{\dot\alpha}(\lambda)] = i\, \bar{S}^{\dot\alpha}(C_S \lambda),$$

and in fact

$$B_S = -\frac{1}{2}\mathbf{1}_{E^*} - K_Q^\vee, \qquad C_S = -\frac{1}{2}\mathbf{1}_{E^*} + K_Q^\vee.$$

Equivalently,

$$[D, S^\alpha(\lambda)] = -\frac{i}{2} S^\alpha(\lambda) - i S^\alpha(K_Q^\vee \lambda),$$

$$[D, \bar{S}^{\dot\alpha}(\lambda)] = -\frac{i}{2} \bar{S}^{\dot\alpha}(\lambda) + i \bar{S}^{\dot\alpha}(K_Q^\vee \lambda).$$

Proof. Because W_S is finite-dimensional, D preserves W_S, and D commutes with the Lorentz action, Schur-type reasoning on the spinor factor gives endomorphisms B_S and C_S with the displayed form.

Apply D to the bridge relation

$$[P_{\beta\dot\gamma}, S^\alpha(\lambda)] = 2\,\delta^\alpha_\beta\,\bar Q_{\dot\gamma}(\lambda^\sharp).$$

Using $[D, P_{\beta\dot\gamma}] = iP_{\beta\dot\gamma}$ and the barred lower-layer formula from Theorem I.1, one gets

$$\begin{aligned}[D,[P_{\beta\dot\gamma}, S^\alpha(\lambda)]] &= [[D, P_{\beta\dot\gamma}], S^\alpha(\lambda)] + [P_{\beta\dot\gamma}, [D, S^\alpha(\lambda)]]\\ &= i[P_{\beta\dot\gamma}, S^\alpha(\lambda)] + i[P_{\beta\dot\gamma}, S^\alpha(B_S\lambda)]\\ &= 2\,\delta^\alpha_\beta\, i\Big(\bar Q_{\dot\gamma}(\lambda^\sharp) + \bar Q_{\dot\gamma}((B_S\lambda)^\sharp)\Big),\end{aligned}$$

while the left-hand side is also

$$[D, 2\,\delta^\alpha_\beta\,\bar Q_{\dot\gamma}(\lambda^\sharp)] = 2\,\delta^\alpha_\beta\Big(\frac{i}{2}\bar Q_{\dot\gamma}(\lambda^\sharp) - i\bar Q_{\dot\gamma}(K_Q\lambda^\sharp)\Big).$$

Cancelling the common factor and using Lemma H.11 through the $\flat/\sharp$ identification gives

$$(B_S\lambda)^\sharp = -\frac{1}{2}\lambda^\sharp - K_Q\lambda^\sharp,$$

which is exactly $B_S = -\frac{1}{2}\mathbf{1}_{E^*} - K_Q^\vee$.

The barred formula is obtained identically from the bridge

$$[P_{\beta\dot\gamma}, \bar S^{\dot\alpha}(\lambda)] = 2\,\delta^{\dot\alpha}_{\dot\gamma} Q_\beta(\lambda^\sharp)$$

and the unbarred lower-layer formula from Theorem I.1, giving $C_S = -\frac{1}{2}\mathbf{1}_{E^*} + K_Q^\vee$. □

Corollary J.1 (Internal action on all four odd families)**.** *Assume Theorem H.16, Theorem H.17, Theorem J.5, Theorem J.6, Lemma J.7, Lemma J.10, and Lemma J.11. Then for every $A \in \mathrm{End}(E)$, $v \in E$, and $\mu \in E^*$,*

$$[T_A, Q_\alpha(v)] = Q_\alpha(Av), \qquad [T_A, \bar Q_{\dot\alpha}(v)] = -\bar Q_{\dot\alpha}(A^\star v),$$

$$[T_A, S^\alpha(\mu)] = -S^\alpha(A^\vee\mu), \qquad [T_A, \bar S^{\dot\alpha}(\mu)] = \bar S^{\dot\alpha}\big((A\mu^\sharp)^\flat\big).$$

In particular, if $\{e_I\}$ is an orthonormal basis of E, $\{e^I\}$ its dual basis, and $R^I{}_J := T_{e_I, e^J}$, then

$$[R^I{}_J, Q^K_\alpha] = \delta^K_J\, Q^I_\alpha, \qquad [R^I{}_J, \bar Q_{\dot\alpha K}] = -\delta^I_K\,\bar Q_{\dot\alpha J},$$

$$[R^I{}_J, S^\alpha{}_K] = -\delta^I_K\, S^\alpha{}_J, \qquad [R^I{}_J, \bar S^{\dot\alpha K}] = \delta^K_J\,\bar S^{\dot\alpha I}.$$

Proof. The first formula is exactly Theorem J.6.

For the barred lower layer, fix $A \in \mathrm{End}(E)$ and $w \in E$. Because $[T_A, P_\mu] = 0$ by linearity from Theorem J.5, and because $[P_\mu, \bar Q_{\dot\alpha}(w)] = 0$, the commutator

$$Y_{\dot\alpha}(w) := [T_A, \bar Q_{\dot\alpha}(w)]$$

is again a translation-invariant barred odd Weyl-spinor symmetry generator. By Theorem H.17, there exists a unique $B_A w \in E$ such that

$$Y_{\dot\alpha}(w) = \bar Q_{\dot\alpha}(B_A w).$$

Now use the HLS anticommutator and $[T_A, P_{\alpha\dot\beta}] = 0$:

$$\begin{aligned}0 &= [T_A, \{Q_\alpha(v), \bar Q_{\dot\beta}(w)\}]\\ &= \{[T_A, Q_\alpha(v)], \bar Q_{\dot\beta}(w)\} + \{Q_\alpha(v), [T_A, \bar Q_{\dot\beta}(w)]\}\\ &= \{Q_\alpha(Av), \bar Q_{\dot\beta}(w)\} + \{Q_\alpha(v), \bar Q_{\dot\beta}(B_A w)\}\\ &= 2\big(\langle Av, w\rangle_Q + \langle v, B_A w\rangle_Q\big)P_{\alpha\dot\beta}.\end{aligned}$$

Choose $(\alpha, \dot\beta)$ with $P_{\alpha\dot\beta} \neq 0$, using Lemma I.2. Then

$$\langle Av, w\rangle_Q + \langle v, B_A w\rangle_Q = 0 \qquad (v, w \in E),$$

so by definition of $A^\star$,

$$B_A = -A^\star.$$

Thus

$$[T_A, \bar Q_{\dot\alpha}(w)] = -\bar Q_{\dot\alpha}(A^\star w).$$

For the unbarred upper layer, fix $\mu \in E^*$ and define

$$X^\alpha(\mu) := [T_A, S^\alpha(\mu)] + S^\alpha(A^\vee \mu).$$

Using $[T_A, P_{\beta\dot\gamma}] = 0$ and the bridge relation from Lemma J.7,

$$[P_{\beta\dot\gamma}, S^\alpha(\mu)] = 2\,\delta^\alpha_\beta\, \bar Q_{\dot\gamma}(\mu^\sharp),$$

we compute

$$\begin{aligned}[P_{\beta\dot\gamma}, X^\alpha(\mu)] &= [T_A, [P_{\beta\dot\gamma}, S^\alpha(\mu)]] + [P_{\beta\dot\gamma}, S^\alpha(A^\vee\mu)] \\ &= 2\,\delta^\alpha_\beta\, [T_A, \bar Q_{\dot\gamma}(\mu^\sharp)] + 2\,\delta^\alpha_\beta\, \bar Q_{\dot\gamma}((A^\vee\mu)^\sharp).\end{aligned}$$

But $(A^\vee\mu)^\sharp = A^\star\mu^\sharp$, so the barred lower-layer formula just proved gives

$$[T_A, \bar Q_{\dot\gamma}(\mu^\sharp)] = -\bar Q_{\dot\gamma}(A^\star\mu^\sharp) = -\bar Q_{\dot\gamma}((A^\vee\mu)^\sharp).$$

Hence

$$[P_{\beta\dot\gamma}, X^\alpha(\mu)] = 0 \qquad (\beta, \dot\gamma).$$

So $X^\alpha(\mu)$ is translation-invariant. By Theorem H.17, there exists a unique $u \in E$ with

$$X^\alpha(\mu) = Q_\alpha(u).$$

Now use Lemma J.10, writing $K_Q = \kappa\,\mathbf{1}_E$ with $\kappa \in i\mathbb{R}$. Then Theorem I.1 and Lemma J.11 give the scalar D-weights

$$[D, Q_\alpha(u)] = \Big(\frac{i}{2} + i\kappa\Big) Q_\alpha(u), \qquad [D, S^\alpha(\mu)] = \Big(-\frac{i}{2} - i\kappa\Big) S^\alpha(\mu).$$

Since $[T_A, D] = 0$, we obtain

$$[D, X^\alpha(\mu)] = \Big(-\frac{i}{2} - i\kappa\Big) X^\alpha(\mu).$$

But $X^\alpha(\mu) = Q_\alpha(u)$, so the same operator would have to carry both eigenvalues $\frac{i}{2} + i\kappa$ and $-\frac{i}{2} - i\kappa$, which differ by i. Therefore $u = 0$, hence $X^\alpha(\mu) = 0$. This proves

$$[T_A, S^\alpha(\mu)] = -S^\alpha(A^\vee\mu).$$

For the barred upper layer, fix $\mu \in E^*$ and define

$$Z^{\dot\alpha}(\mu) := [T_A, \bar S^{\dot\alpha}(\mu)] - \bar S^{\dot\alpha}\big((A\mu^\sharp)^\flat\big).$$

Again using $[T_A, P_{\beta\dot\gamma}] = 0$ and the second bridge relation from Lemma J.7,

$$[P_{\beta\dot\gamma}, \bar S^{\dot\alpha}(\mu)] = 2\,\delta^{\dot\alpha}_{\dot\gamma}\, Q_\beta(\mu^\sharp),$$

we get

$$\begin{aligned}[P_{\beta\dot\gamma}, Z^{\dot\alpha}(\mu)] &= [T_A, [P_{\beta\dot\gamma}, \bar S^{\dot\alpha}(\mu)]] - [P_{\beta\dot\gamma}, \bar S^{\dot\alpha}((A\mu^\sharp)^\flat)] \\ &= 2\,\delta^{\dot\alpha}_{\dot\gamma}\,[T_A, Q_\beta(\mu^\sharp)] - 2\,\delta^{\dot\alpha}_{\dot\gamma}\,Q_\beta(((A\mu^\sharp)^\flat)^\sharp) \\ &= 2\,\delta^{\dot\alpha}_{\dot\gamma}\,Q_\beta(A\mu^\sharp) - 2\,\delta^{\dot\alpha}_{\dot\gamma}\,Q_\beta(A\mu^\sharp) \\ &= 0.\end{aligned}$$

So $Z^{\dot\alpha}(\mu)$ is translation-invariant. By Theorem H.17, there exists $u \in E$ such that

$$Z^{\dot\alpha}(\mu) = \bar Q_{\dot\alpha}(u).$$

Using again the scalar D-weights,

$$[D, \bar Q_{\dot\alpha}(u)] = \Big(\frac{i}{2} - i\kappa\Big)\bar Q_{\dot\alpha}(u), \qquad [D, \bar S^{\dot\alpha}(\mu)] = \Big(-\frac{i}{2} + i\kappa\Big)\bar S^{\dot\alpha}(\mu),$$

and $[T_A, D] = 0$, we get

$$[D, Z^{\dot\alpha}(\mu)] = \Big(-\frac{i}{2} + i\kappa\Big)Z^{\dot\alpha}(\mu).$$

Again the two eigenvalues differ by i, so $u = 0$. Therefore

$$[T_A, \bar S^{\dot\alpha}(\mu)] = \bar S^{\dot\alpha}\big((A\mu^\sharp)^\flat\big).$$

Finally, specializing to $A = e_I \otimes e^J$ gives the displayed basis formulas. □

Lemma J.12 (Internal $\star$ on the multiplicity block)**.** *The HLS Hermitian form on E defines an involution*

$$A \longmapsto A^\star \qquad \text{on } \operatorname{End}(E)$$

by

$$\langle Av, w\rangle_Q = \langle v, A^\star w\rangle_Q.$$

For rank-one operators,

$$(v \otimes \lambda)^\star = \lambda^\sharp \otimes v^\flat.$$

The $\star$-skew part is

$$\mathfrak{u}(E) := \{A \in \operatorname{End}(E) : A^\star = -A\} \cong \mathfrak{u}(N),$$

and its commutator subalgebra is

$$\mathfrak{su}(E) \cong \mathfrak{su}(N).$$

Proof. For rank-one operators one computes

$$\langle (v \otimes \lambda)w, x\rangle_Q = \lambda(w)\,\langle v, x\rangle_Q = \langle w, (\lambda^\sharp \otimes v^\flat)x\rangle_Q,$$

which gives $(v\otimes\lambda)^\star = \lambda^\sharp \otimes v^\flat$. The final two statements are the standard linear algebra of a finite-dimensional Hermitian space. □

Corollary J.2 (Correct barred mixed-bracket formula)**.** *Assume Theorem J.6, Lemma J.9, Lemma J.8, and Lemma J.12. Then*

$$\{\bar Q_{\dot\alpha}(v), \bar S^{\dot\beta}(\lambda)\} = \lambda(v)\,\bar L^{\dot\beta}{}_{\dot\alpha} - \delta^{\dot\beta}_{\dot\alpha}\,T_{(v\otimes\lambda)^\star}.$$

Proof. Write the barred same-chirality mixed anticommutator in the form

$$\{\bar{Q}_{\dot\alpha}(v), \bar{S}^{\dot\beta}(\lambda)\} = \lambda(v)\, \bar{L}^{\dot\beta}{}_{\dot\alpha} + \delta^{\dot\beta}_{\dot\alpha}\, B_{v,\lambda}.$$

Apply the super-Jacobi identity to the triple $(Q_\gamma(w), \bar{Q}_{\dot\alpha}(v), \bar{S}^{\dot\beta}(\lambda))$. Because $\{Q, \bar{S}\} = 0$, the right-hand side reduces to the P–$\bar{S}$ bridge term; using the HLS anticommutator and the normalized bridge relation of Lemma J.7 gives

$$[B_{v,\lambda}, Q_\gamma(w)] = Q_\gamma\big((v \otimes \lambda)^\star w\big).$$

By Theorem J.6, the internal operator $T_{(v\otimes\lambda)^\star}$ has the same action on the full lower odd layer. Set

$$U := B_{v,\lambda} + T_{(v\otimes\lambda)^\star}.$$

Then U acts trivially on the lower layer. Since U is degree-zero internal, it commutes with translations, Lorentz generators, and D; by Lemma J.9, it therefore acts trivially on the entire odd closure. Thus, inside the generated degree-zero internal block,

$$B_{v,\lambda} = -\,T_{(v\otimes\lambda)^\star},$$

which is the displayed formula. □

Theorem J.7 (Closure-generated internal saturation)**.** *Assume Theorem J.6, Lemma J.9, and Lemma J.12. Then the internal degree-zero block is a faithful copy of* $\mathrm{End}(E)$*: for all* $A, B \in \mathrm{End}(E)$*,*

$$R_{\mathrm{cl},\mathbb{C}} = \{T_A : A \in \mathrm{End}(E)\}, \qquad [T_A, T_B] = T_{[A,B]}.$$

Under this identification, the $\star$*-skew part is* $\mathfrak{u}(E) \cong \mathfrak{u}(N)$*, and its commutator subalgebra is* $\mathfrak{su}(E) \cong \mathfrak{su}(N)$*.*

Proof. Theorem J.6 identifies every rank-one operator $v \otimes \lambda$ with $T_{v,\lambda}$, and such rank-one operators span $\mathrm{End}(E)$. Hence

$$R_{\mathrm{cl},\mathbb{C}} = \{T_A : A \in \mathrm{End}(E)\}.$$

For the commutator law, let $A, B \in \mathrm{End}(E)$. Then for every $w \in E$,

$$[[T_A, T_B], Q_\alpha(w)] = Q_\alpha([A, B]w) = [T_{[A,B]}, Q_\alpha(w)].$$

The difference $[T_A, T_B] - T_{[A,B]}$ is again degree-zero internal, so Lemma J.9 shows that it vanishes on the odd closure. Thus $[T_A, T_B] = T_{[A,B]}$ in the generated internal block.

The $\star$-skew part and its commutator subalgebra are exactly $\mathfrak{u}(E)$ and $\mathfrak{su}(E)$ by Lemma J.12. □

Theorem J.8 (The S-$\bar{S}$ anticommutator)**.** *Assume Theorem J.1, Theorem J.5, Corollary J.2, Theorem J.6, and Corollary I.1. Then on* $\mathcal{D}_{\mathrm{sc}}$ *one has*

$$\{S^\alpha(\lambda), \bar{S}^{\dot\beta}(\mu)\} = 2\,\langle \mu^\sharp, \lambda^\sharp\rangle_Q\, K^{\alpha\dot\beta},$$

with the coefficient fixed by the normalization conventions of Lemma J.7. Moreover,

$$\{S, S\} = \{\bar{S}, \bar{S}\} = 0.$$

Proof. By Corollary I.1, the upper odd generators have exact D-weight $-\frac{1}{2}$. Therefore $\{S^\alpha(\lambda), \bar{S}^{\dot\beta}(\mu)\}$ is an even operator of D-weight -1 and Lorentz type $(\frac{1}{2}, \frac{1}{2})$. The internal block has weight 0 and no vector piece, so inside the even block there is only one possible generator of this type, namely $K^{\alpha\dot\beta}$. Hence there exists a bilinear scalar $c(\lambda, \mu)$ such that

$$\{S^\alpha(\lambda), \bar{S}^{\dot\beta}(\mu)\} = c(\lambda, \mu)\, K^{\alpha\dot\beta}.$$

Apply $[P_{\gamma\dot\delta}, -]$ to both sides. Using the normalized bridge relations of Lemma J.7 gives

$$[P_{\gamma\dot\delta}, \{S^\alpha(\lambda), \bar{S}^{\dot\beta}(\mu)\}] = 2\delta^\alpha_\gamma \{\bar{Q}_{\dot\delta}(\lambda^\sharp), \bar{S}^{\dot\beta}(\mu)\} + 2\delta^{\dot\beta}_{\dot\delta} \{S^\alpha(\lambda), Q_\gamma(\mu^\sharp)\}.$$

Now insert the barred mixed formula and the unbarred mixed formula:

$$\{\bar{Q}_{\dot{\delta}}(\lambda^\sharp), \bar{S}^{\dot{\beta}}(\mu)\} = \mu(\lambda^\sharp)\, \bar{L}^{\dot{\beta}}{}_{\dot{\delta}} - \delta^{\dot{\beta}}_{\dot{\delta}}\, T_{(\lambda^\sharp \otimes \mu)^\star},$$

$$\{S^\alpha(\lambda), Q_\gamma(\mu^\sharp)\} = \mu(\lambda^\sharp)\, L_\gamma{}^\alpha + \delta^\alpha_\gamma\, T_{\mu^\sharp \otimes \lambda}.$$

But

$$(\lambda^\sharp \otimes \mu)^\star = \mu^\sharp \otimes \lambda,$$

so the internal terms cancel exactly. Therefore

$$[P_{\gamma\dot{\delta}}, \{S^\alpha(\lambda), \bar{S}^{\dot{\beta}}(\mu)\}] = 2\,\langle \mu^\sharp, \lambda^\sharp \rangle_Q \big(\delta^\alpha_\gamma \bar{L}^{\dot{\beta}}{}_{\dot{\delta}} + \delta^{\dot{\beta}}_{\dot{\delta}} L_\gamma{}^\alpha\big),$$

and comparison with the conformal commutator $[P_{\gamma\dot{\delta}}, K^{\alpha\dot{\beta}}]$ fixes the scalar and yields the displayed formula.

For same-chirality upper anticommutators, $\{S, S\}$ and $\{\bar{S}, \bar{S}\}$ have D-weight -1 but transform in same-chirality spinor-square representations rather than vector type, while the internal block has Lorentz type $(0, 0)$. Hence both vanish. □

Corollary J.3 (Internal generators commute with K)**.** *Under the hypotheses of Theorem J.8,*

$$[T_A, K_\mu] = 0 \qquad (A \in \mathrm{End}(E)).$$

Hence the internal operators commute with the full even conformal algebra.

Proof. Fix a rank-one operator $A = v \otimes \lambda$. Apply $[K^{\gamma\dot{\delta}}, -]$ to the unbarred mixed-bracket formula

$$\{Q_\alpha(v), S^\beta(\lambda)\} = \lambda(v) L_\alpha{}^\beta + \delta^\beta_\alpha T_A.$$

Since $[K, S] = 0$, the left-hand side becomes

$$\{[K^{\gamma\dot{\delta}}, Q_\alpha(v)], S^\beta(\lambda)\} = 2\,\delta^\gamma_\alpha \{\bar{S}^{\dot{\delta}}(v^\flat), S^\beta(\lambda)\}.$$

By Theorem J.8, this is exactly the conformal commutator with the $\lambda(v) L_\alpha{}^\beta$ term, leaving no internal remainder. Therefore

$$[K^{\gamma\dot{\delta}}, T_A] = 0.$$

Since the rank-one operators span $\mathrm{End}(E)$, linearity gives $[K_\mu, T_A] = 0$ for every $A \in \mathrm{End}(E)$. □

Corollary J.4 (Normalized dilatation has the standard K-commutator)**.** *Assume Corollary I.1 and Corollary J.3. Then*

$$[D_{\mathrm{std}}, K_\mu] = -iK_\mu.$$

Proof. By Definition B.19,

$$[D, K_\mu] = -iK_\mu.$$

Since $R_{K_Q} = T_{K_Q}$ is internal, Corollary J.3 gives

$$[R_{K_Q}, K_\mu] = 0.$$

Therefore

$$[D_{\mathrm{std}}, K_\mu] = [D - iR_{K_Q}, K_\mu] = [D, K_\mu] - i[R_{K_Q}, K_\mu] = -iK_\mu.$$ □

Remark J.1 (Basis form of the internal generators)**.** Let $\{e_I\}$ be an orthonormal basis of E, let $\{e^I\}$ be the dual basis, and define

$$Q^I_\alpha := Q_\alpha(e_I), \qquad \bar{Q}_{\dot\alpha I} := \bar{Q}_{\dot\alpha}(e_I), \qquad S^\alpha{}_I := S^\alpha(e^I), \qquad \bar{S}^{\dot\alpha I} := \bar{S}^{\dot\alpha}(e^I),$$

$$R^I{}_J := T_{e_I, e^J}.$$

Then

$$[R^I{}_J, Q^K_\alpha] = \delta^K_J\, Q^I_\alpha, \qquad [R^I{}_J, \bar{Q}_{\dot\alpha K}] = -\delta^I_K\, \bar{Q}_{\dot\alpha J},$$

$$[R^I{}_J, S^\alpha{}_K] = -\delta^I_K\, S^\alpha{}_J, \qquad [R^I{}_J, \bar{S}^{\dot\alpha K}] = \delta^K_J\, \bar{S}^{\dot\alpha I}.$$

The trace part

$$r := \sum_I R^I{}_I$$

spans the $\mathfrak{u}(1)$ factor of $\mathfrak{u}(N)$. By Lemma J.10,

$$T_{K_Q} = \kappa r,$$

so the shift $D_{\mathrm{std}} - D = -i\kappa r$ lies in the central trace direction.

4. Full superconformal realization on the same UV Hilbert space

a. Normalized dilatation convention. From this point onward, once Corollary I.1, Corollary J.3, and Corollary J.4 are available, we write D for the normalized operator D_{std}. Because

$$D_{\mathrm{std}} - D = -i\kappa r$$

is a central internal trace generator, this replacement preserves all conformal commutators and all commutators with the internal block. It also changes the degree-zero splitting only by a basis change in the abelian span of the conformal trace direction and the internal trace direction. Consequently the block-matrix identification of the final superalgebra is unchanged.

Theorem J.9 (Explicit bracket table on the common core)**.** *Assume Theorem J.1, Theorem J.4, Theorem J.5, Theorem J.6, Corollary J.1, Corollary J.2, Theorem J.7, Corollary I.1, Theorem J.8, and Corollary J.3, and Corollary J.4. Then on the common invariant core $\mathcal{D}_{\mathrm{sc}}$ the generators*

$$P_\mu,\ M_{\mu\nu},\ D,\ K_\mu,\ Q_\alpha(v),\ \bar{Q}_{\dot\alpha}(v),\ S^\alpha(\lambda),\ \bar{S}^{\dot\alpha}(\lambda),\ T_{v,\lambda}$$

satisfy the following relations:

(i) the even conformal generators satisfy the standard conformal relations from Definition B.19, while the internal generators satisfy

$$[T_{v,\lambda}, T_{w,\mu}] = T_{\lambda(w)v,\mu} - T_{w,\mu(v)\lambda},$$

which is the faithful $\mathrm{End}(E)$ *commutator of the internal block $A \mapsto T_A$, and*

$$[T_{v,\lambda}, P_\mu] = [T_{v,\lambda}, M_{\mu\nu}] = [T_{v,\lambda}, D] = [T_{v,\lambda}, K_\mu] = 0;$$

(ii) the even conformal algebra acts on the odd generators by the standard two-layer conformal-spinor representation:

$$[P, Q] = 0, \qquad [K, S] = 0, \qquad [D, Q] = \frac{i}{2} Q, \qquad [D, S] = -\frac{i}{2} S,$$

together with the barred analogues and the normalized bridging relations of Lemma J.7; the internal algebra acts by the rank-one formulas from Theorem J.6 and Corollary J.1;

(iii) the nonzero odd anticommutators are exactly

$$\{Q_\alpha(v), \bar{Q}_{\dot\beta}(w)\} = 2\,\langle v, w\rangle_Q\, P_{\alpha\dot\beta},$$

$$\{S^\alpha(\lambda), \bar{S}^{\dot\beta}(\mu)\} = 2\,\langle \mu^\sharp, \lambda^\sharp\rangle_Q\, K^{\alpha\dot\beta},$$

$$\{Q_\alpha(v), S^\beta(\lambda)\} = \lambda(v)\, L_\alpha{}^\beta + \delta_\alpha^\beta\, T_{v,\lambda},$$

$$\{\bar{Q}_{\dot\alpha}(v), \bar{S}^{\dot\beta}(\lambda)\} = \lambda(v)\, \bar{L}^{\dot\beta}{}_{\dot\alpha} - \delta_{\dot\alpha}^{\dot\beta}\, T_{(v\otimes\lambda)^\star},$$

while

$$\{Q, Q\} = \{\bar{Q}, \bar{Q}\} = \{S, S\} = \{\bar{S}, \bar{S}\} = 0, \qquad \{Q, \bar{S}\} = \{\bar{Q}, S\} = 0.$$

Proof. Item (i): the even conformal commutators are those of Definition B.19, now with the normalized generator $D = D_{\mathrm{std}}$, with the D-K relation supplied by Corollary J.4. The internal degree-zero block is the faithful copy $A \mapsto T_A$ of $\mathrm{End}(E)$ constructed above. For $A, B \in \mathrm{End}(E)$, Jacobi on the lower layer gives

$$[[T_A, T_B], Q_\alpha(w)] = Q_\alpha([A, B]w).$$

By Theorem J.7, this is exactly the commutator law $[T_A, T_B] = T_{[A,B]}$, whose rank-one form is the displayed formula. All internal generators commute with P_μ, $M_{\mu\nu}$, and the normalized D by construction, and with K_μ by Corollary J.3.

Item (ii): the lower and upper exact half-weight relations are those of Corollary I.1. The bridge relations are those of Lemma J.7. The internal action formulas are the faithful $A \mapsto T_A$ action on the lower layer together with the rank-one formulas of Corollary J.1.

Item (iii): the Q-$\bar{Q}$ relation is the HLS seed relation. The S-$\bar{S}$ relation is Theorem J.8. The unbarred mixed bracket is Theorem J.5, and the barred mixed bracket is Corollary J.2. The same-chirality upper anticommutators vanish by Theorem J.8, while the cross-chiral mixed anticommutators vanish by Theorem J.5. Thus every graded bracket closes on $\mathcal{D}_{\mathrm{sc}}$ and no extra odd or even generators appear. □

Theorem J.10 (Same-$\mathcal{H}$ realization of the full 4D superconformal algebra)**.** *Assume Theorem J.9. Then on the same UV Hilbert space $\mathcal{H}$ there exist densely defined operators*

$$P_\mu,\ M_{\mu\nu},\ D,\ K_\mu,\ Q^I_\alpha,\ \bar{Q}_{\dot\alpha I},\ S^\alpha{}_I,\ \bar{S}^{\dot\alpha I},\ R^I{}_J$$

with common invariant core $\mathcal{D}_{\mathrm{sc}}$ such that:

(i) the even operators realize the UV conformal algebra together with the internal algebra $\mathfrak{u}(N)$;

(ii) the odd operators realize the two weight layers of the standard 4D superconformal odd module, with $Q, \bar{Q}$ at weight $+\frac{1}{2}$ and $S, \bar{S}$ at weight $-\frac{1}{2}$;

(iii) all graded brackets close on $\mathcal{D}_{\mathrm{sc}}$;

(iv) the resulting operator Lie superalgebra is the full 4D superconformal algebra.

More precisely, by the standard four-dimensional superconformal classification picture [17, 34], for $N \neq 4$ the resulting algebra is $\mathfrak{su}(2,2|N)$, while in the $N = 4$ case the standard simple algebra is obtained after quotienting the central $\mathfrak{u}(1)$, giving $\mathfrak{psu}(2,2|4)$.

Proof. Let

$$V := S_+ \oplus S_- \qquad (\dim_{\mathbb{C}} V = 4).$$

Let the odd multiplicity space be E. The bracket table gives the complexified conformal block as the V-block $\mathfrak{sl}(V) \cong \mathfrak{sl}(4, \mathbb{C})$, the internal block as the faithful E-block $\mathrm{End}(E)$, and the odd generators as the off-diagonal blocks

$$Q, \bar{S} \in \mathrm{Hom}(E, V), \qquad S, \bar{Q} \in \mathrm{Hom}(V, E).$$

With the mixed-bracket formulas, the odd-odd anticommutators are exactly the block compositions whose $\mathrm{End}(V)$ component gives the conformal generators and whose $\mathrm{End}(E)$ component gives the internal operators T_A. Therefore the complexified generated superalgebra is the standard block-matrix algebra

$$\mathfrak{sl}(V|E) \cong \mathfrak{sl}(4|N).$$

The real form is the standard one: the conformal block is $\mathfrak{su}(2,2)$ by the Hilbert-space adjoint on the conformal generators, the odd generators pair as $Q^\dagger = \bar{Q}$ and $S^\dagger = \bar{S}$, and the internal real block is the $\star$-skew part $\mathfrak{u}(E) \cong \mathfrak{u}(N)$ supplied by the HLS Hermitian form on E. These are exactly the standard reality conditions for $\mathfrak{su}(2,2|N)$. For $N = 4$, the scalar superidentity has zero supertrace and is central, so quotienting it gives $\mathfrak{psu}(2,2|4)$. This proves items (i)–(iv). $\square$

This establishes the same-$\mathcal{H}$ superconformal closure.

Appendix K: Subprogram 4c: Maximality of supersymmetry at the UV fixed point

This subprogram is an optional strengthening of the superconformal closure obtained in Subprogram 4b. We take that closure as imported input and ask what additional assumptions would force maximal supersymmetry at the UV fixed point. The answer is conditional: under a short list of extra assumptions about non–gravitationality, interaction, and symmetry selection, the superconformal symmetry must be $\mathcal{N} = 4$ (i.e. $N = 4$ copies of Weyl supercharges).

1. Inputs and goal

We begin by collecting the earlier outputs used in this optional strengthening.

- Subprogram 1b produced the Wightman vacuum representation $(\mathcal{H}, \pi, U, \Omega)$ with locality, positivity, and positive–energy Poincaré covariance (Theorem D.1).
- Subprogram 3c produced the HLS odd charge module on that UV boundary representation, together with the operator implementation of the odd symmetry derivation (Theorem H.16 and Proposition D.1).
- Subprogram 4a showed that on the common UV sector, any nontrivial odd charge Q^I_α has dilatation weight $\Delta(Q^I_\alpha) = \frac{1}{2}$ (Theorem I.1).
- Subprogram 4b established the same-$\mathcal{H}$ superconformal closure of the odd sector: the generators $P_\mu, M_{\mu\nu}, D, K_\mu, Q, \bar{Q}, S, \bar{S}, R$ close on a common invariant core to the full 4D superconformal algebra (Theorem J.10). At Main level the same conclusion is packaged in Corollary H.5.

In the present subprogram we:

(1) import from Subprogram 4b the resulting 4D superconformal algebra and write N for its odd multiplicity parameter;

(2) impose that the theory is non–gravitational and has a unique stress tensor, together with standard representation–theoretic input, to show that $N \leq 4$;

(3) add interaction and nonfactorization assumptions so that N measures the intrinsic supersymmetry of the interacting UV fixed point rather than of a theory enlarged by decoupled free sectors;

(4) assume a maximality principle and deduce that, within this optional strengthening, the emergent supersymmetry is maximal, i.e. $N = 4$.

Everything beyond the imported superconformal closure of Subprogram 4b is therefore assumption-driven.

2. Imported superconformal algebra from Subprogram 4b

Proposition K.1 (Imported algebraic form of the superconformal symmetry)**.** *Use Theorem J.10 as imported input. Then on the same UV Hilbert space the resulting 4D superconformal algebra is of standard type: for some integer $N \geq 1$, it is $\mathfrak{su}(2,2|N)$ for $N \neq 4$, while in the $N = 4$ case the standard simple form is obtained after quotienting the central $\mathfrak{u}(1)$, giving $\mathfrak{psu}(2,2|4)$. Throughout this section we write N for this imported multiplicity parameter. When convenient, we speak informally of the $N = 4$ case as $\mathfrak{su}(2,2|4)$, since the central $\mathfrak{u}(1)$ does not affect the arguments below.*

Proof. This is exactly the algebraic form isolated in Theorem J.10; see also Corollary H.5 for the corresponding Main packaging. The Kac–Nahm classification explains why this is the standard 4D superconformal family, but that classification is not rederived here. □

At this stage N is an imported integer label. The purpose of the remaining subsections is to examine what additional assumptions would bound or determine it.

3. Stress tensor multiplet and absence of dynamical gravity

We now impose that our theory is purely field–theoretic, with a unique stress tensor and no dynamical gravity.

Definition K.1 (Stress tensor and its multiplet)**.** Let $T_{\mu\nu}(x)$ be the unique conserved, symmetric, traceless stress tensor of the boundary Wightman theory constructed in Subprogram 1b. In the presence of the imported superconformal symmetry of Proposition K.1, the *stress tensor multiplet* is the smallest unitary positive–energy representation of $\mathfrak{su}(2,2|N)$ on $\mathcal{H}$ whose component fields include $T_{\mu\nu}$ and close under the action of all supercharges $Q^I_\alpha, \bar{Q}_{\dot\alpha I}$ and $S^\alpha{}_I, \bar{S}^{\dot\alpha I}$.

The existence and structure of this multiplet is standard: it contains $T_{\mu\nu}$, the supercurrents $\mathcal{J}^\mu{}_\alpha{}^I, \bar{\mathcal{J}}^\mu{}_{\dot\alpha I}$, the R–symmetry currents, and typically a scalar primary at the bottom, subject to unitarity bounds.

Assumption K.1 (No dynamical gravity (NG))**.** *The boundary theory is non–gravitational: it has a unique conserved stress tensor $T_{\mu\nu}$ generating translations and does not contain any additional massless spin–2 gauge field (graviton) or distinct stress tensors associated with decoupled sectors.*

In the representation theory of superconformal algebras, one typically distinguishes between:

- *matter* or *Yang–Mills* multiplets, which contain fields of spin ≤ 1 (plus the stress tensor in a specific short multiplet), and
- *supergravity multiplets*, which necessarily include a massless spin–2 gauge field (graviton) and additional gauge fields fixed by supersymmetry.

We use the following standard structural input.

Assumption K.2 (Spin–2 saturation for $N > 4$)**.** *In four dimensions, any unitary positive–energy representation of $\mathfrak{su}(2,2|N)$ with more than 16 real supercharges (i.e. $N > 4$) whose component fields include a unique conserved stress tensor necessarily coincides with a supergravity multiplet and thus contains a dynamical graviton. In other words, for $N > 4$ there are no purely "matter" stress tensor multiplets compatible with Assumption K.1.*

Assumption K.2 summarises the standard representation–theoretic lore familiar from the classification of 4D supergravity multiplets: once the number of supercharges exceeds 16, the stress tensor cannot sit in a purely matter multiplet.

Lemma K.1 (Bound on the number of supercharges)**.** *Under Assumptions K.1 and K.2, the superconformal algebra $\mathfrak{g}_{\rm SC}$ constructed in Subprogram 4b has at most 16 real Poincaré supercharges, i.e.*

$$N \leq 4. \tag{K1}$$

Proof. By Proposition K.1, $\mathfrak{g}_{\rm SC}$ is a superconformal algebra of type $\mathfrak{su}(2,2|N)$ for some $N \geq 1$, with the stress tensor $T_{\mu\nu}$ sitting in a stress tensor multiplet in the sense of Definition K.1. Suppose, for contradiction, that $N > 4$. Then by Assumption K.2, this multiplet necessarily contains a dynamical graviton and realises a supergravity multiplet, contradicting Assumption K.1 that our theory is non–gravitational. Hence $N \leq 4$. □

Thus, after imposing non–gravitationality and the standard structure of superconformal multiplets, the only candidates are $N = 1, 2, 3, 4$.

4. Interacting SK fixed points and exclusion of decoupled free sectors

We now sharpen the interpretation of N so that it counts supersymmetries of the *interacting* universality class singled out by the SK RG structure, not of arbitrary decoupled free factors.

Assumption K.3 (Interacting SK fixed point)**.** *The UV fixed point under consideration is* interacting *in the following sense:*

(a) the SK RG monotone $\mathcal{M}$ of Subprogram 3b admits the following strengthening at the UV fixed point: it is strictly monotone along nontrivial RG flows, additive under decoupled tensor products, and has a nondegenerate local extremum there (in particular, no flat directions corresponding to decoupled generalized free sectors);

(b) the GNS Hilbert space $\mathcal{H}$ is generated, as a representation of $\mathfrak{g}_{\mathrm{SC}}$, by the orbit of local operators in the stress tensor multiplet and their descendants; in particular, there is no orthogonal direct summand carrying a decoupled free superconformal theory with an independent stress tensor.

In addition we exclude higher–spin enhanced free theories.

Assumption K.4 (No higher–spin enhancement)**.** *The UV fixed point carries no infinite tower of conserved higher–spin currents beyond the stress tensor and its superconformal descendants. Equivalently, it is not a free or generalized free CFT in the sense of the higher–spin symmetry no–go theorems.*

Assumption K.4 is motivated by the results showing that in unitary CFTs in $d > 2$, an infinite tower of higher–spin symmetries implies that the theory is essentially free.

Lemma K.2 (Intrinsic nature of N for interacting SK fixed points)**.** *Under Assumptions K.3 and K.4, the integer N appearing in Proposition K.1 is an intrinsic invariant of the interacting SK universality class: it cannot be increased by tensoring with decoupled free superconformal sectors without violating the standing assumptions.*

Proof. Suppose we attempt to pass from our interacting fixed point $\mathcal{C}$ with superconformal symmetry $\mathfrak{su}(2,2|N)$ to a product theory $\mathcal{C} \otimes \mathcal{F}$, where $\mathcal{F}$ is a decoupled free SCFT carrying additional supersymmetry $\mathfrak{su}(2,2|N')$ with $N' > 0$. Then:

(1) The product theory has two independent, conserved, symmetric stress tensors, $T^{(\mathcal{C})}_{\mu\nu}$ and $T^{(\mathcal{F})}_{\mu\nu}$, whose sum generates the total translation operator. This violates the uniqueness of the stress tensor in Assumption K.1, unless one of the factors is trivial.

(2) Alternatively, if one attempts to keep only the diagonal combination of the two stress tensors while retaining both sets of odd generators as symmetries of the product, the additional higher–spin currents in the free factor $\mathcal{F}$ violate Assumption K.4.

(3) Finally, the SK RG monotone $\mathcal{M}$ constructed in Subprogram 3b is assumed in Assumption K.3(a) to be additive over tensor products of decoupled sectors and to have a nondegenerate UV extremum; this rules out a nontrivial direct product decomposition of the fixed point.

In all cases, raising the effective number of supercharges by adding a decoupled free factor contradicts the assumptions. It follows that the value of N determined by Proposition K.1 and Lemma K.1 is an invariant of the interacting SK fixed point. □

Combined with Lemma K.1, we thus have:

$$1 \;\leq\; N \;\leq\; 4, \tag{K2}$$

for the interacting, non–gravitational UV fixed point singled out by our SK construction.

5. Maximality principle and selection of $N = 4$

We now articulate the maximality principle that connects the SK RG structure to a preference for maximal supersymmetry.

The remainder of this subsection is explicitly conditional. We do not attempt to derive the maximality principle or the existence statement below inside the present appendix; we record only what they imply once the imported superconformal algebra and the previous assumptions are in place.

Assumption K.5 (Maximality of symmetry at SK fixed points)**.** *Consider the class of interacting, non–gravitational, SK–admissible UV fixed points in four dimensions satisfying the standing Wightman and SK assumptions and equipped with the two–time BRST/RG structure of Subprograms 1a and 3b. Among all such fixed points compatible with a given set of macroscopic constraints (number of dimensions, spin range, locality, etc.), the SK RG dynamics selects those with the* maximal *amount of unbroken symmetry, in the sense that:*

(a) the SK RG monotone $\mathcal{M}$ attains local extrema on fixed points whose symmetry algebra is maximal within their basin of attraction in theory space, and

(b) for fixed macroscopic data, increasing the number of supercharges N (within the allowed algebraic range) cannot be compensated by adding further continuous symmetries without changing the universality class.

Assumption K.5 is an additional assumption of this optional strengthening. It promotes the observed tendency of RG flows to land on highly symmetric fixed points, together with the SK construction of odd symmetries from monotones, into a symmetry-selection rule. It is not derived here.

Assumption K.6 (Existence of interacting SCFTs for all $N \leq 4$)**.** *For each $N \in \{1, 2, 3, 4\}$ there exist interacting, non–gravitational, unitary four–dimensional superconformal field theories with symmetry $\mathfrak{su}(2, 2|N)$ satisfying the standing assumptions (in particular, with a unique stress tensor and compact R–symmetry).*

Assumption K.6 is motivated by the known families of $\mathcal{N} = 1, 2, 4$ SCFTs and the recent constructions of genuinely $\mathcal{N} = 3$ theories; we do not require a complete classification, only that the possible N in the allowed range are all realised.

Proposition K.2 (Maximal supersymmetry: $N = 4$)**.** *Let $\mathcal{C}$ be an interacting, non–gravitational, SK–admissible UV fixed point in four dimensions, equipped with the two-time RG structure and SK monotone of Subprogram 3b, and suppose Assumptions K.1, K.2, K.3, K.4, K.6, and Assumption K.5 hold. Then the superconformal symmetry of $\mathcal{C}$ is maximally supersymmetric:*

$$\mathfrak{g}_{\mathrm{SC}} \cong \mathfrak{su}(2, 2|4), \tag{K3}$$

i.e. the boundary theory has $\mathcal{N} = 4$ superconformal symmetry.

Proof. By Proposition K.1 and Lemma K.1, the superconformal symmetry is of imported type $\mathfrak{su}(2, 2|N)$ with $1 \leq N \leq 4$. By Lemma K.2, N is an intrinsic invariant of the interacting SK universality class. By Assumption K.6, there exist interacting SCFTs realising each N in this range; in particular, there exist candidate fixed points with superconformal symmetry $\mathfrak{su}(2, 2|4)$.

Assumption K.5 then asserts that, within the class of SK–admissible fixed points compatible with a given set of macroscopic constraints, the RG dynamics selects fixed points with maximal symmetry. Specialising to the superconformal symmetries under discussion, this implies that the emergent algebra must correspond to the largest available value of N, namely $N = 4$. Hence $\mathfrak{g}_{\mathrm{SC}} \cong \mathfrak{su}(2, 2|4)$. □

Remark K.1 (Projective form and R–symmetry)**.** The even part of $\mathfrak{su}(2, 2|4)$ is $\mathfrak{so}(4, 2) \oplus \mathfrak{su}(4) \oplus \mathfrak{u}(1)$. In many physical applications the central $\mathfrak{u}(1)$ acts trivially and one works with the projective algebra $\mathfrak{psu}(2, 2|4)$. Our arguments are insensitive to this subtle distinction, which can be fixed later by imposing the absence of conserved currents for the central $\mathfrak{u}(1)$.

6. Summary of Subprogram 4c

We summarise the logical content of this subprogram.

- We import from Subprogram 4b the same-$\mathcal{H}$ realization of a 4D superconformal algebra, written in standard form as $\mathfrak{su}(2, 2|N)$ (with the projective quotient in the $N = 4$ case) (Proposition K.1).
- Under the additional non–gravitationality and spin–2 saturation assumptions, the number of supercharges is bounded by $N \leq 4$ (Lemma K.1).

- Under the additional interaction, nonfactorization, and no–higher–spin assumptions, N is an intrinsic invariant of the interacting SK universality class and cannot be raised by tensoring with decoupled free factors (Lemma K.2).
- Finally, under the additional maximality principle and the existence of interacting SCFTs for each $N \leq 4$, the emergent symmetry at the UV fixed point must be maximally supersymmetric, $\mathfrak{su}(2,2|4)$ (Proposition K.2).

This subsection therefore gives a conditional route from the imported superconformal closure of Subprogram 4b to maximal $\mathcal{N} = 4$ symmetry under the added assumptions above. The conjecture below records the corresponding cosmology-theory interpretation suggested by this optional strengthening.

Conjecture K.1 (String realisation of a $\mathbf{2T}_{\mathrm{Main}}$ cosmology theory)**.** *A cosmology theory that implements the 2T-Main interface* $\mathbf{2T}_{\mathrm{Main}}$ *of Definition B.20 is realised by the* $\mathcal{N} = 4$ *superconformal field theory with symmetry algebra* $\mathfrak{su}(2,2|4)$*, together with its type IIB superstring dual on* $\mathrm{AdS}_5 \times S^5$*.*

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